Quantitative Model of Default

Juan M. Sanchez

February 28, 2018

Washington University in St. Louis

Literature: First Generation Models

- Arellano, C. 2008, . 'Default Risk and Income Fluctuations in Emerging Economies'. American Economic Review.

- Juan C. Hatchondo & L. Martinez & H. Sapriza, 2010. "Quantitative properties of sovereign default models: solution methods," RED.

- Chatterjee Satyajit & Dean Corbae & Makoto Nakajima & José-Víctor Ríos-Rull, (2007). "A Quantitative Theory of Unsecured Consumer Credit with Risk of Default," Econometrica.

- Livshits, Igor , and James MacGee, Michèle Tertilt, 2007. "Consumer Bankruptcy: A Fresh Start," The American Economic Review.

- Simple 2x2 example in my paper title "The IT Revolution and the Unsecured Credit Market," (Economic Inquiry, 2017).

- "Bankruptcy and Delinquency in a Model of Unsecured Debt," (forthcoming, May 2018), with Kartik Athreya, Xuan S. Tam, and Eric R. Young. International Economic Review.

- "Labor market upheaval, default regulations, and consumer debt," (January 2015), with Kartik Athreya, Xuan S. Tam, and Eric R. Young. Review of Economic Dynamics

Literature: Exogenous Long Maturity Bonds

- Satyajit Chatterjee & Burcu Eyigungor, 2012. ”Maturity, Indebtedness, and Default Risk,” American Economic Review, American Economic Association, vol. 102(6), pages 2674-99, October.

- Hatchondo, Juan Carlos & Martinez, Leonardo, 2009. ”Long-duration bonds and sovereign defaults,” Journal of International Economics, Elsevier, vol. 79(1), pages 117-125, September.

- Leonardo Martinez & Juan Carlos Hatchondo & Juan M. Sanchez. ”Mortgage Defaults,” Journal of Monetary Economics, 2017.

- Satyajit Chatterjee & Burcu Eyigungor. ”A Quantitative Analysis of the US Housing and Mortgage Markets and the Foreclosure Crisis,” Working Paper 11-26, July 2011.

Literature: Endogenous Maturity

- Arellano, C. and Ramanarayanan, A. (2012). 'Default and the Maturity Structure in Sovereign Bonds'. Journal of Political Economy, volume 120, no. 2, 187-232.

- Juan Carlos Hatchondo & Leonardo Martinez & Cesar Sosa-Padilla, 2014."Debt Dilution and Sovereign Default Risk," Department of Economics Working Papers 2014-06, McMaster University.

- Sanchez, Juan M. & Sapriza, Horacio & Yurdagul, Emircan, 2014. "Sovereign Default and Maturity Choice" Journal of Monetary Economics, forthcoming.

-

Two-by-two model

- Preview of the main driving forces at work in the full model using a simple example.
- Risk-free interest rate i and wages w are given.
- Individuals are born with assets a_1 and live for 2 periods.
- They are endowed with efficiency units of labor, l_n, which can take 2 values, $\{l_L, l_H\}$.
- The transition probability between state L and H is π_{LH}. Persistence implies $\pi_{HH} > \pi_{LH}$.
- A proportion of income τ is lost after bankruptcy.

Individuals' problem

- Lifetime utility of an individual born with assets a_1 and income $y_{t,n} = wl_{t,n}$ is

$$U(a_1, y_{1,n}) = \max_{a_2 \in \mathbf{A}} \underbrace{u(y_{1,n} + a_1 - q(a_2, n; a_1)a_2)}_{\text{first period utility}}$$

$$\text{high income} \Rightarrow + \beta \pi_{n,H} \max\{u(y_{2,H} + a_2), u(y_{2,H}(1-\tau))\}$$

$$\text{low income} \Rightarrow + \beta \pi_{n,L} \max\{\underbrace{u(y_{2,L} + a_2)}_{\text{pay back}}, \underbrace{u(y_{2,L}(1-\tau))}_{\text{default}}\},$$

Individuals' problem

- Lifetime utility of an individual born with assets a_1 and income $y_{t,n} = wl_{t,n}$ is

$$U(a_1, y_{1,n}) = \max_{a_2 \in \mathbf{A}} \underbrace{u(y_{1,n} + a_1 - q(a_2, n; a_1)a_2)}_{\text{first period utility}}$$

$$\text{high income} \Rightarrow + \beta \pi_{n,H} \max\{u(y_{2,H} + a_2), u(y_{2,H}(1-\tau))\}$$

$$\text{low income} \Rightarrow + \beta \pi_{n,L} \max\{\underbrace{u(y_{2,L} + a_2)}_{\text{pay back}}, \underbrace{u(y_{2,L}(1-\tau))}_{\text{default}}\},$$

where a discount price **q** indicates that a borrower must pay one unit of consumption next period for a loan of size **q** today.

Bankruptcy decision

Bankruptcy decision

- The bankruptcy decision can be characterized by

$$\text{Default} \Rightarrow \quad \underbrace{y_{2,n} + a_2}_{\text{income - debt}} \quad < \quad \underbrace{y_{2,n}(1-\tau),}_{\text{income after punishment}}$$

Bankruptcy decision

- The bankruptcy decision can be characterized by

$$\text{Default} \Rrightarrow \quad \underbrace{y_{2,n} + a_2}_{\text{income - debt}} \; < \; \underbrace{y_{2,n}(1-\tau),}_{\text{income after punishment}}$$

$$\text{Pay back} \Rrightarrow \quad \text{otherwise.}$$

- This determines threshold levels of assets for each level of income,

$$\underline{a}_L = -\tau y_{2,L},$$
$$\underline{a}_H = -\tau y_{2,H}.$$

where $\underline{a}_L > \underline{a}_H$.

- Then individuals

$$\text{pay back:} \quad \text{if } a_2 > \underline{a}_L, \text{ or}$$
$$\text{if } \underline{a}_L > a_2 > \underline{a}_H \text{ and } y_2 = y_H$$

Bankruptcy decision

- The bankruptcy decision can be characterized by

$$\text{Default} \Rrightarrow \quad \underbrace{y_{2,n} + a_2}_{\text{income - debt}} \; < \; \underbrace{y_{2,n}(1-\tau),}_{\text{income after punishment}}$$

$$\text{Pay back} \Rrightarrow \quad \text{otherwise.}$$

- This determines threshold levels of assets for each level of income,

$$\underline{a}_L = -\tau y_{2,L},$$
$$\underline{a}_H = -\tau y_{2,H}.$$

where $\underline{a}_L > \underline{a}_H$.

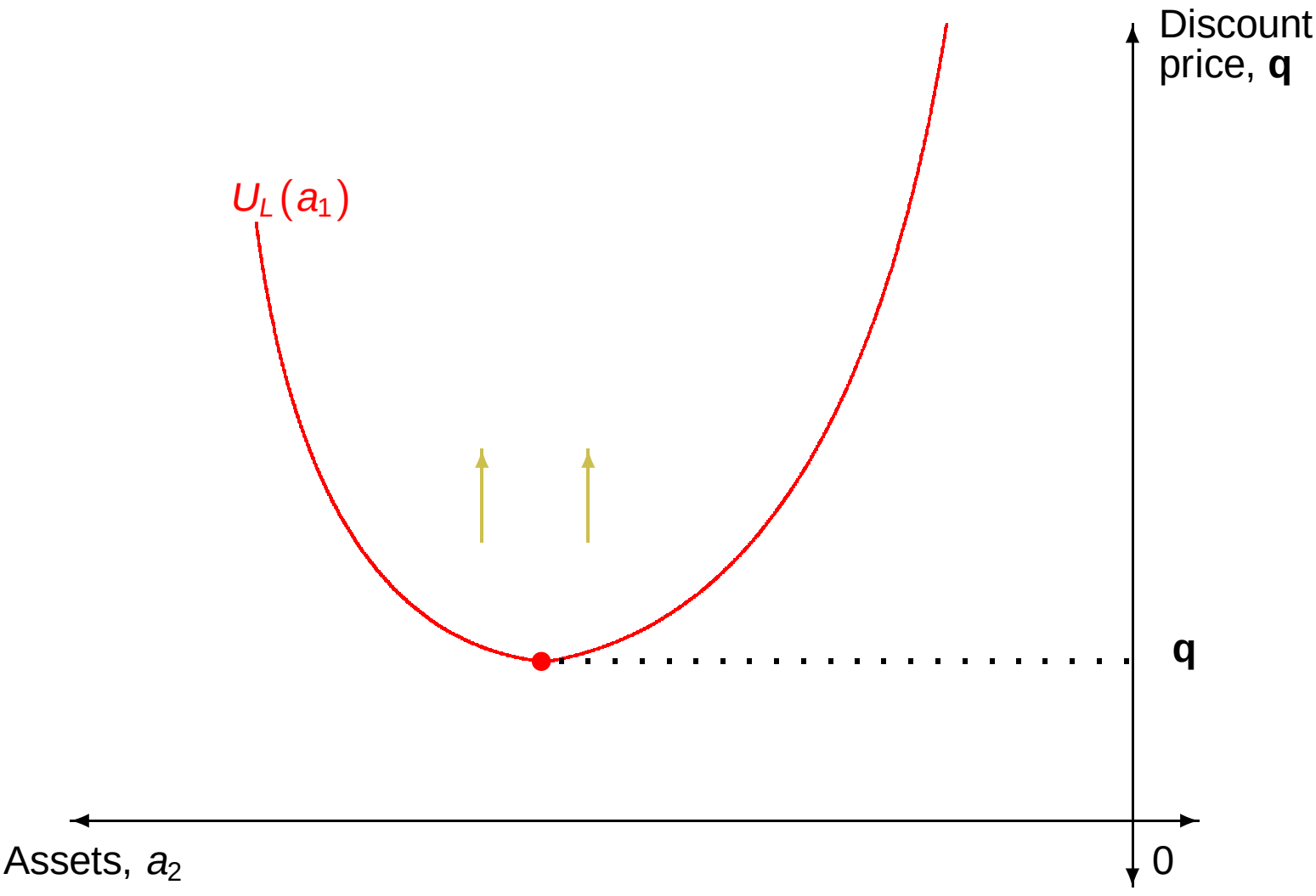
- Then individuals

$$\begin{array}{ll} \text{pay back:} & \text{if } a_2 > \underline{a}_L, \text{ or} \\ & \text{if } \underline{a}_L > a_2 > \underline{a}_H \text{ and } y_2 = y_H \\ \text{default:} & \text{if } \underline{a}_L > a_2 > \underline{a}_H \text{ and } y_2 = y_L, \text{ or} \\ & \text{if } \underline{a}_H > a_2. \end{array}$$

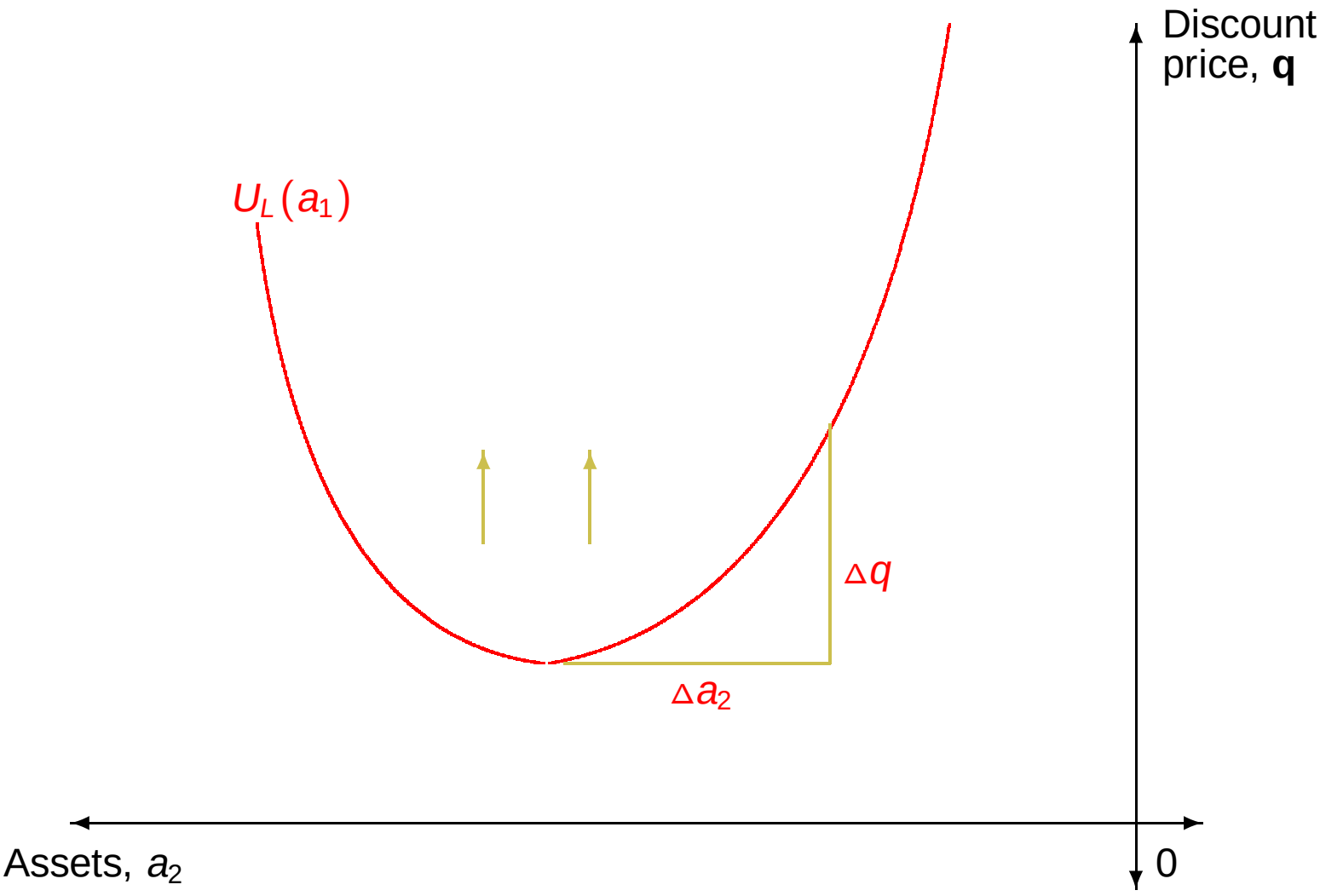
Indifference curves



Indifference curves



Indifference curves



Zero profits and discount prices

Zero profits and discount prices

- Lenders expected profits are

$$\underbrace{q(a_2, n)a_2}_{\text{amount borrowers receive}} - \underbrace{\Pr(\text{repayment} \mid a_2, n)a_2(1+i)^{-1}}_{\text{discounted funds lenders expect to recover}},$$

Zero profits and discount prices

- Lenders expected profits are

$$\underbrace{q(a_2, n)a_2}_{\text{amount borrowers receive}} - \underbrace{\Pr(\text{repayment} \mid a_2, n)a_2(1+i)^{-1}}_{\text{discounted funds lenders expect to recover}},$$

- Then, taking into account the default decision, zero-expected-profit discount prices are

$$q(a_2, n) = \begin{cases} (1+i)^{-1} \\ \pi_{n,H}(1+i)^{-1} \\ 0 \end{cases}$$

Zero profits and discount prices

- Lenders expected profits are

$$\underbrace{q(a_2, n)a_2}_{\text{amount borrowers receive}} - \underbrace{\Pr(\text{repayment} \mid a_2, n)a_2(1+i)^{-1}}_{\text{discounted funds lenders expect to recover}},$$

- Then, taking into account the default decision, zero-expected-profit discount prices are

$$q(a_2, n) = \begin{cases} (1+i)^{-1} & \text{if } \underline{a}_L \leq a_2, \\ \pi_{n,H}(1+i)^{-1} & \\ 0 & \end{cases}$$

Zero profits and discount prices

- Lenders expected profits are

$$\underbrace{q(a_2, n)a_2}_{\text{amount borrowers receive}} - \underbrace{\Pr(\text{repayment} \mid a_2, n)a_2(1+i)^{-1}}_{\text{discounted funds lenders expect to recover}},$$

- Then, taking into account the default decision, zero-expected-profit discount prices are

$$q(a_2, n) = \begin{cases} (1+i)^{-1} & \text{if } \underline{a}_L \leq a_2, \\ \pi_{n,H}(1+i)^{-1} & \text{if } \underline{a}_H \leq a_2 < \underline{a}_L, \\ 0 & \end{cases}$$

Zero profits and discount prices

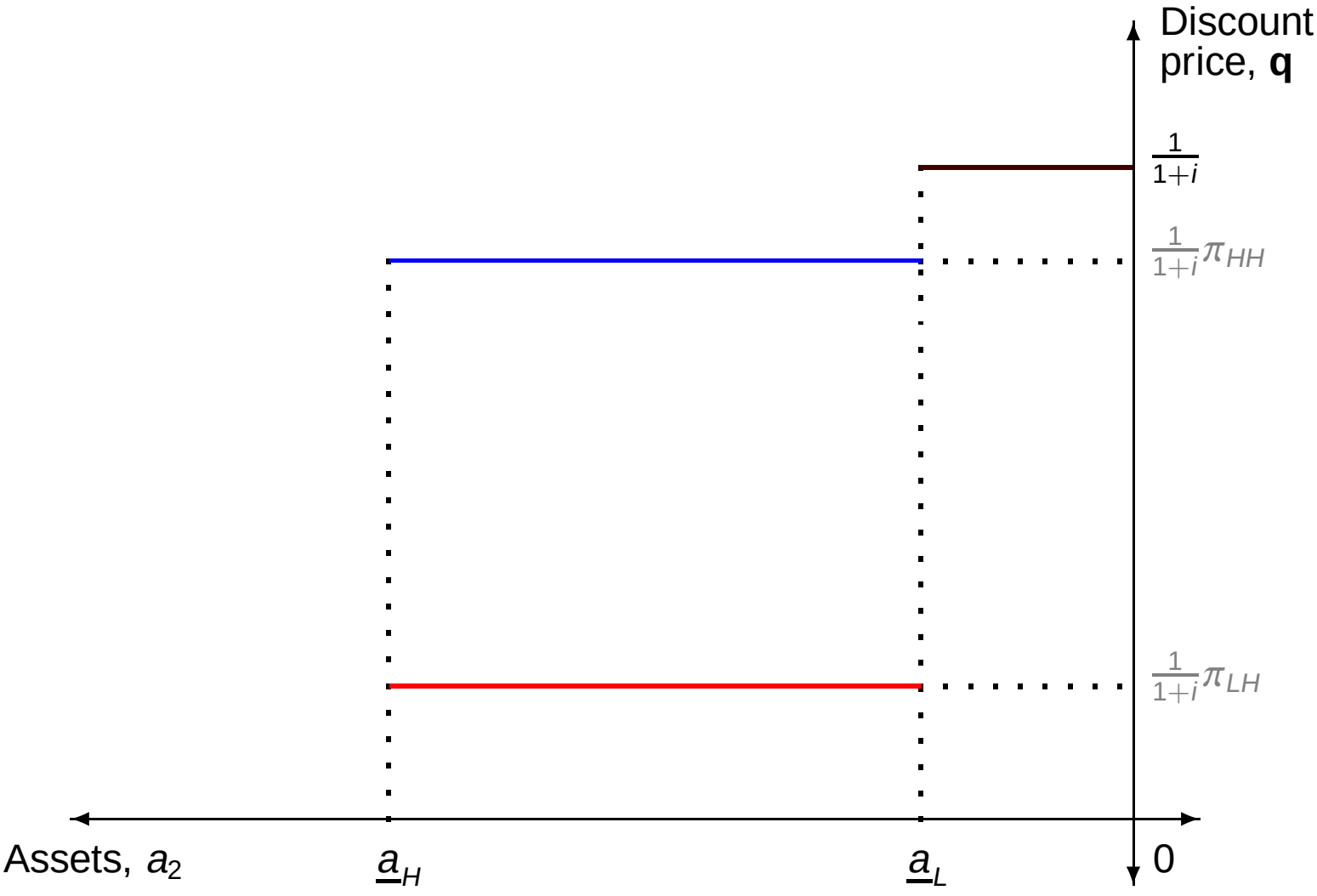
- Lenders expected profits are

$$\underbrace{q(a_2, n)a_2}_{\text{amount borrowers receive}} - \underbrace{\Pr(\text{repayment} \mid a_2, n)a_2(1+i)^{-1}}_{\text{discounted funds lenders expect to recover}},$$

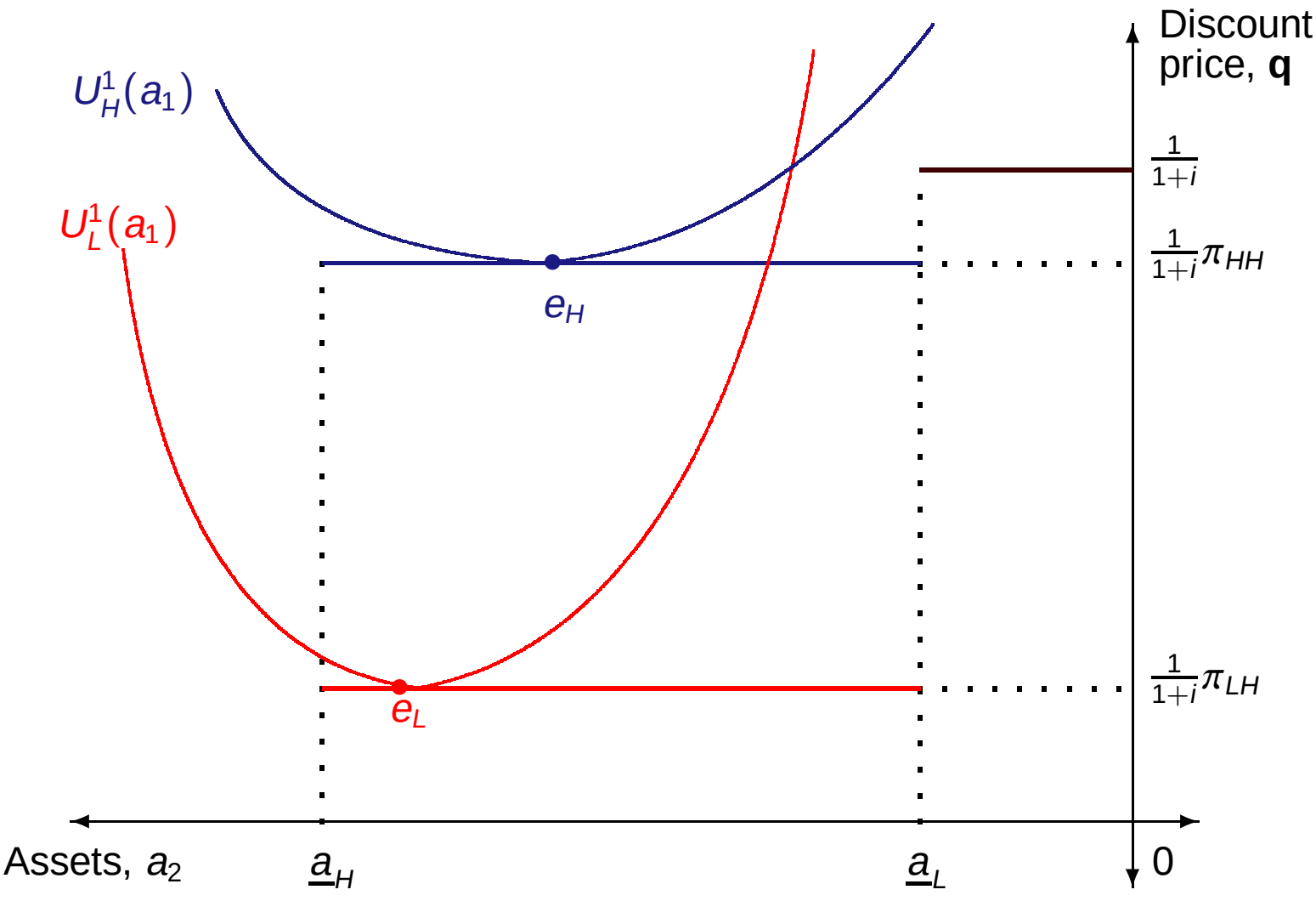
- Then, taking into account the default decision, zero-expected-profit discount prices are

$$q(a_2, n) = \begin{cases} (1+i)^{-1} & \text{if } \underline{a}_L \leq a_2, \\ \pi_{n,H}(1+i)^{-1} & \text{if } \underline{a}_H \leq a_2 < \underline{a}_L, \\ 0 & \text{if } a_2 < \underline{a}_H. \end{cases}$$

Equilibrium price



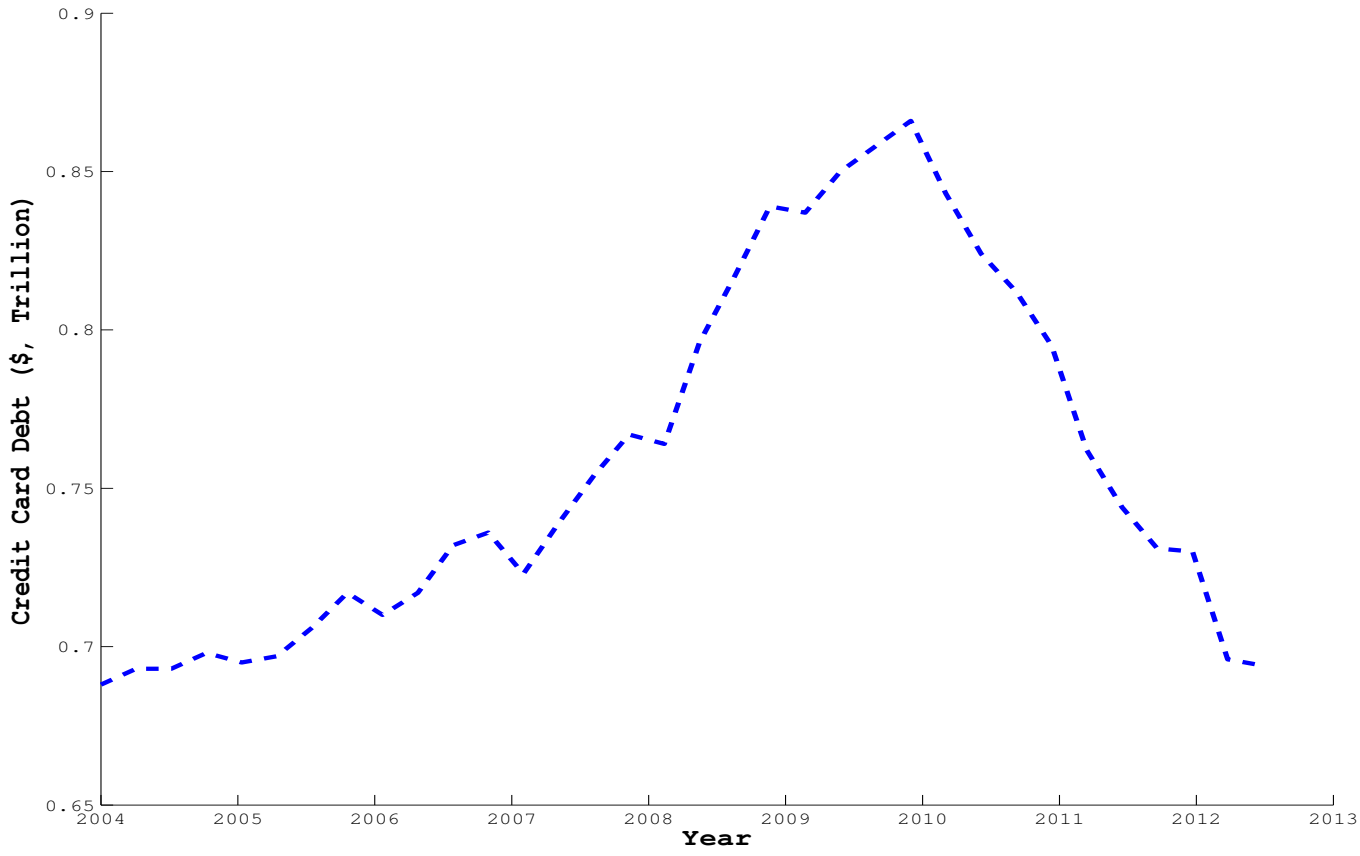
Equilibrium allocation



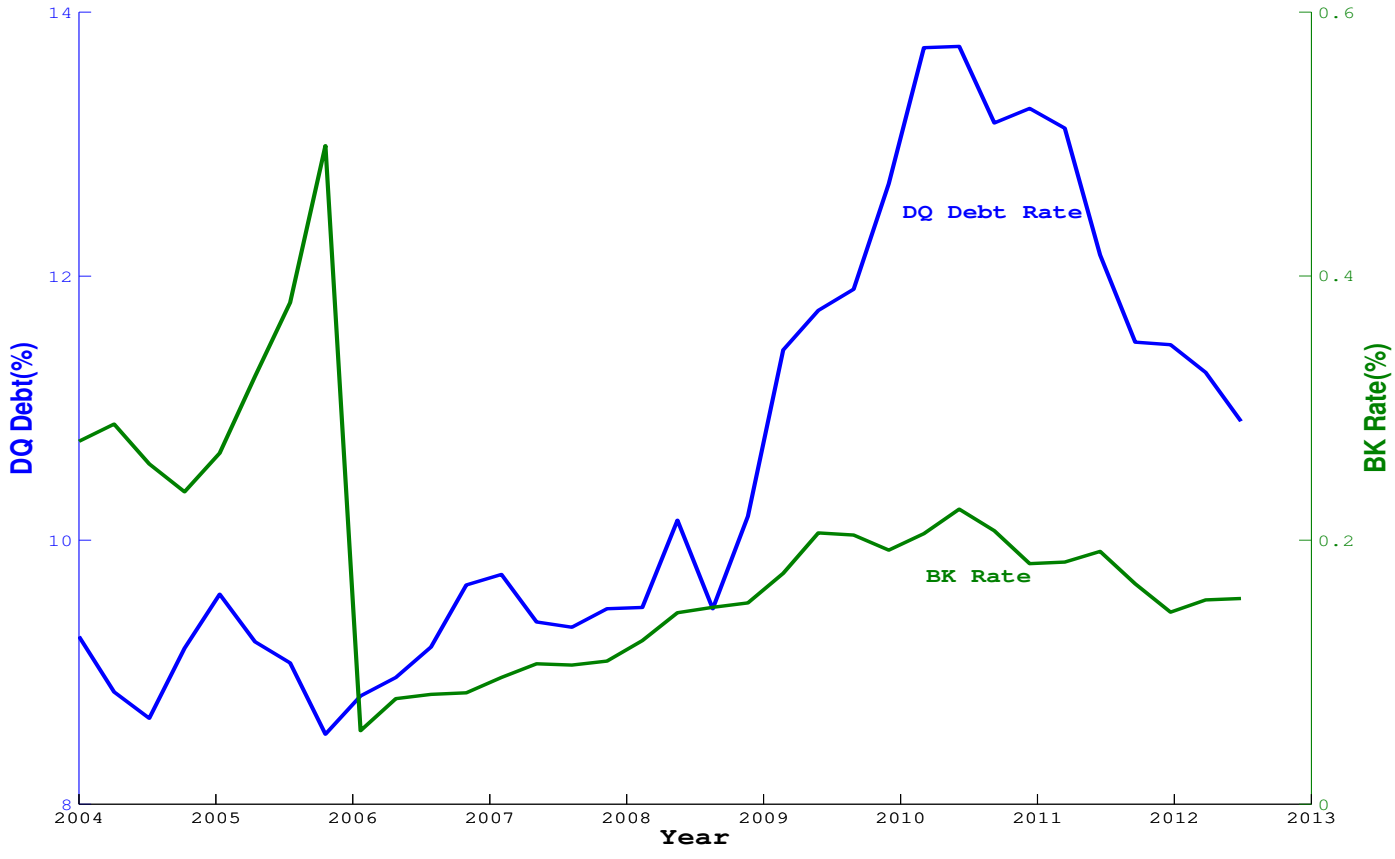
Labor Market Upheaval, Default Regulations, and Consumer Debt

- Joint with Kartik Athreya, Xuan S. Tam, and Eric R. Young
- Great recession:
  - Big drop in consumer debt
  - Big rise in informal default (delinquency)
  - Muted rise in formal default (bankruptcy)

The Facts: Deleveraging



The Facts: Default

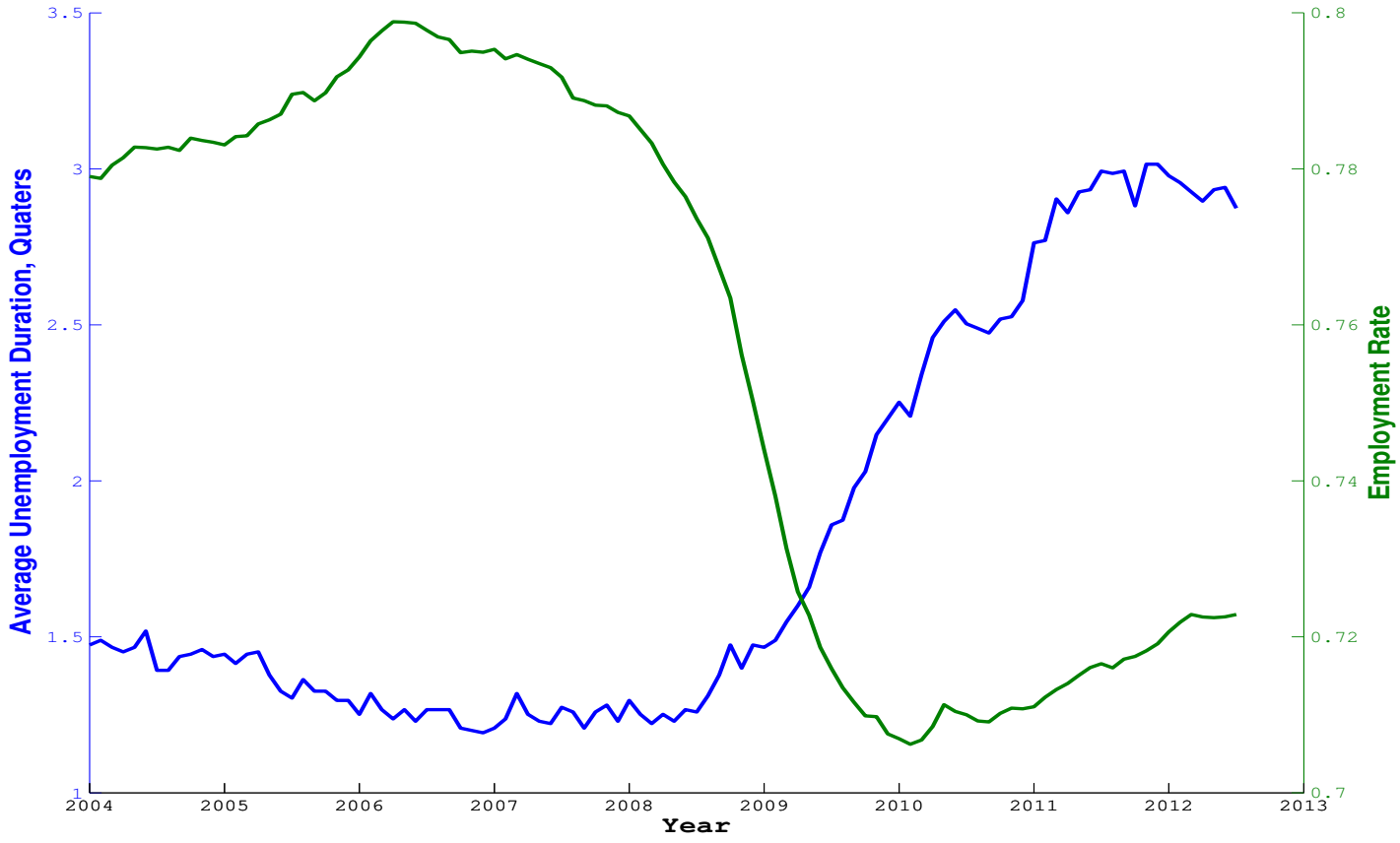


Why?

Two things happened around the same time:

- → Since 2008, big changes in labor market risk
- → In 2005, change in bankruptcy costs: "BAPCPA"

Thing 1: Labor Market Upheaval



Thing 2: Bankruptcy Reform

- In late 2005, BAPCPA roughly doubled bankruptcy filing cost
- Seems to have come as a surprise, at least as of more than one quarter prior to change

This paper

Use our model of "Bankruptcy and Delinquency .." (2012) to answer:

- How did labor market deterioration in the GR affect the path of consumer debt and default?
- How did the bankruptcy reform of 2005 matter for this?

Hint: Morgan (2012) empirical analysis suggests we observed far fewer bankruptcies than we should have, given labor markets.

Model Framework

- Life cycle model with uninsurable idiosyncratic earnings risk:
  - Deterministic education and lifecycle component
  - Persistent component
  - Transitory component
  - Job offers and option to reject
  - (We'll simplify the notation for income process in what follows)
- Individuals can default on debt in two ways:
  - Bankruptcy: incur filing costs and high utility cost, debts are eliminated
  - Delinquency: incur lower utility cost, debt is reset

Optimal Behavior of an indebted household

- $v_{j,e}(b,y) = \max\left\{ \underbrace{v_{j,e}^{d=0}(b,y)}_{\text{solvent}}, \underbrace{v_{j,e}^{d=1}(y)}_{\text{delinquent}}, \underbrace{v_{j,e}^{d=2}(y)}_{\text{bankrupt}} \right\}$

Value Function, Solvent

- $v_{j,e}^{d=0}(b,y) = \max_{b'} \left\{ u(c) + \beta \sum_{y'} \pi(y'|y)\, v_{j+1,e}(b',y') \right\}$
- subject to

$$c + q_{j,e}(b',y)b' = b + y$$

Value Function, DQ

- $v_{j,e}^{d=1}(y) = u(c) - \psi_D + \beta \sum_{y'} \pi(y'|y)\, v_{j+1,e}(h_{j+1,e}(y), y')$
- subject to

$$c = y$$

- key object is $h_{j,e}(\cdot)$, as explained below

Value Function, BK

- $v_{j,e}^{d=2}(y) = u(c) - \psi_B + \beta \sum_{y'} \pi(y'|y)\, v_{j+1,e}(0, y')$
- subject to

$$c = y - \Delta(y)$$

- $\Delta(y)$ will differ by employment status b/c filing costs can be waived

Delinquency and Debt

- If faced with a delinquent borrower, optimizing lenders solve:

$$h_{j,e}(y) = \arg\max_{b} \{-b q_{j,e}(b, y)\}$$

- Price of face value b of new debt:

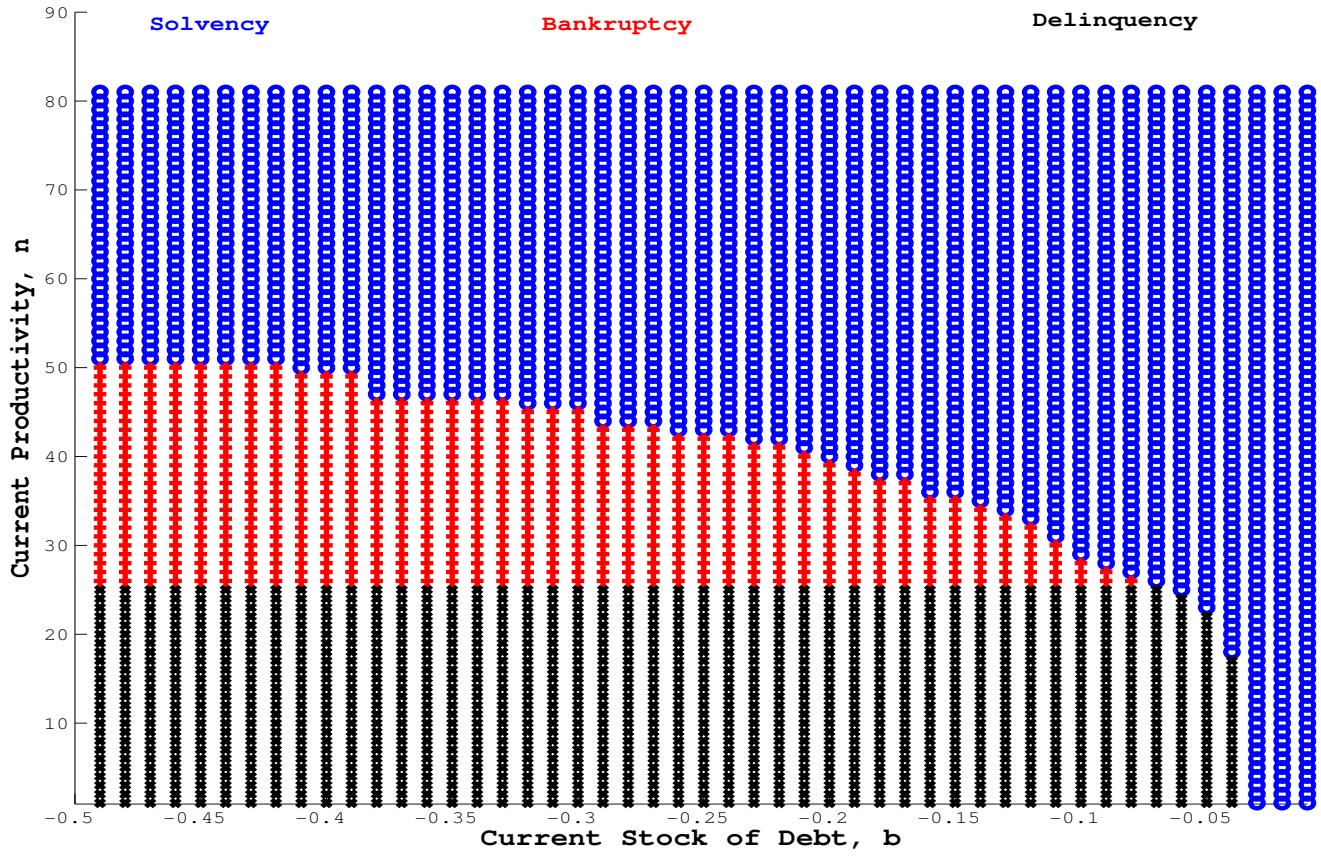
$$q_{j,e}(b, y) = \frac{\sum_{y'} \pi(y'|y)\, Q(b, y')}{1 + r + \phi}$$

where

$$Q(b, y') = \underbrace{1\left(d_{j+1,e}\left(b, y'\right) = 0\right)}_{\text{solvent}} +$$

$$\underbrace{1\left(d_{j+1,e}\left(b, y'\right) = 1\right)\left[\frac{q_{j+1,e}\left(h_{j+1,e}\left(y'\right), y'\right) h_{j+1,e}\left(y'\right)}{b}\right]}_{\textit{delinquent}}$$

Repayment Decisions–Persistent shocks



Calibration: Earnings Risk and Social Insurance

- Quarterly model–key for capturing delinquency
- Follow Low, Meghir, Pistaferri (2010, AER):
  - Wage risk and employment risk
  - UI, DI, Food stamps
  - Wages and employment both risky, depend on education
  - Workers matched with firms, quality specific to current match.

Calibration strategy

- Earnings Risk and Social Insurance taken from Low, Meghir, Pistaferri (2010, AER):
- Other parameters directly taken from data
  - Risk-free interest rate $r = 0.375\%$
  - Lending cost $\phi = 0.75\%$
  - BK filing fee for employed $\$1,200$
  - BK filing fee for unemployed $\$600$
- Risk aversion $\gamma = 2.0$
- Remaining parameters calibrated to match specific targets.

Calibrated Parameters and Targets

Discount factor	β	0.957
Non-pecuniary cost BK	ψ_B	1.786
Non-pecuniary cost DQ	ψ_D	0.104

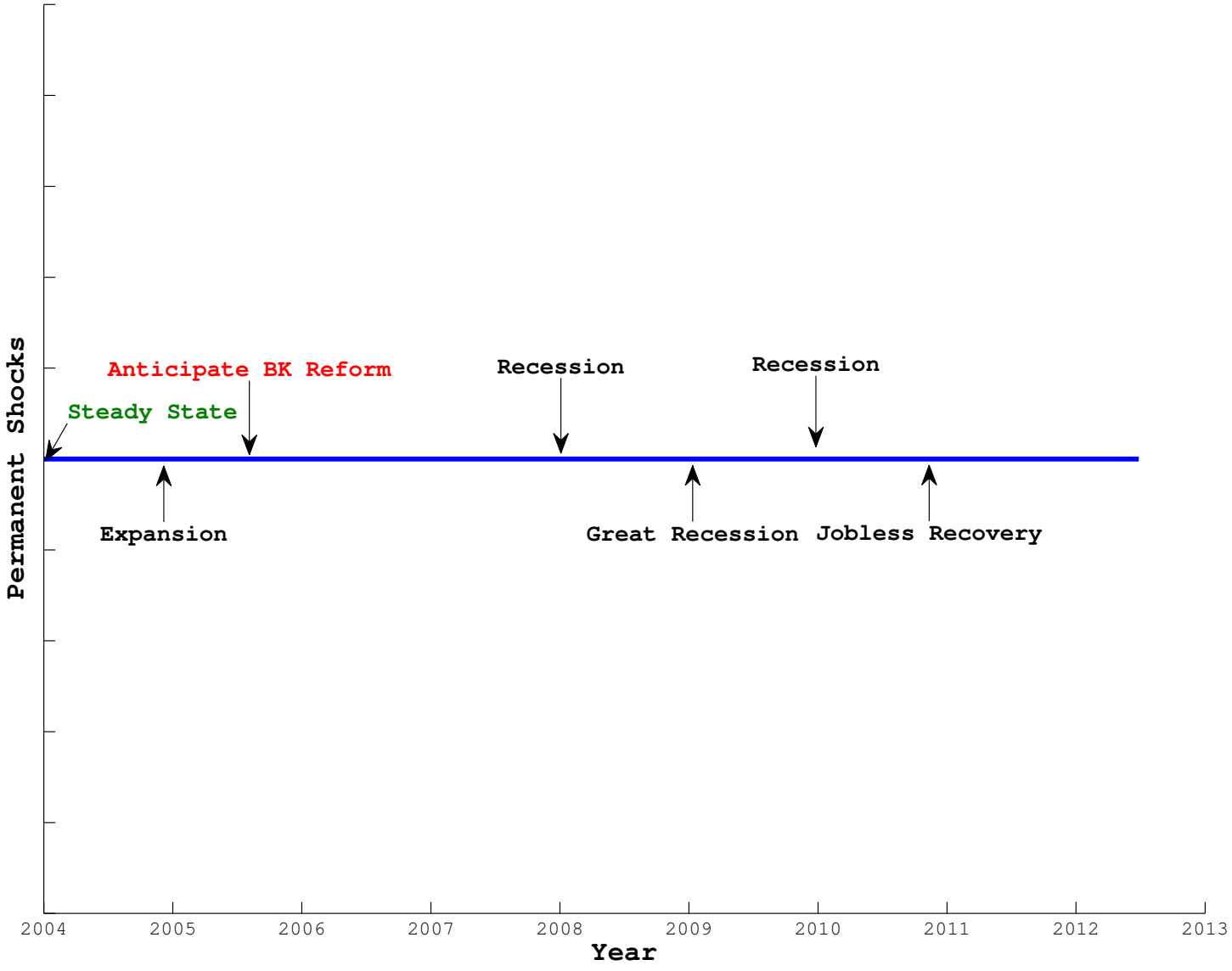
	Data	Model
Share of debt in 90+ DQ, %	8.9	7.8
Bankruptcy rate, %	0.26	0.26
Mean (assets/income)	4.07	3.09

The Experiment

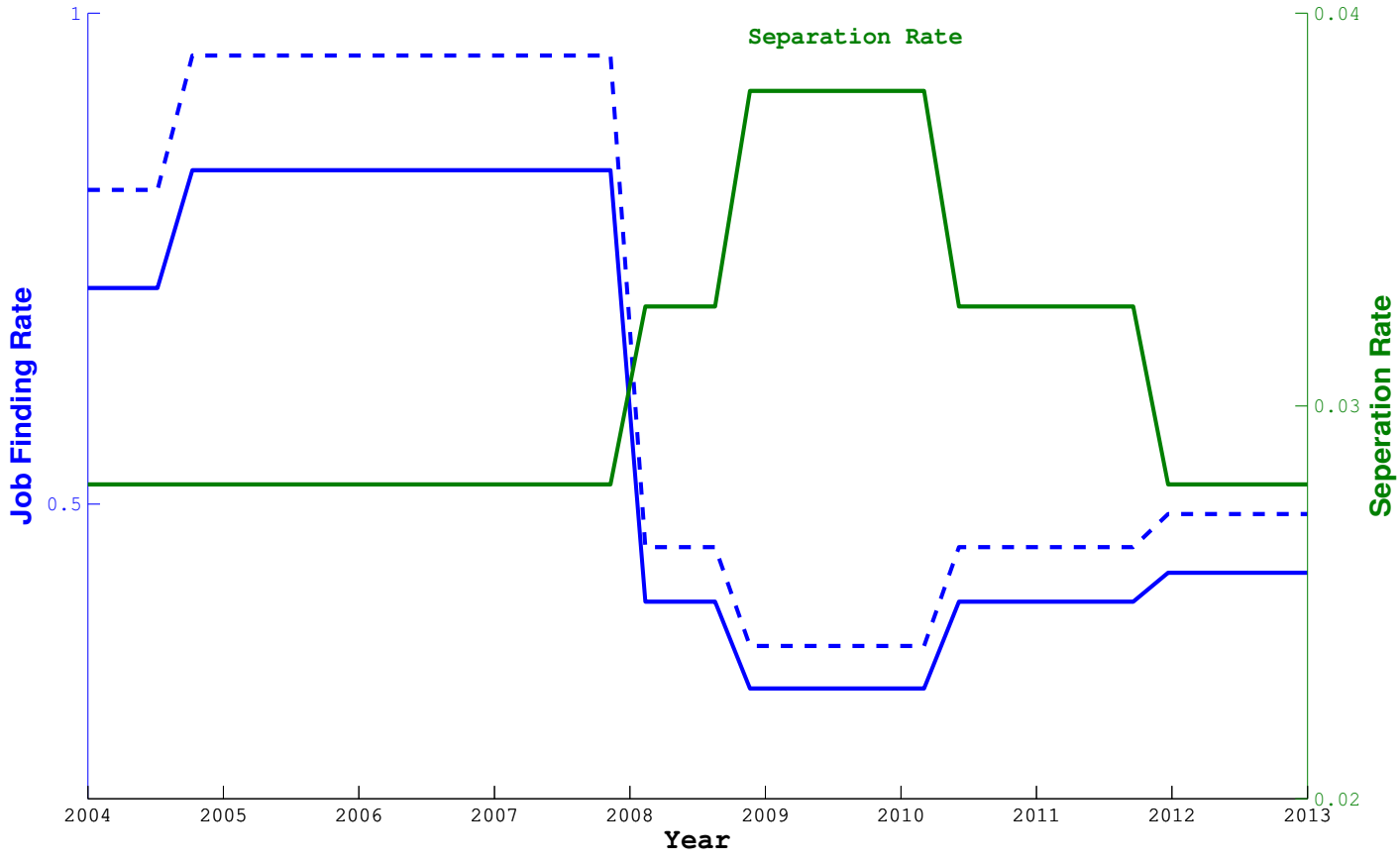
Our approach:

- Step 1: Feed in labor market upheaval, by setting
  - job separation rates
  - job finding rates
- Aggregate shocks are not unexpected.
  - Transition matrix calibrated to expected duration of agg. states.
- Step 2: Solve for paths of debt, delinquency, bankruptcy
- Step 3: Counterfactuals:
  - Evaluate Step 2 with and without 2005 BK reform
  - Evaluate Step 2 with and without labor market shocks
- Agents learn about the reform one period ahead.

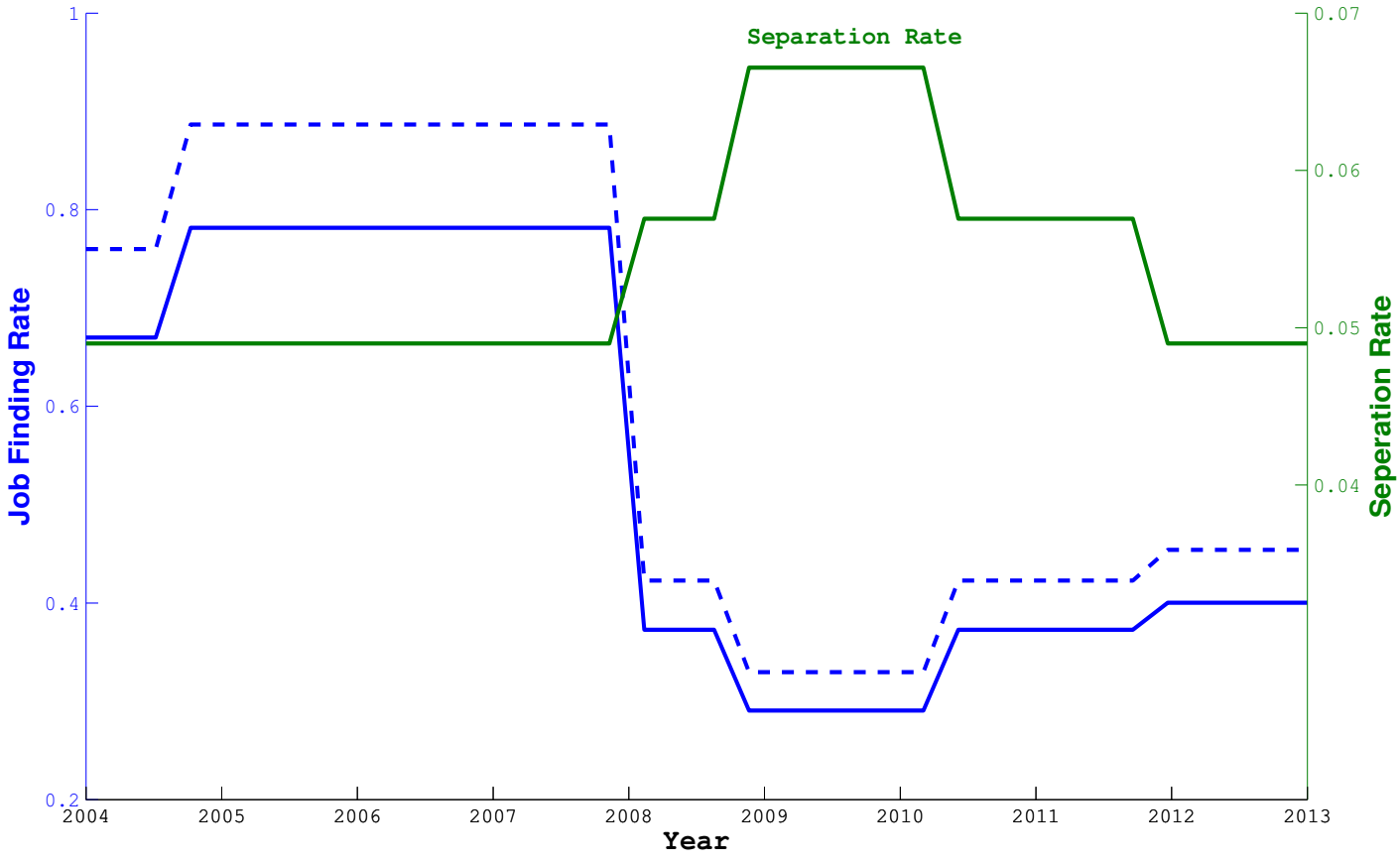
Model Approximation, shocks



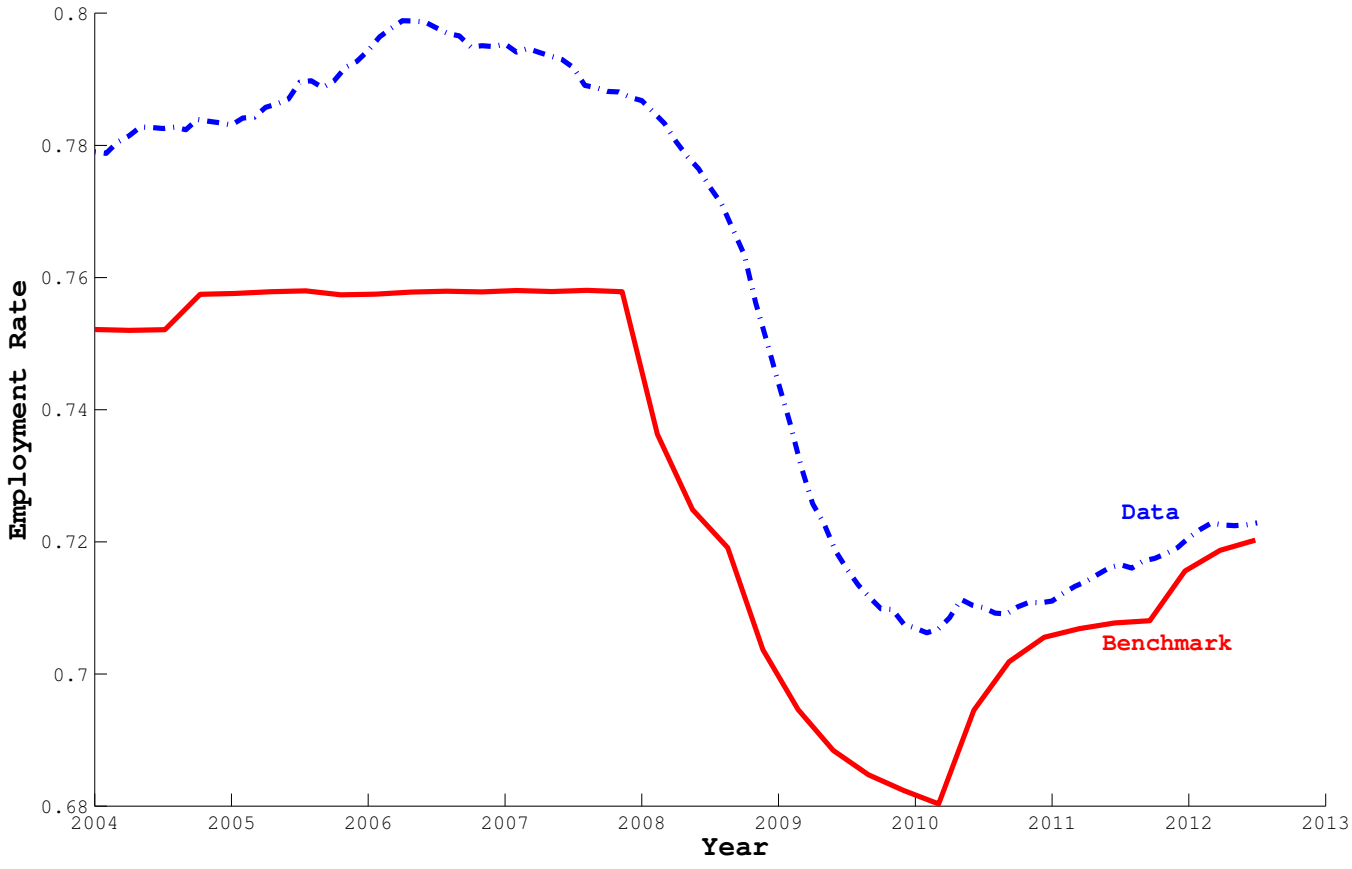
The Shocks


Separation Rate
Job Finding Rate
Seperation Rate
Year
1
0.5
0.04
0.03
0.02
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013

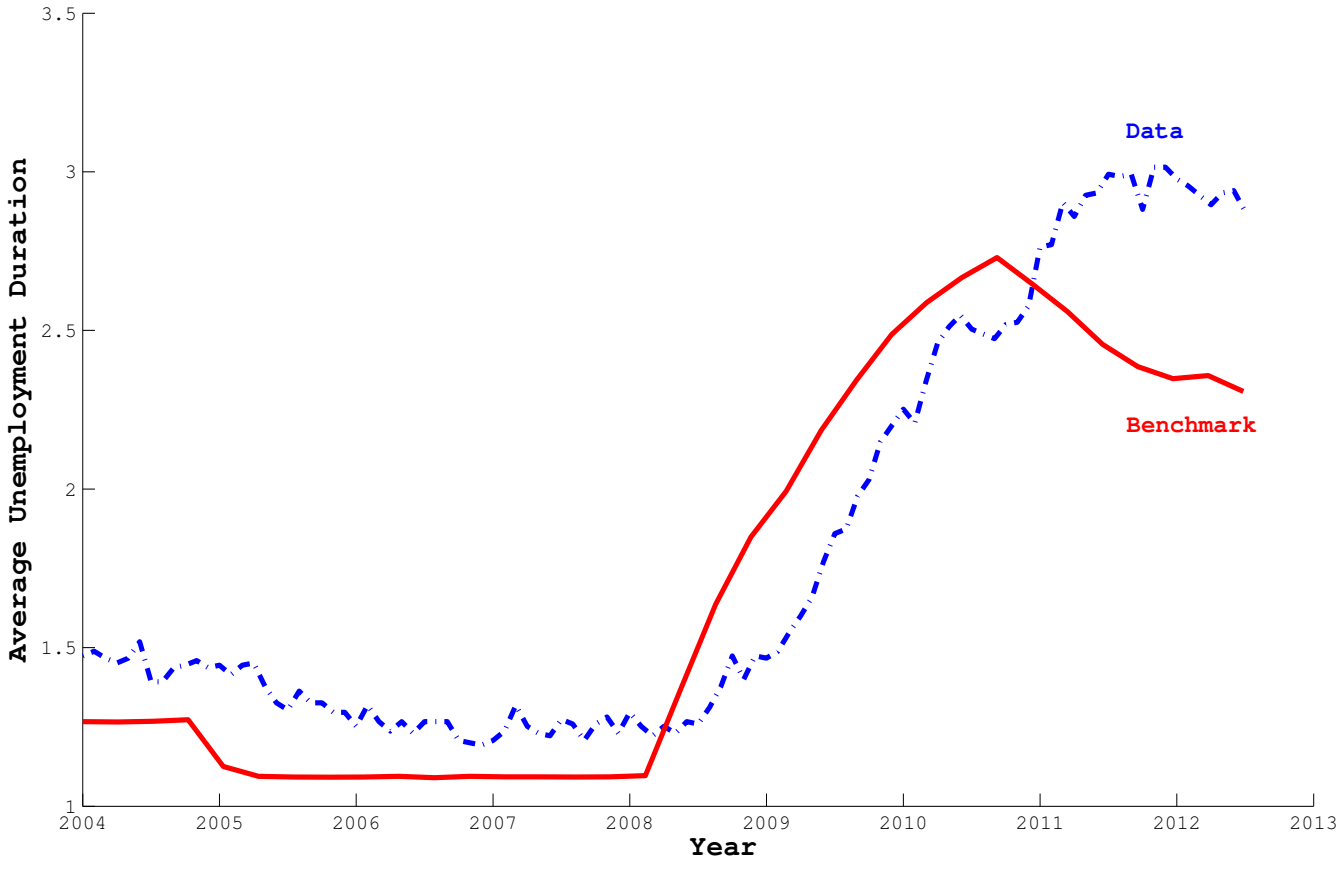
The Shocks: Low Education



Tracking Employment Rates



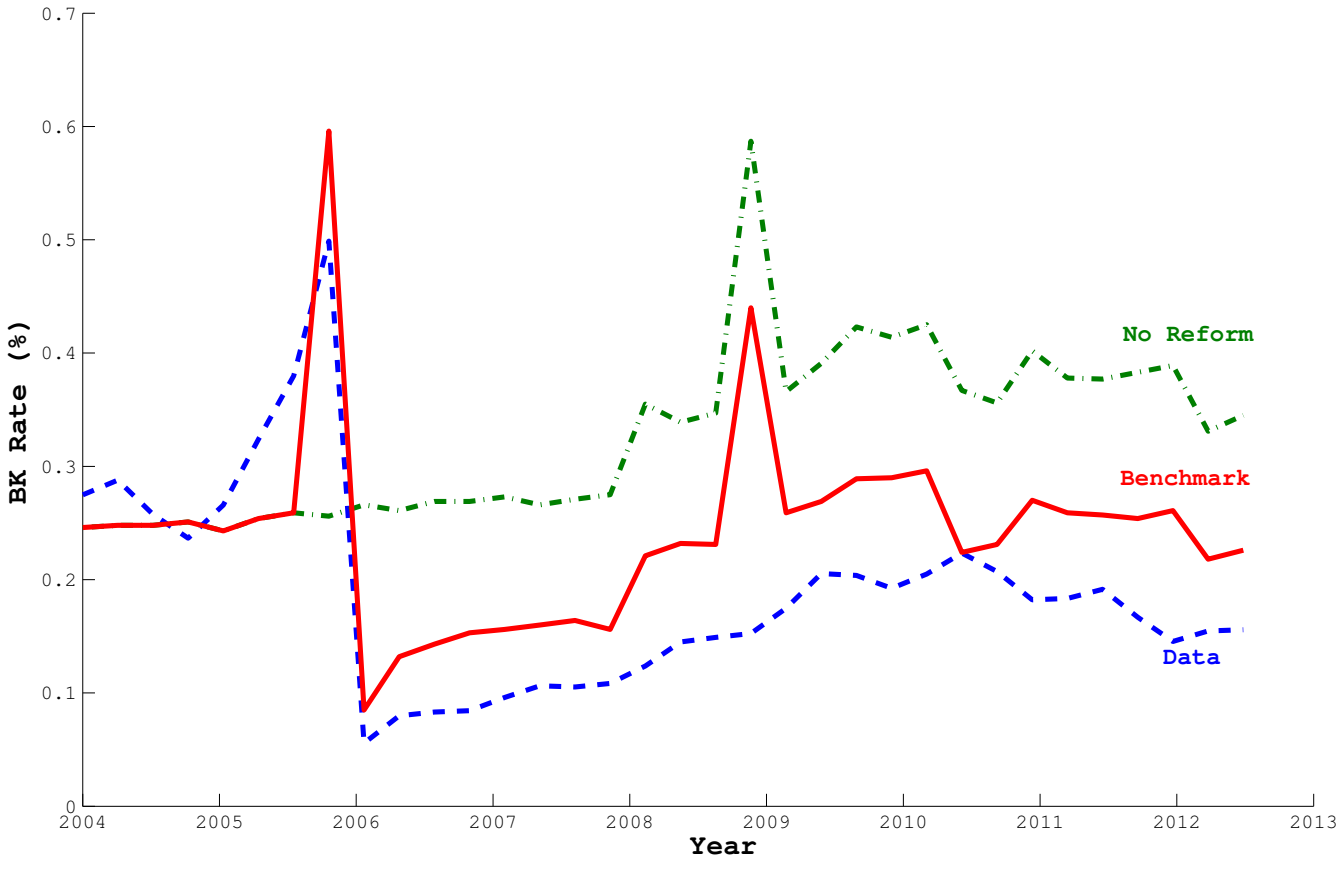
Tracking Unemployment Duration



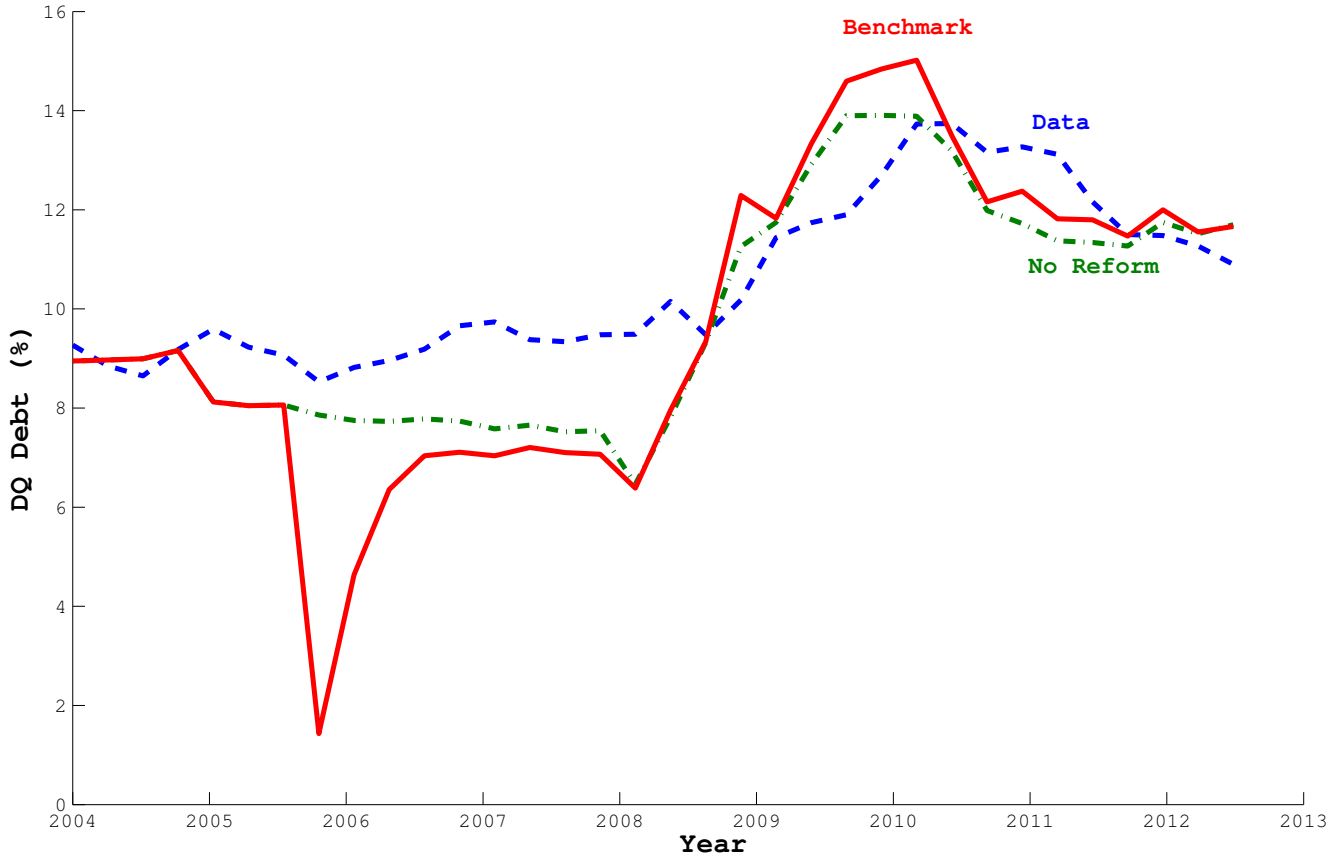
Findings: 2004Q1 vs. 2012Q2

	Reform		**No Reform**	
	2004Q1	2012Q2	2004Q1	2012Q2
DQ Debt Rate	8.95%	11.67%	8.95%	11.70%
BK Rate	0.25%	0.23%	0.25%	0.345%
Frac Borr	15.27%	12.86%	15.27%	11.81%
Relative Debt Size to 04Q1	1.000	0.886	1.000	0.762

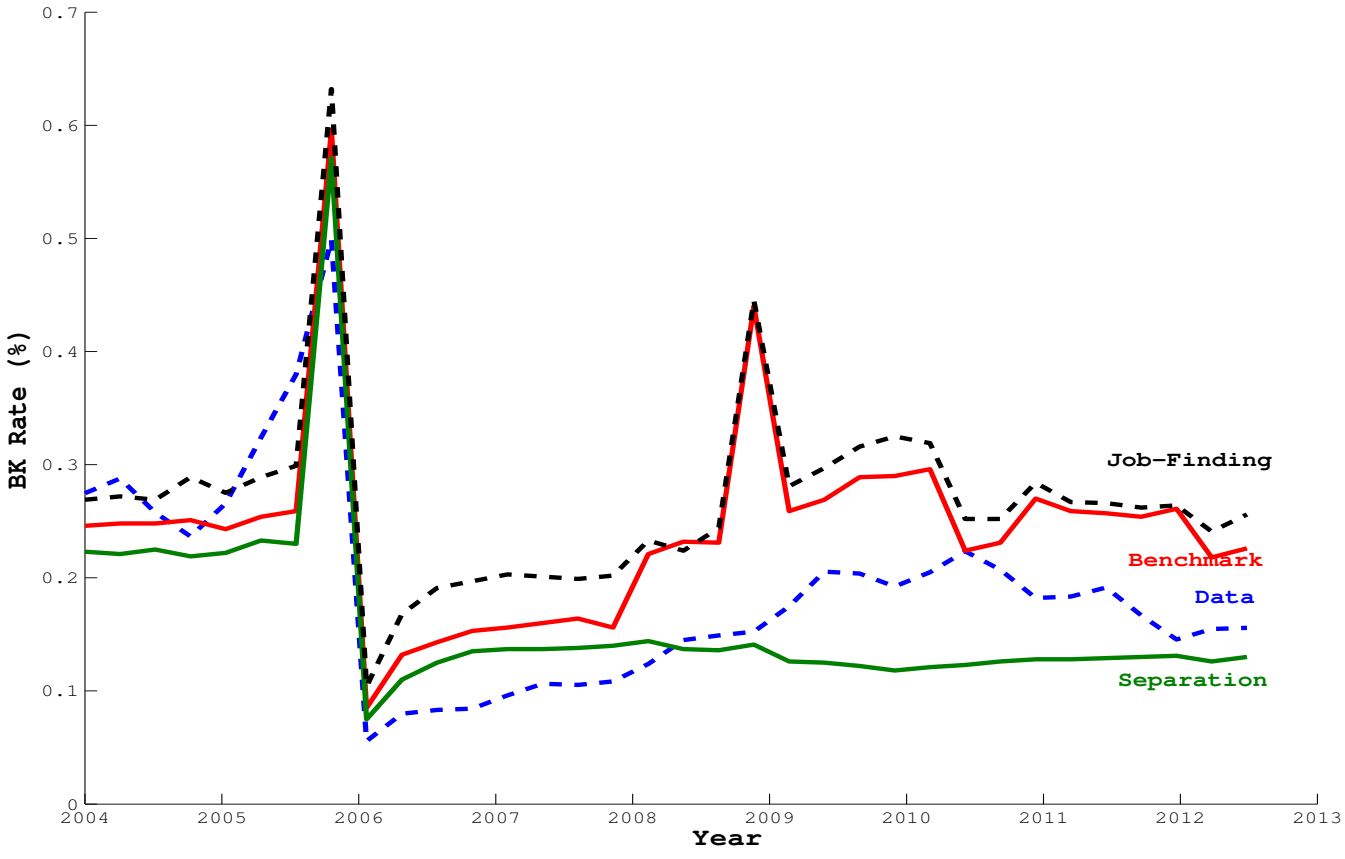
BK Reform Mattered for Observed Bankruptcy



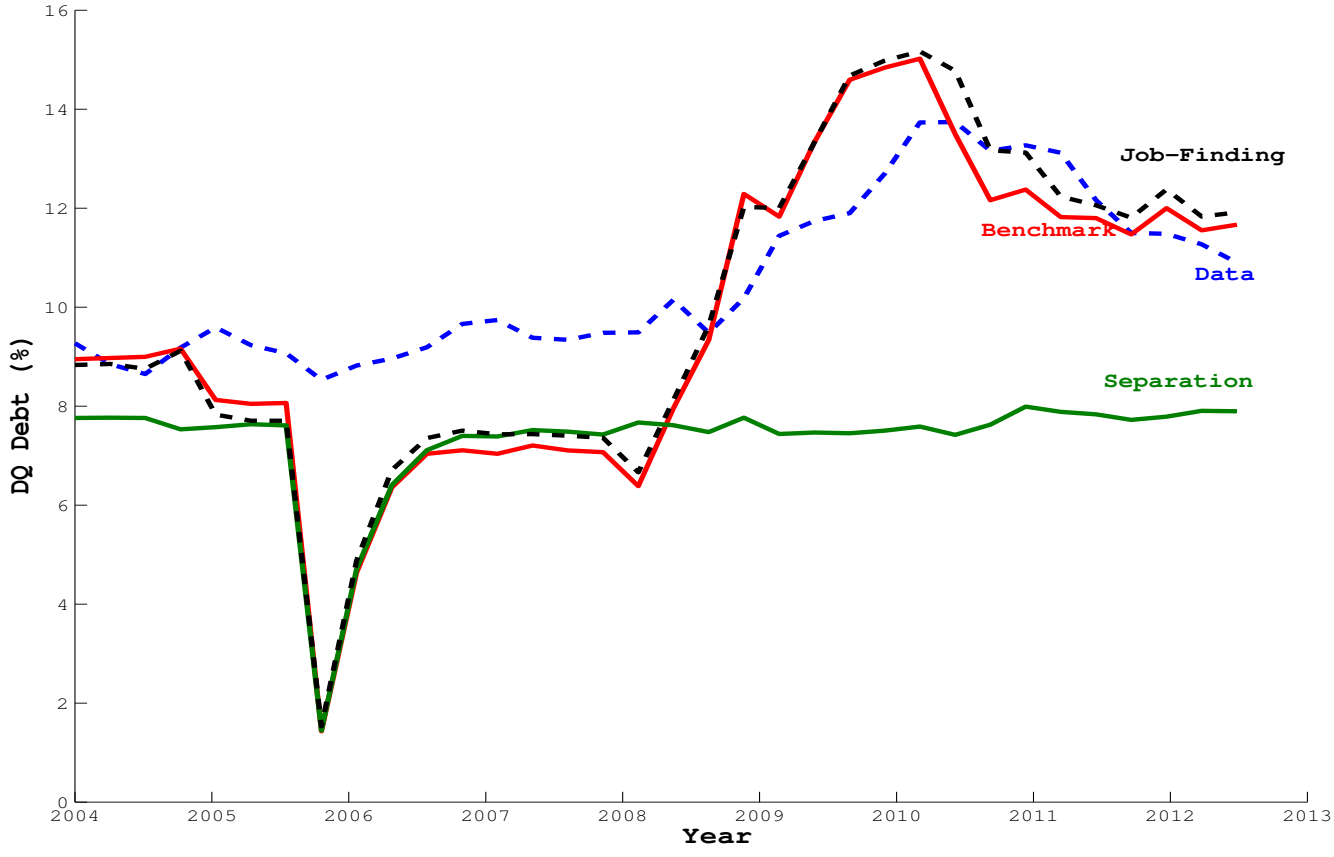
For delinquency, labor markets deterioration key, not the BK reform



For bankruptcy, job finding rate is central, not separation



For DQ, movements in job finding rate are the key



Conclusions

- We asked: How did labor markets and BK law influence credit use and default patterns in the GR?
- Provided a model of consumer credit use and default during Great Recession
  - Changes in job finding rates key for default and debt paths
  - Bankruptcy reform did in fact lower filing rates, given the evolution of labor market risk

Mortgage Defaults and Prudential Regulations in a Standard Incomplete Market Model

- Joint with Leo Martinez and Juan C Hatchondo.
- Motivation
  - Mortgage defaults received a lot of attention since 2008.
    - Recently, 24 countries adopted policies to prevent defaults.
  - Policies often discussed and adopted are:
    - Loan-to-value limits
    - Recourse

This paper

- Builds a quantitative theory of mortgage default.
  - Add housing, mortgages, and default to the SIM model.
- Compares the predictions of the model with data.
  - Demand for housing, home owners' leverage, mortgage defaults.
- Study how recourse and LTV limits affect:
  - housing demand, use of mortgages, and mortgage defaults both in the long run and after large declines in house prices.

Overview of results

- Model consistent with key features of the data.
- Recourse has hump-shaped effect on the default rate and has large positive effects on housing demand.
- LTV limits lower default abruptly with negligible negative effects on the demand for housing.
- Combination of recourse and LTV limits lowers the default rate while increasing the demand for housing.
- Together, these policies reduce the sensitivity of the default rate to declines in house prices.

Model

- Incomplete markets and partial equilibrium.
- Households live up to T periods and work until age $R \leq T$.
- Consume non-durable goods and housing.
- Each household lives in a house and may own up to one house.
- Households earn rate r with savings and finance housing with mortgages.
- Households face idiosyncratic earning and house-price risk.

Risk

- Household's i earnings process has 4 components

$$y_{i,t}(age_i) = \exp(\underbrace{f(age_i)}_{\text{life-cycle}} + \underbrace{s_i}_{\text{permanent}} + \underbrace{z_{i,t}}_{\text{persistent}} + \underbrace{\varepsilon_{i,t}}_{\text{transitory}}),$$

  where $z_{i,t} = \rho_z z_{i,t-1} + e_{i,t}$ and the shocks follow Gaussian distributions.

- Post-retirement income depends on the last realization of z.

- Household who lives in location i face housing prices:

$$log(p_{i,t+1}) = (1-\rho_p) log(\underbrace{\bar{p}}_{\text{mean price}}) + \rho_p log(p_{i,t}) + \nu_{i,t+1}.$$

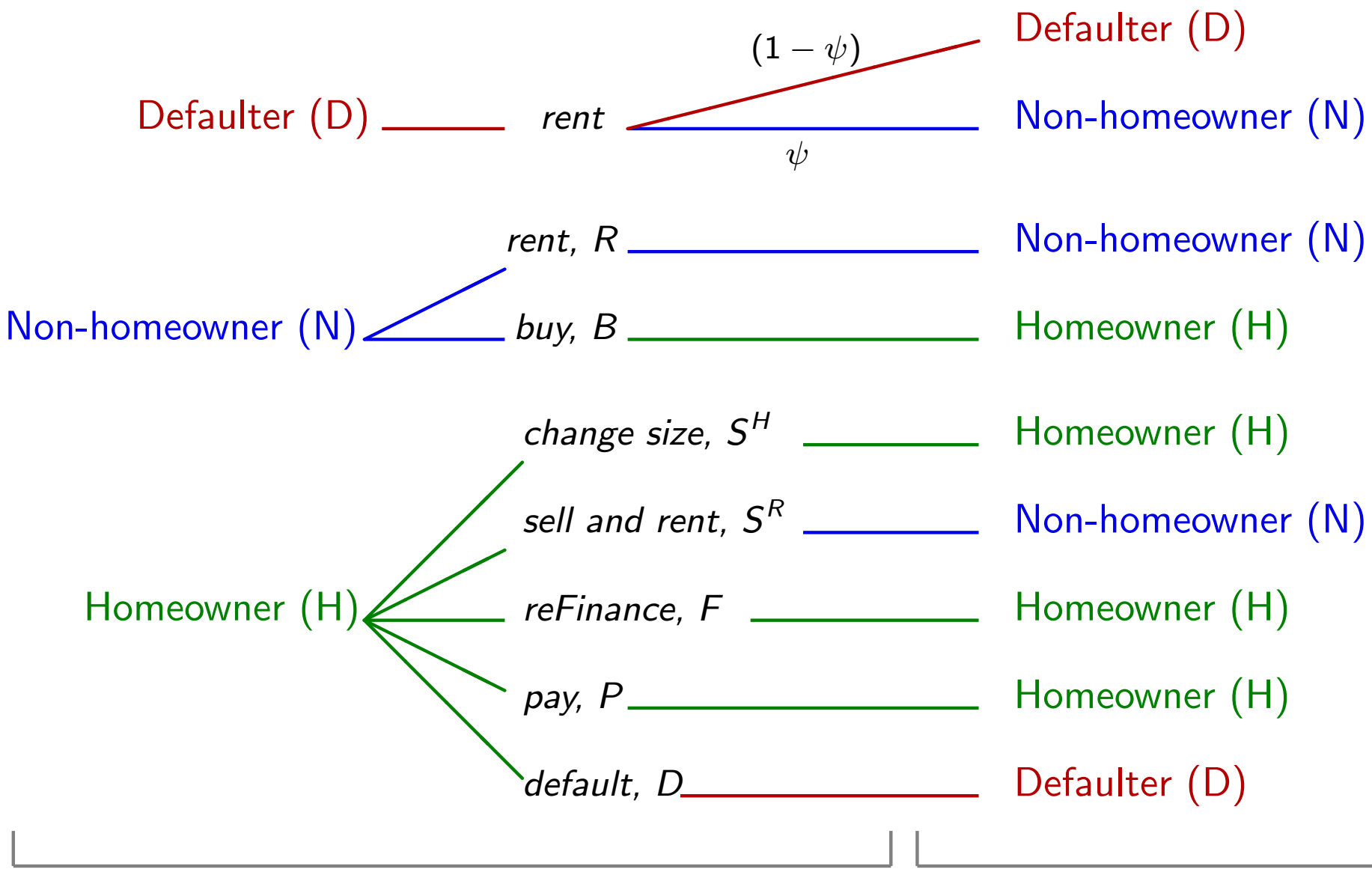
- Price of house with housing services h is $= ph$.

- Shocks to earnings, e, and house prices, ν, are correlated.

Mortgage contracts

- A mortgage for a household of age t is a promise to:
  - Make payments for remaining years of life, $n = T - t$, or
  - But household can prepay its debt in every period.
- Households cannot commit to repay the mortgage.
- A defaulting household hands in the house to its lender.
  - The lender sells the house with a discount.
  - The household must rent for a stochastic number of periods.
- At origination, households choose the payments size , b.
  - Mortgage payments decay at the rate δ (inflation)
- Wealth (financial and housing) is annuitized to deal with mortality risk (not in today's presentation)

Households' choices



Bellman Equations, Lenders

- Risk-neutral lenders make zero expected profits.
  - They face a given interest rate r
- A household promising to pay b' obtains $b'q(b', a', z, p, h', n)$.
- Zero-expected discounted profits implies that q is

$$q(b', a', z, p, h', n) = \frac{q_{\text{prepay}} + q_{\text{default}} + q_{\text{pay}}}{1 + r}$$

- $q_{\text{prepay}} = \mathbb{E}\left[I_{\text{prepay}}(b', w', z', p', h', n-1)q^*(n-1)|z, p\right]$
- $q_{\text{default}} = \mathbb{E}\left[I_{\text{default}}\,(b', w', z', p', h', n-1)p'h'(1-\overline{\xi}_S)|z, p\right]/b'$
- $q_{\text{pay}} = \mathbb{E}\left[I_{\text{pay}}(b', w', z', p', h', n-1)\left(1+(1-\delta)\,q\left(b'\,(1-\delta), a'', z', p', h', n-1\right)\right)|z, p\right]$

where a'' is the next-period optimal saving choice of a household who pays his mortgage.

Bellman Equations, Home-owner

Household **changing the house size**

$$S^H(b, w, z, p, h, n) = \max_{\{b' \geq 0, a' \geq 0, h'\}} u\left(c, h'\right) + \beta \mathbb{E}\left[H(b', w', z', p', h', n-1) \mid z, p\right]$$

subject to

$$c = \underbrace{\left[b' q(b', a', z, p, h', n) - q^*(n) b\right]}_{\text{extra amount borrowed}} - \underbrace{p\left[\left(1 + \xi_B\right) h' - h\left(1 - \xi_S\right)\right]}_{\text{extra cost of the house}} + w - \left(I_{b'>0} \xi_M + \xi_P\right) - \frac{a'}{1+r}$$

$b' q(b', a', z, p, h', n) / p h' \leq 1$, **no unsecured borrowing**

Calibration

- Calibrate the model using data for the U.S..
- Feed into the model stochastic processes for households' y and p estimated using micro data.
- All but 4 parameter values are taken from previous studies
- Parameters:
  - size of the rental house, h^R,
  - discount factor, β,
  - housing utility α,
  - probability of regaining access to the market after a default, ψ.
- Targets:
  - home ownership rate
  - median financial savings / median income
  - median price / median income ratio
  - median down-payment

Parameters and fit

Variables	Data	Model
Homeownership rate	0.66	0.66
Median Price/ Median income	2.80	2.91
Median (saving/income)	0.85	0.77
Median down payment	0.18	0.17
Parameter	**Value**	**Definition**
h^R	1.43	Size rental house
α	0.9	Housing utility
β	0.946	Discount factor
ψ	0.667	Probability default ends

Who does what?

- Households that default have sufficient negative equity and low income.

Mean of Key Statistics

Decision	Equity/Price	Income
Default	**-0.26**	**0.88**
Pay	0.32	1.44
Sell	0.31	1.91
Refinance	0.44	1.83

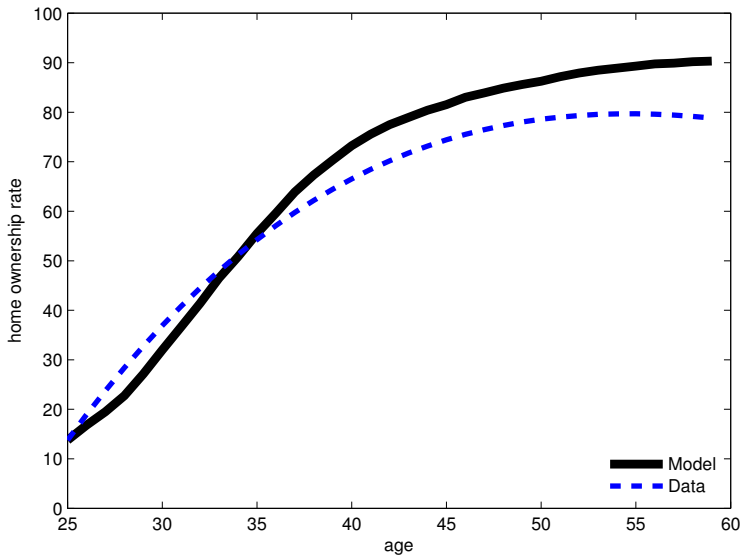
Use of mortgages and default

- The use of mortgages and default replicates the data well, although it was not targeted.

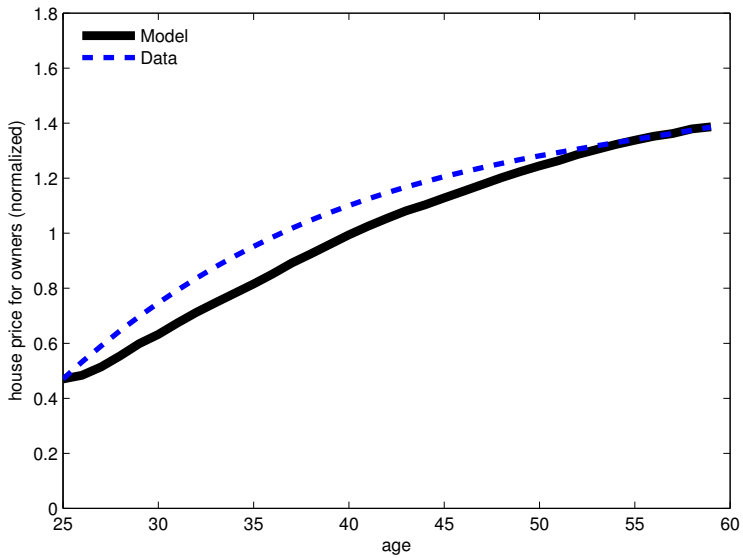
Variables	Data	Model
Homeowners with mortgages	0.82	0.83
Median payment / Median Income	0.15	0.12
Default rate (%)	0.50	0.59

Demand for housing over the life cycle

- Life-cycle profiles of ownership and house value are also closely replicated, although they were not targeted.



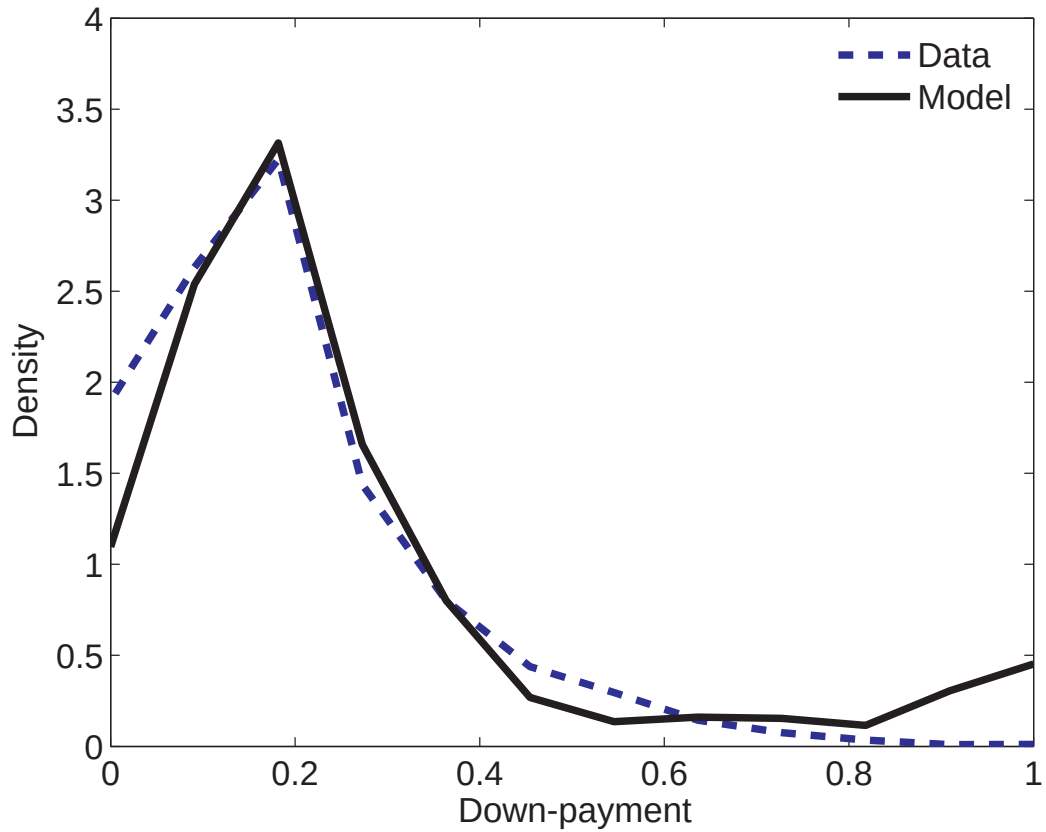
Homeownership rate



Average price of houses

Distribution of down payments

- The dispersion of use of mortgages to buy houses replicates data on combined mortgages.



The empirical distribution is constructed using data presented by [Paniza Bontas, 2010].

Policy experiments

- Use the model as a laboratory to study:

    1. How (steady state) default rates (and other key statistics) change with LTV limits and recourse rules.

    2. The effect of those policies after large shocks to house prices.

Recourse

- Defaulting household must use all financial wealth plus current income above a threshold ϕ for deficiency payments,

$$\Phi(\cdot) = \min\{w - \phi\tilde{y}, \text{defficiency}\}$$

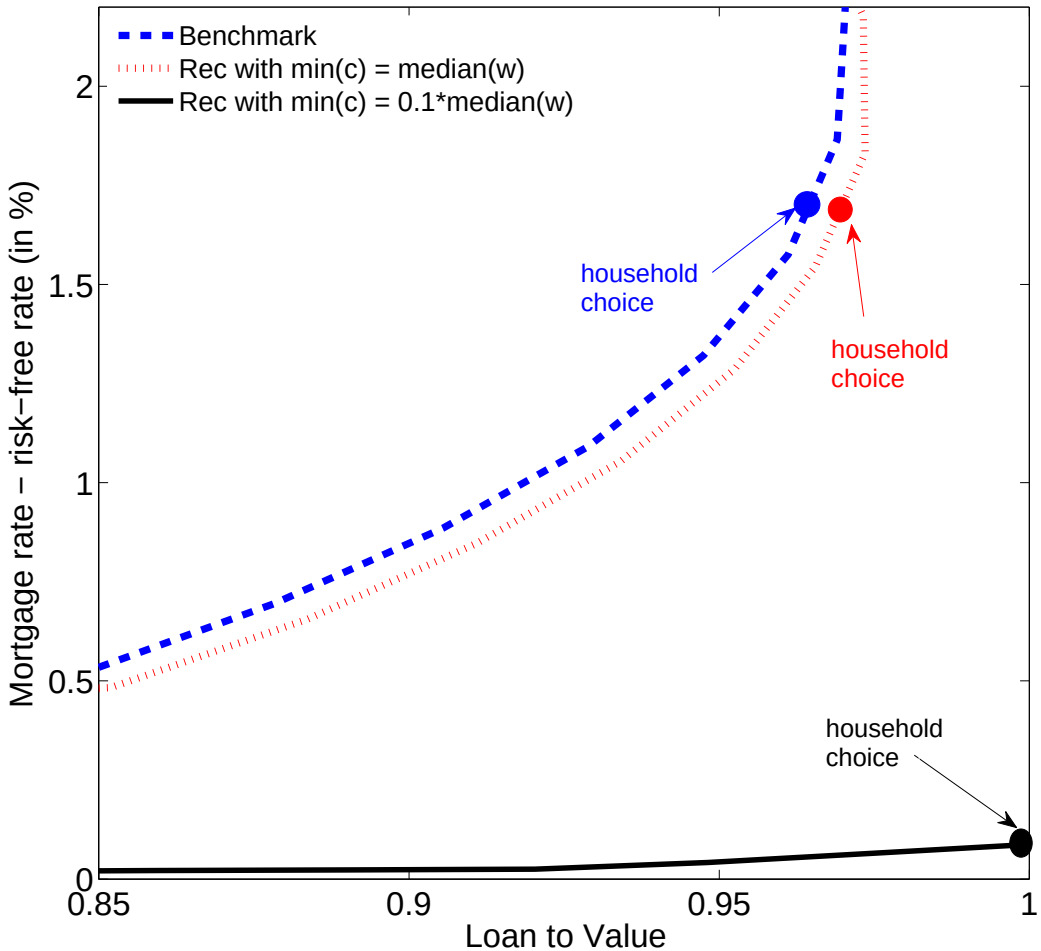
where $\tilde{y}$ represents the median income.

Recourse

		Recourse, $\phi =$					
Variable	Benchmark	4	2	1	0.5	0.1	0.01
Home ownership rate	0.66	0.67	0.70	0.73	0.78	0.80	0.74
Median down payment	0.17	0.15	0.12	0.04	0.00	0.00	0.00
Default rate, (%)	**0.59**	**0.75**	**0.77**	**0.79**	**0.58**	**0.06**	**0.00**
Mean (equity/price), mort.	0.24	0.19	0.12	0.07	0.04	0.03	0.07

- The effect on default is hump shaped.
- Large effects on ownership because of lower down payments.

Recourse lowers the borrowing cost ... And induce more borrowing!



Imposing LTV limits

- Study the effect of imposing a limit on down payments.
- Now, b' must satisfy

$$\frac{b' q(b', a', z, p, h', n)}{p h'} \leq \text{LTV limit}$$

Imposing LTV limits

Variable	(LTV$\leq$ 100%) Benchmark	LTV limit at 90%	85%	80%
Home ownership rate	0.66	0.66	0.66	0.64
Median downpayment	0.17	0.16	0.16	0.20
Default rate (%)	**0.59**	**0.39**	**0.21**	**0.10**
Mean (equity/price), mortgagees	0.24	0.24	0.26	0.29

- Large reduction in defaults, because equity increases.
- Mild effect on housing demand (at least for limits above 85%)

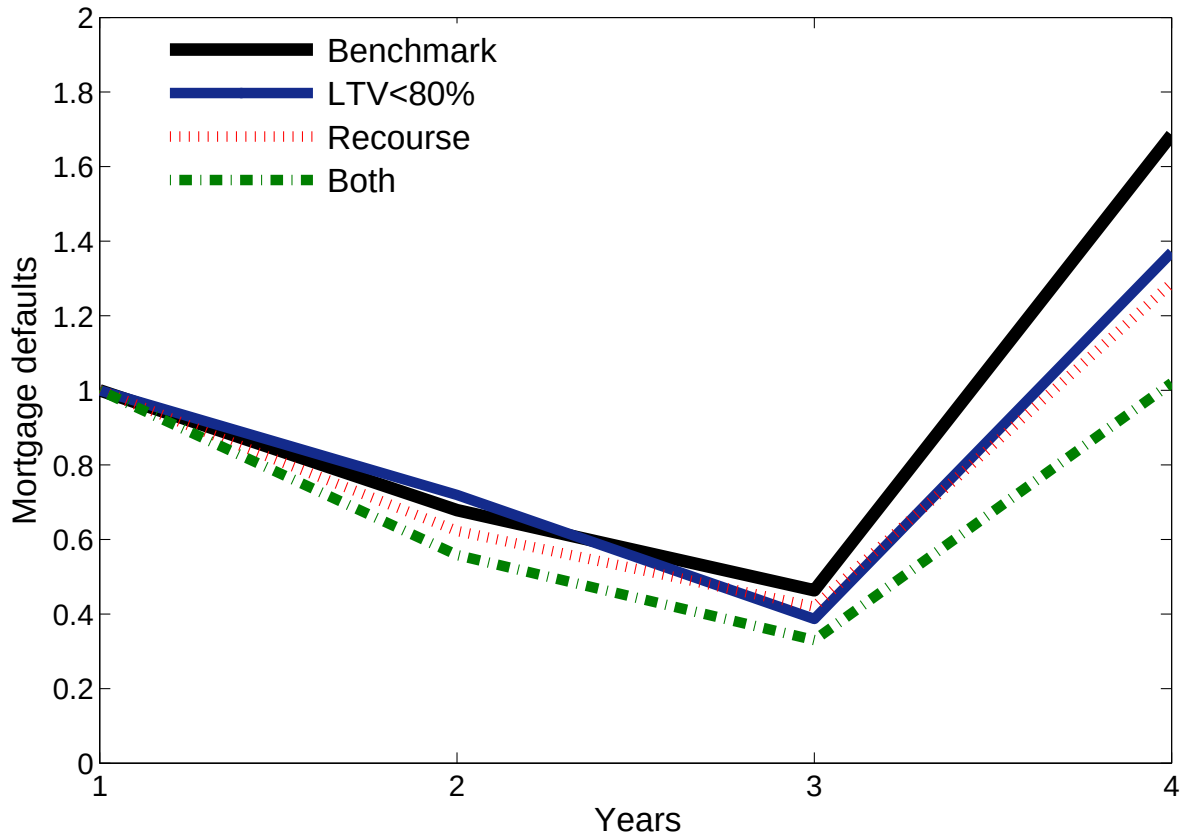
Adopting both Recourse and LTV limits

Variable	Benchmark	Recourse $\phi = 1$ & $LTV = 80\%$
Home ownership rate	**0.66**	**0.68**
Median downpayment	0.17	0.20
Default rate, (%)	**0.59**	**0.04**
Mean (equity/price), mortgagees	0.24	0.25

- Reduces defaults while increasing the demand for housing.

Large swings in the average price of housing

- Prudential policies reduce the sensitivity of default to fluctuations in house prices.



Welfare gains of a reform

- Welfare gains the reform depend on current state of household.
- Also, prices of previously issued debt will change with the reform.
- Compute welfare gains when the change in the value of debt is distributed **lump-sum** across household.

Welfare gains	LTV ≤ 80	Recourse	Both
5th percentile, CE %	-0.23	-0.81	-0.91
10th percentile, CE %	-0.14	-0.33	-0.47
25th percentile, CE %	-0.06	0.07	-0.00
50th percentile, CE %	**-0.02**	**0.37**	**0.27**
75th percentile, CE %	-0.01	0.90	0.57
90th percentile, CE %	-0.00	1.49	1.01
95th percentile, CE %	0.00	2.32	1.75

Conclusions

- Extend the SIM model to incorporate housing and mortgages.
- Model predictions are consistent with key features of the data.
- Recourse has hump-shaped effect on the default rate and increases significantly the demand for housing.
- LTV limits lower the default rate with relatively mild effects on the demand for housing.
- Combination of LTV limits and recourse lower the default rates while increasing the demand for housing.
- Together, these policies keep defaults low even after large house-price declines.

Quantitative properties of sovereign default models: solution methods

Juan C. Hatchondo, Leonardo Martinez and Horacio Sapriza

Review of Economic Dynamics, 2010, vol. 13, issue 4, 919-933

EMERGING ECONOMY MODELS

- Quantitative studies of business cycle in emerging markets feature a state-dependent **interest rate scheme**.
  - The interest rate scheme can be exogenously given o it can be endogenous.
  - It is endogenous in the models of sovereign default.
  - Different from standard RBC model.
- We show that the numerical error implied by the discrete state space technique may be non-negligible.
- Miscalculation of the **interest rate** volatility and the correlations of the **interest rate** with other macroeconomic variables.

EXAMPLE

- Consider the following dynamic optimization problem:

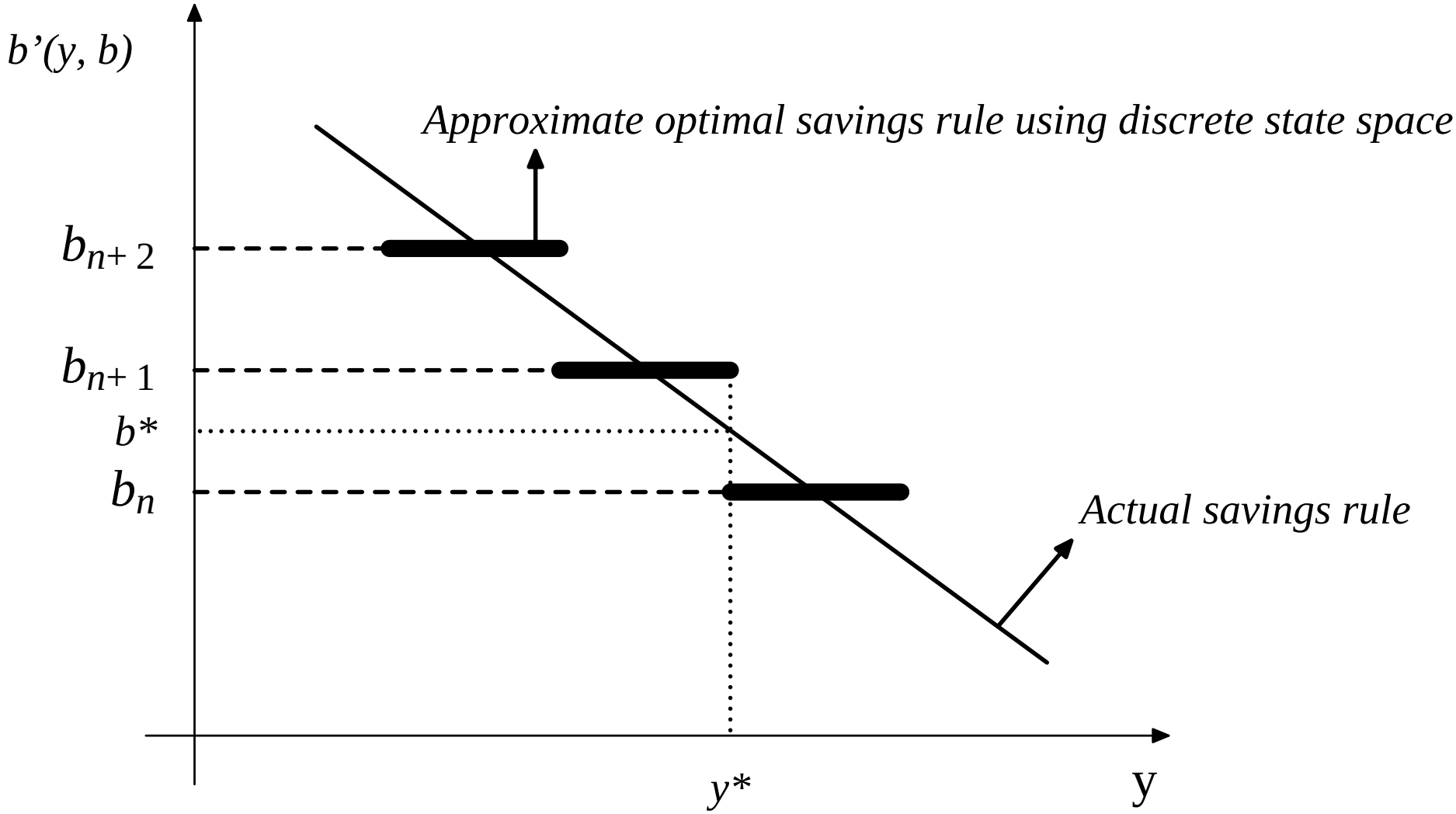
$$V(y,b) = \underset{b'}{Max}\left\{u\left(y+b-\frac{b'}{(1+r(y,b'))}\right)+\beta\int V(y',b')\,F(dy'\mid y)\right\}$$

- $y =$ income
- $b =$ bond position
- $r =$ interest rate
- Assume

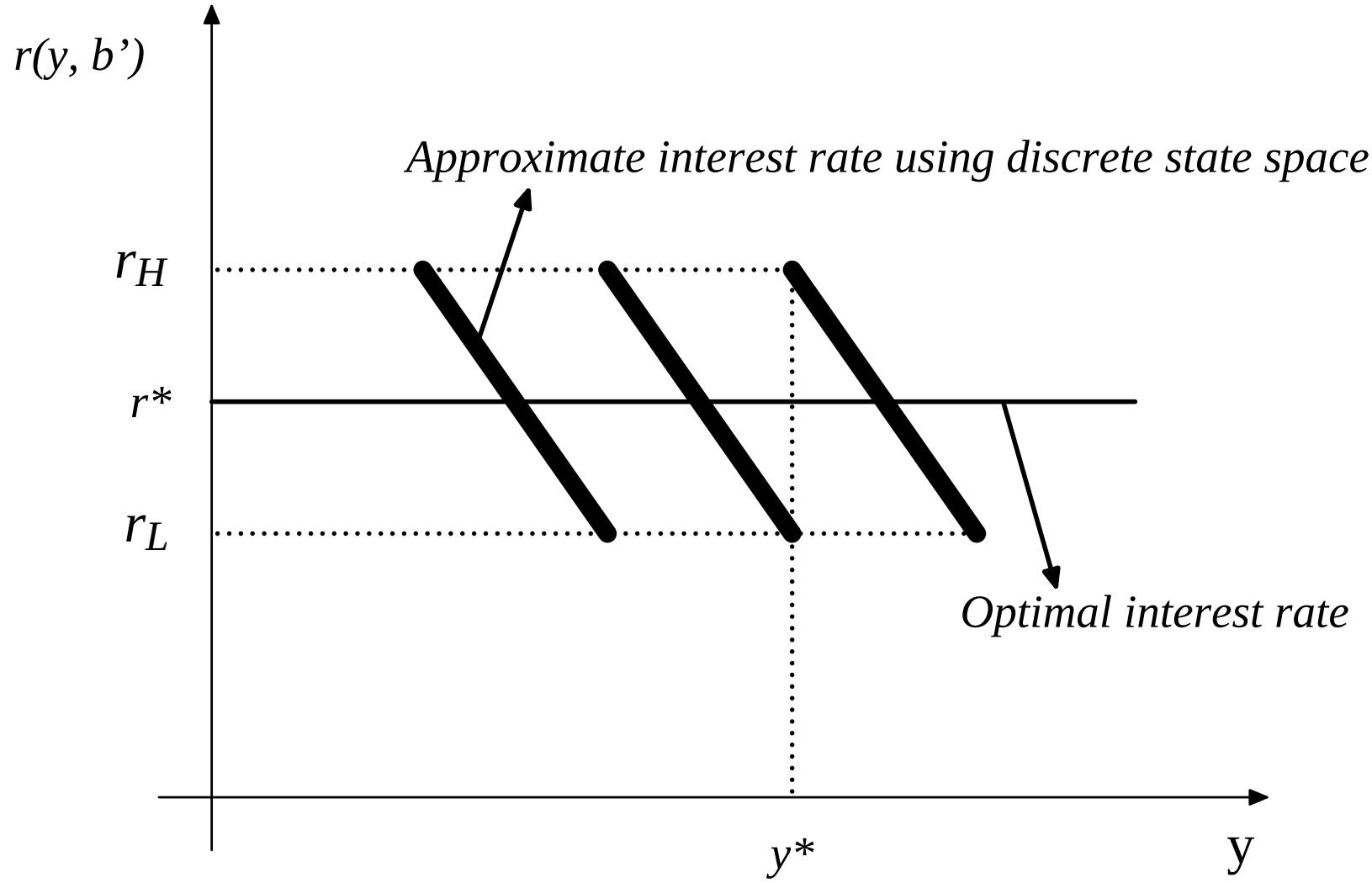
$$\frac{\partial r}{\partial y} < 0$$

$$\frac{\partial r}{\partial b'} < 0 (\text{ More borrowing } \Rightarrow \text{ higher } r)$$

EXAMPLE: OPTIMAL SAVINGS



EXAMPLE: INTEREST RATE PAID IN EQUILIBRIUM



QUANTITATIVE IMPORTANCE

- We show that this problem may be quantitatively important
- We solve the stylized model of **sovereign default** presented in Aguiar and Gopinath (JIE, 2006).
  - Based on Eaton and Gersovitz (REStud, 1981)
  - Extensions by: Arellano (2005), Arellano and Ramanarayanan (2006), Bai and Zhang (2005), Cuadra and Sapriza (2006a, 2006b), Lizarazo (2005a, 2005b), and Yue (2005).

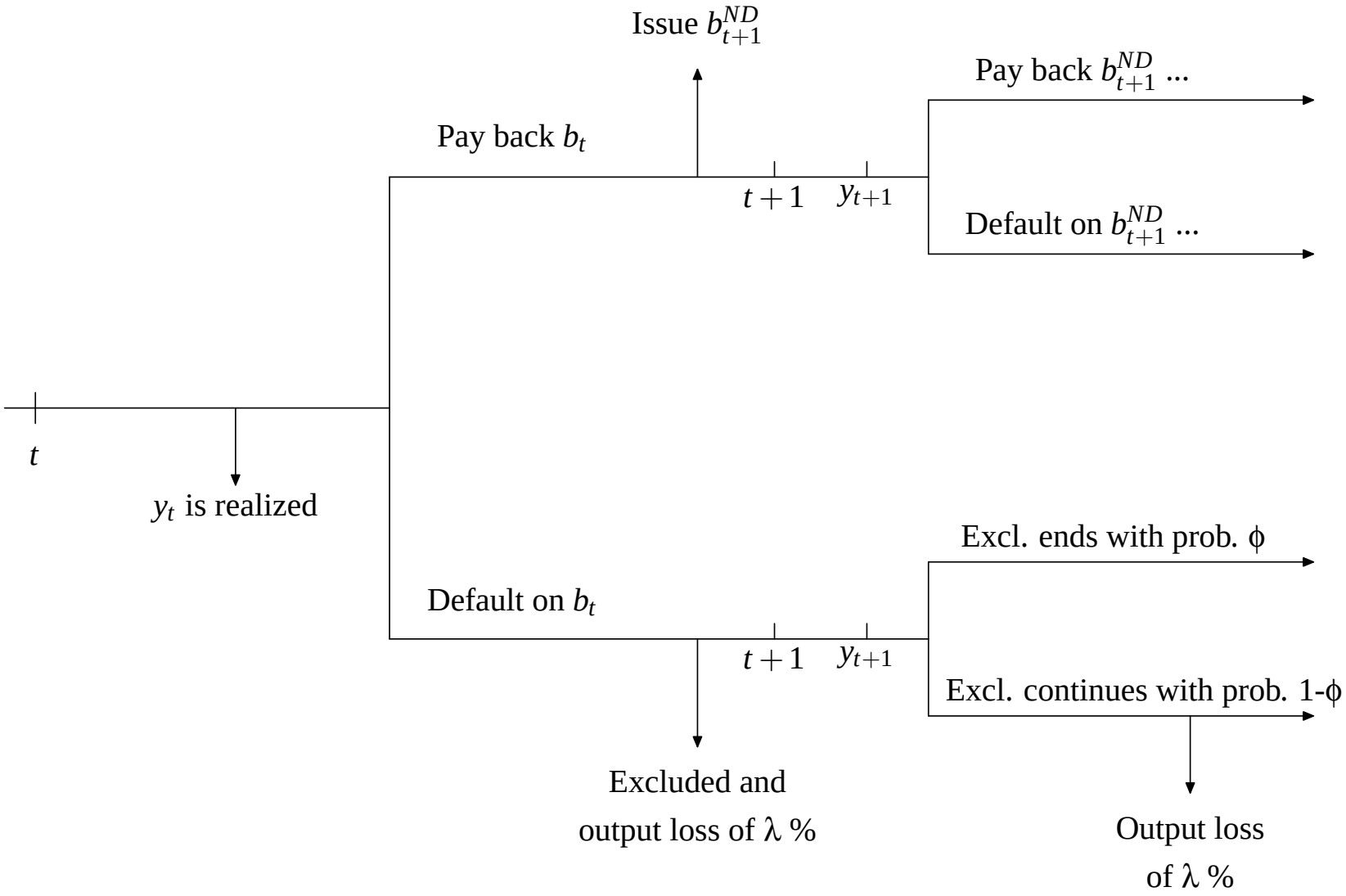
THE MODEL

- Benevolent government decides how much to borrow and whether to default or not.
- Government's objective:

$$Max\ E_t \left[\sum_{s=0}^{\infty} \beta^s u\left(c_{t+s}\right) \right]$$

- Instrument: issuance of one-period bonds.
- Bonds are auctioned in a competitive market and priced by risk neutral lenders.
- Endowment economy.
- Two cost of defaulting: output loss and exclusion from capital markets for a stochastic number of periods.

Timing


Issue b_{t+1}^{ND}
Pay back b_{t+1}^{ND} ...
Pay back b_t
$t+1$
y_{t+1}
Default on b_{t+1}^{ND} ...
t
y_t is realized
Excl. ends with prob. ϕ
Default on b_t
$t+1$
y_{t+1}
Excl. continues with prob. 1-ϕ
Excluded and
output loss of λ %
Output loss
of λ %

THE ENDOWMENT PROCESS

$$y_t = e^{z_t}\Gamma_t,$$

where the transitory component

$$z_t = (1-\rho_z)\,\mu_z + \rho_z z_{t-1} + \varepsilon_t^z$$

$|\rho_z| < 1$, and $\varepsilon_t \sim N(0, \sigma_z^2)$; and

$$\Gamma_t = g_t \Gamma_{t-1},$$

where

$$ln(g_t) = (1-\rho_g)\,(ln(\mu_g) - m) + \rho_g ln(g_{t-1}) + \varepsilon_t,$$

$|\rho_g| < 1$, $\varepsilon_t \sim N\left(0, \sigma_g^2\right)$, and $m = \frac{1}{2}\frac{\sigma_g^2}{1-\rho_g^2}$.

RECURSIVE FORMULATION (1)

$$V(b,z,\Gamma,g) = \max_{d\in\{0,1\}} \left\{(1-d)\,V_0(b,z,\Gamma,g) + dV_1(z,\Gamma,g)\right\},$$

where

- Value function under default

$$\begin{aligned} V_1(z,\Gamma,g) &= u(y(1-\lambda)) + ... \\ &\quad \beta\left[\int\int \left[\phi V(0,z',g'\Gamma,g') + (1-\phi)\,V_1(z',g'\Gamma,g')\right] F_z\left(dz' \mid z\right) F_g\left(dg' \mid g\right)\right], \end{aligned}$$

RECURSIVE FORMULATION (2)

- Value function under no default:

$$V_0(b,z,\Gamma,g) = \max_{b'} \left\{ \begin{array}{c} u(y+b-q(b',z,\Gamma,g)\,b') + ... \\ \beta\left[\int\int V(b',z',g'\Gamma,g')F_z(dz' \mid z)\,F_g(dg' \mid g)\right] \end{array} \right\}$$

- Bond price:

$$q(b',z,\Gamma,g) = \frac{1}{1+r}\left[1 - \underbrace{\int\int d(b',z',g'\Gamma,g')\,F_z(dz' \mid z)\,F_g(dg' \mid g)}_{Pr(\text{government defaults tomorrow on b'}|z,g,\Gamma)}\right]$$

EQUILIBRIUM DEFINITION

A recursive equilibrium for this small open economy consists of

1. a set of value functions $V(b, z, \Gamma, g)$, $V_1(z, \Gamma, g)$, and $V_0(b, z, \Gamma, g)$,
2. a set of policies for asset holdings $b'(b, z, \Gamma, g)$ and default decisions $d(b, z, \Gamma, g)$,
3. and a bond price function $q(b', z, \Gamma, g)$, such that
   (a) $V(b, z, \Gamma, g)$, $V_1(z, \Gamma, g)$, and $V_0(b, z, \Gamma, g)$ satisfy the functional equations above;
   (b) the default policy $d(b, z, \Gamma, g)$ and the policy for asset holdings $b'(b, z, \Gamma, g)$ solve the problems above;
   (c) the bond price function $q(b', z, \Gamma, g)$ is as defined above.

AG's PARAMETER VALUES (1)

Table 2. Model specific parameter values

	Model I	Model II
σ_z	3.4%	0
ρ_z	0.9	NA
σ_g	0	3%
ρ_g	NA	0.17

AG's PARAMETER VALUES (2)

Table 1. Parameter values common to both models		
Risk aversion	σ	2
Interest rate	r	1%
Loss of output	λ	2%
Mean growth rate	μ_g	1.006
Mean (log) transitory productivity	μ_z	(-1/2)σ_z^2
Discount factor	β	0.8

Discrete state space

- Define grid for endowment shocks and bond positions

$$(x_1, ..., x_{N_x})$$

$$(b_1, ..., b_{N_b})$$

- x denotes shocks to the endowment.
- Optimization problem:

$$V_1(b_i, x_j) = \underset{i' \in \{1,...,N_b\}}{Max} u\left(y(x_j) + b_i - q\left(x_j, b_{i'}\right) b_{i'}\right) + \beta \sum_{j'} \pi_{j,j'} V\left(b'_{i'}, x_{j'}\right)$$

What do we do?

- Define grid for endowment and bond positions

$$\left(x_1, ..., x_{\hat{N}_x}\right)$$

$$\left(b_1, ..., b_{\hat{N}_b}\right)$$

- Approximate value functions V_1 and V_0 using Chebychev polynomials.
- for $(x, b) \in [\underline{x}, \bar{x}] \times \left[\underline{b}, \bar{b}\right]$

$$V_0(x, b) = \sum_{i=1}^{\hat{N}_x} \sum_{j=1}^{\hat{N}_b} a_{i,j}^1 T_i(x) T_j(b)$$

$$V_1(x) = \sum_{i=1}^{\hat{N}_x} a_i^0 T_i(x)$$

where

$$T_i(x) = cos\left(i \times cos^{-1}\left(2\frac{x - \underline{x}}{\bar{x} - \underline{x}} - 1\right)\right)$$

- To extrapolate: use 2nd. order Taylor approximation around the closest grid point.

SIMULATION RESULTS

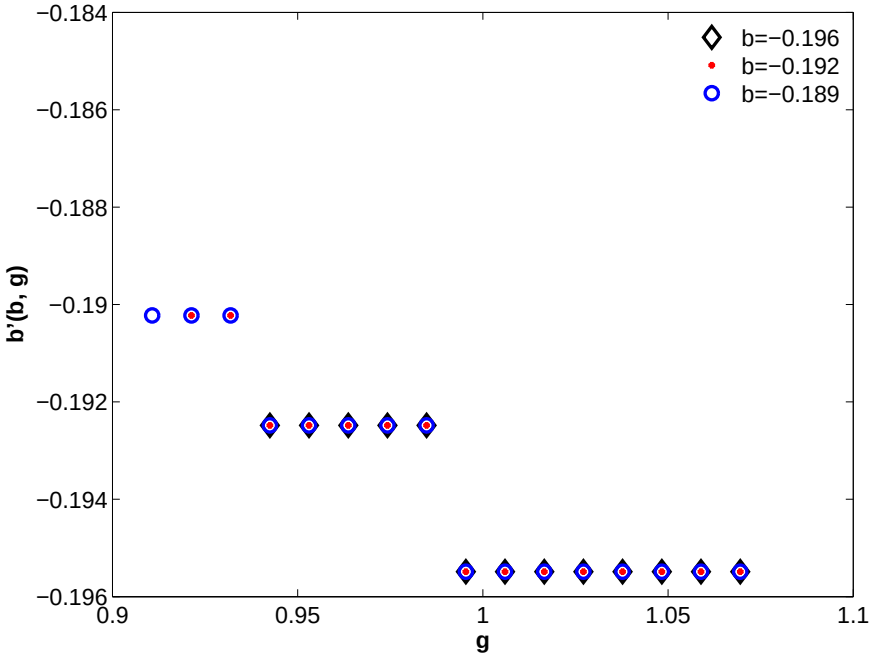
	Model I AG	Model I Interp.	Model II AG	Model II Interp.	Argentina Data
$\sigma(y)$	4.32	4.34	4.45	4.42	4.08
$\sigma(c)$	4.37	4.47	4.71	4.67	4.85
$\sigma(TB/Y)$	**0.17**	0.48	0.95	0.92	1.36
$\sigma(R_s)$	0.04	0.01	0.32	0.07	**3.17**
$\rho(c, y)$	0.99	0.99	0.98	0.98	0.96
$\rho(TB/Y, y)$	-0.33	-0.3	-0.19	-0.19	-0.89
$\rho(R_s, y)$	0.51	-0.53	-0.03	0.08	**-0.59**
$\rho(R_s, TB/Y)$	-0.21	0.73	0.11	0.01	**0.68**
Default rate (per 10,000)	2	8	23	22	75
Mean debt output ratio (%)	27	25	19	19	
Maximum R_s (basis point)	23	27	151	97	

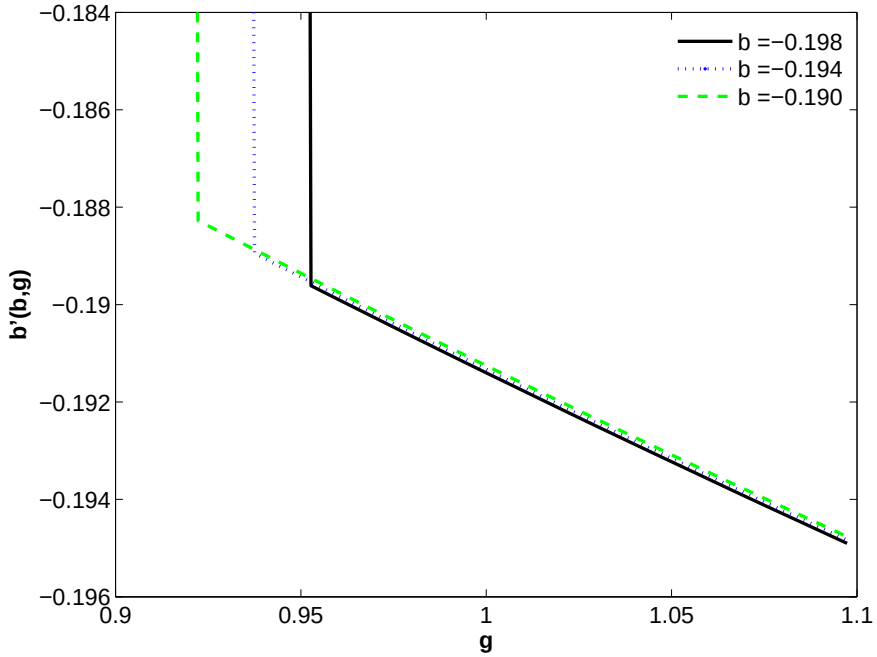
- Average of 500 samples. Each sample contains 500 periods.

Model I: simulation results with different grid specifications

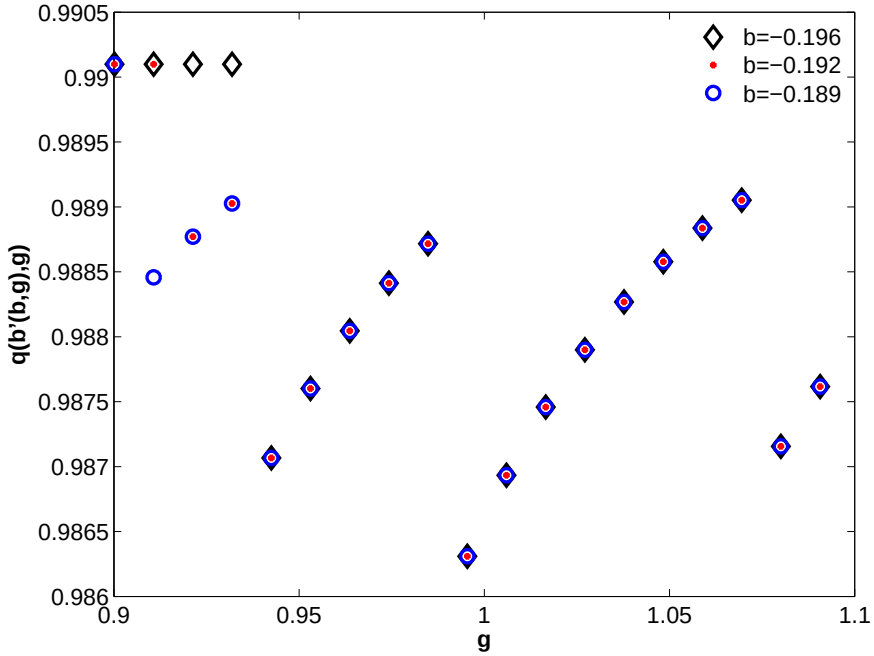
	AG	(400,75,**5**)	(600,25,**5**)	(400,25,**12**)	(400,75,**12**)	(600,25,**12**)
$\sigma(y)$	4.32	4.27	4.30	4.54	4.33	4.53
$\sigma(c)$	4.37	4.31	4.34	4.67	4.45	4.66
$\sigma\left(TB/Y\right)$	**0.17**	**0.17**	**0.16**	**0.52**	**0.44**	**0.52**
$\sigma\left(R_s\right)$	0.04	0.07	0.06	0.26	0.11	0.26
$\rho\left(c,y\right)$	0.99	0.99	0.99	0.99	0.99	0.99
$\rho\left(TB/Y,y\right)$	-0.33	-0.35	-0.35	-0.29	-0.31	-0.29
$\rho\left(R_s,y\right)$	0.51	0.30	0.52	-0.24	0.014	-0.23
$\rho\left(R_s,TB/Y\right)$	-0.21	-0.22	-0.35	-0.03	-0.08	-0.03
Mean debt to output	27	27	27	25	25	25
Rate of default	**2**	**2**	**2**	**8**	**6**	**8**

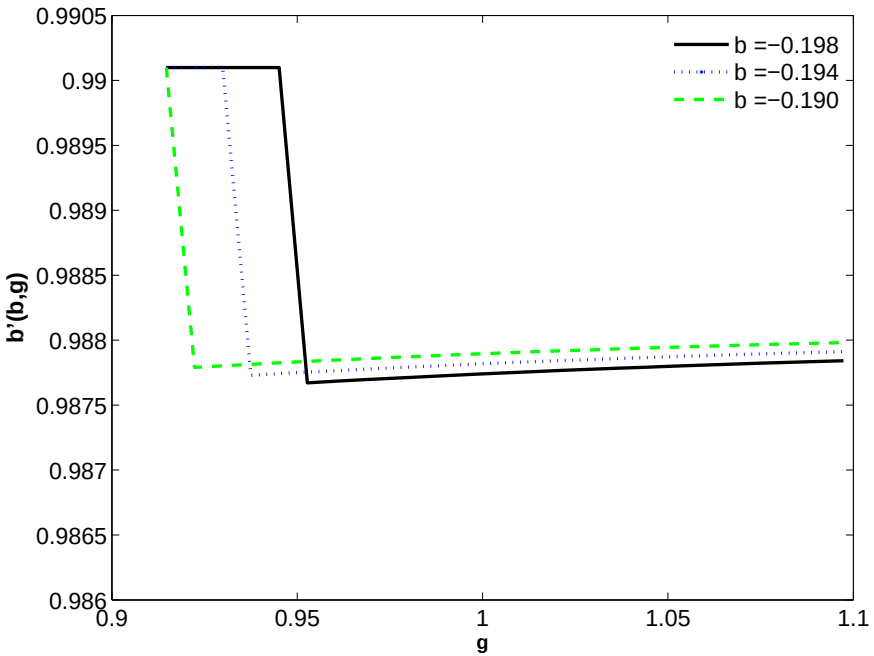
Optimal savings in Model II





Equilibrium bond price in Model II





Default regions in Model II

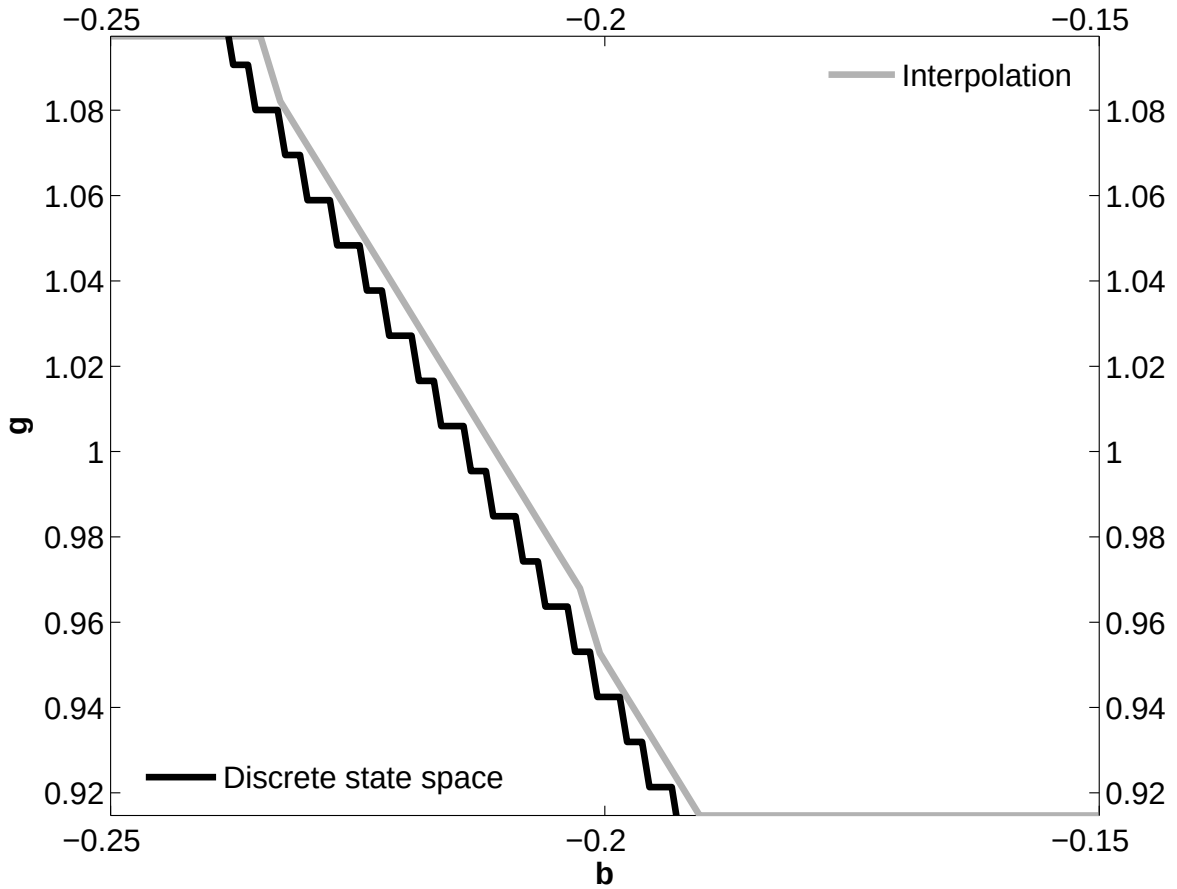
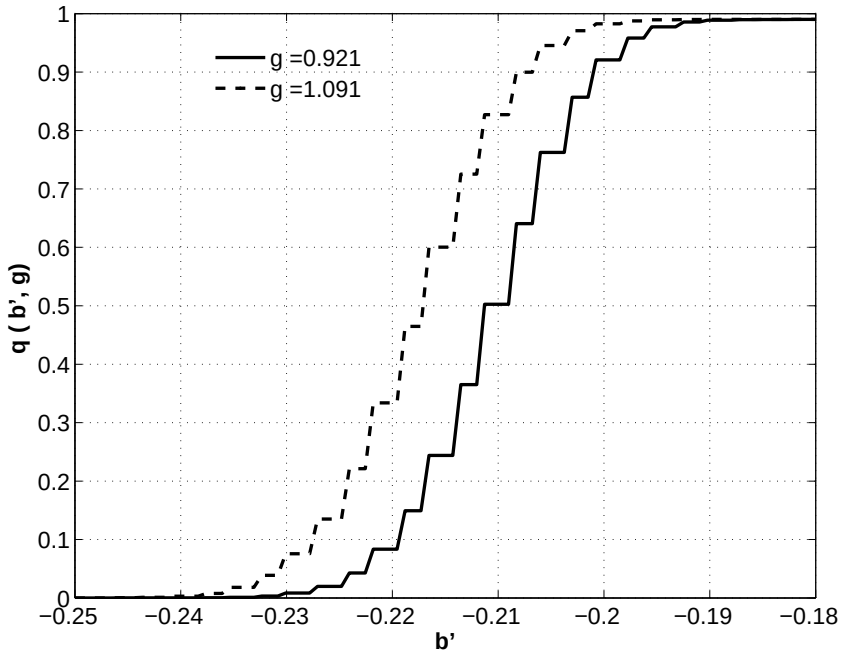
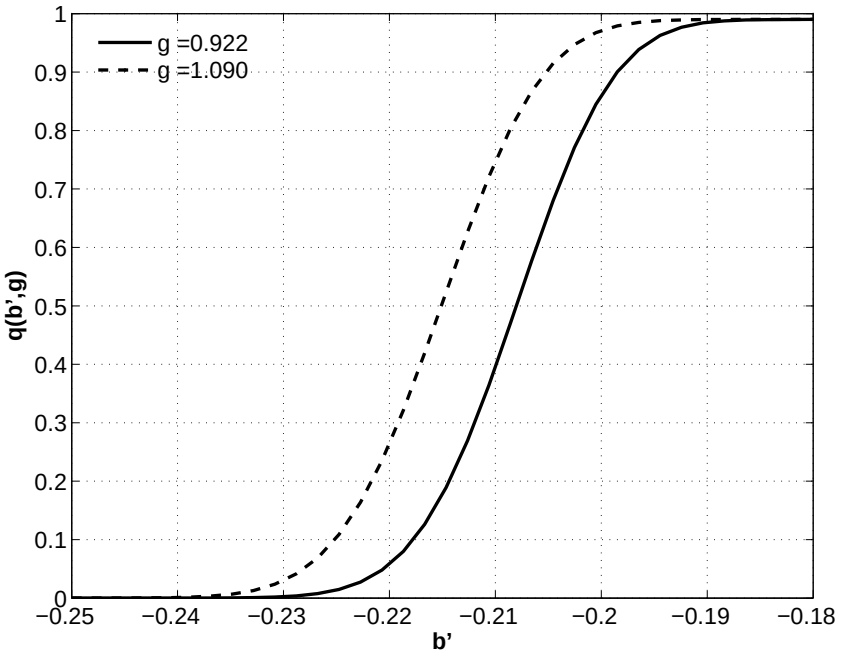


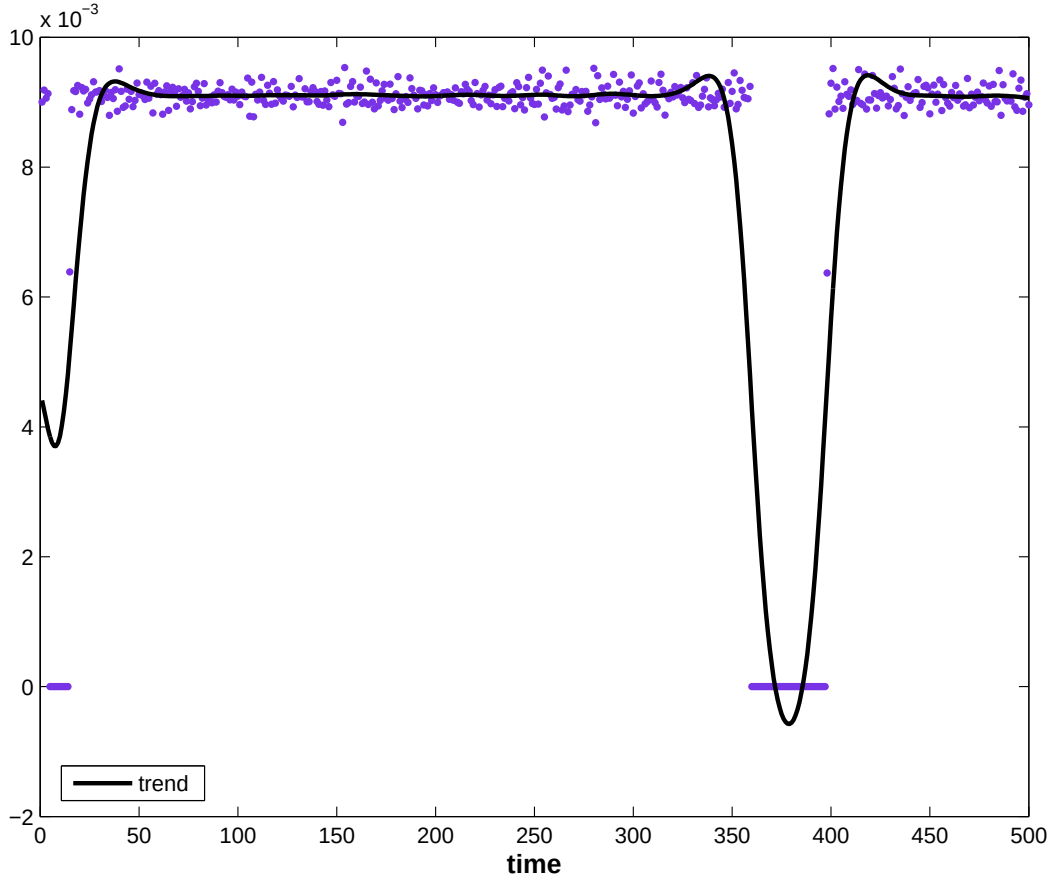
Figure 1: **Default regions computed using the discrete state space and interpolation methods.**

Bond price functions in Model II

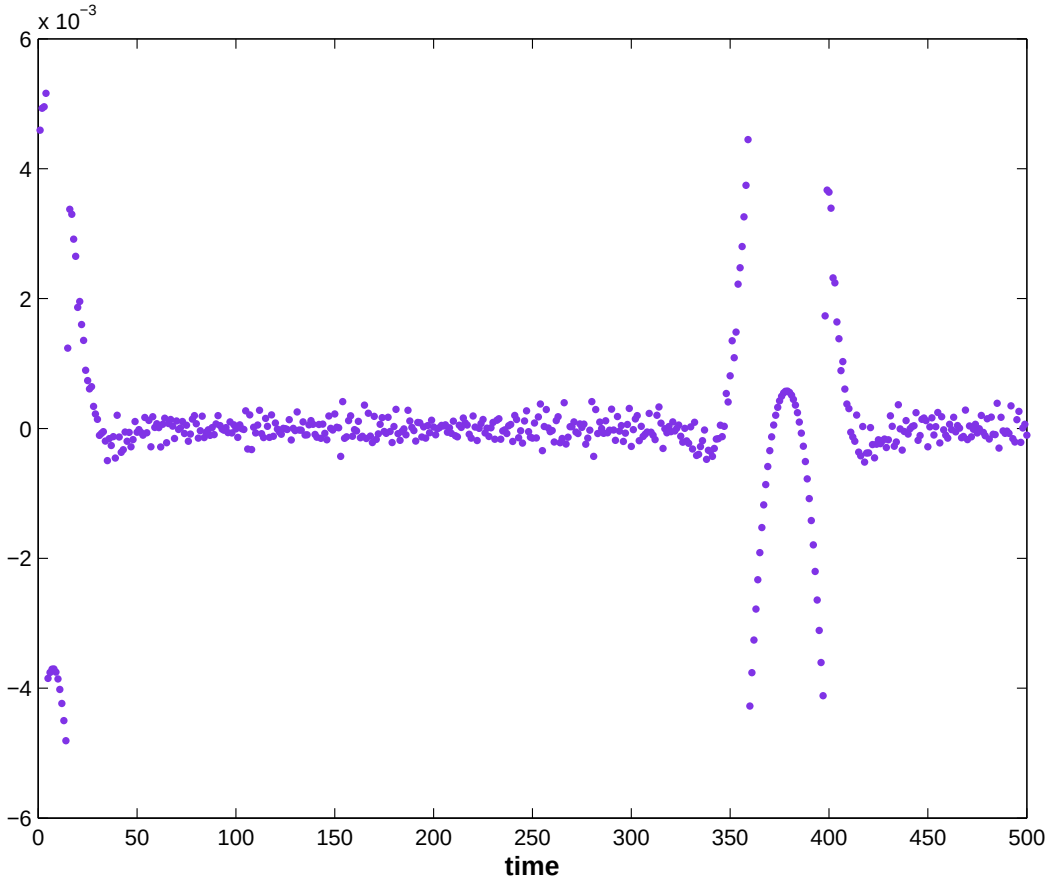




Outliers: a sample with two default episodes



De-trended sample of spreads



Results without outliers

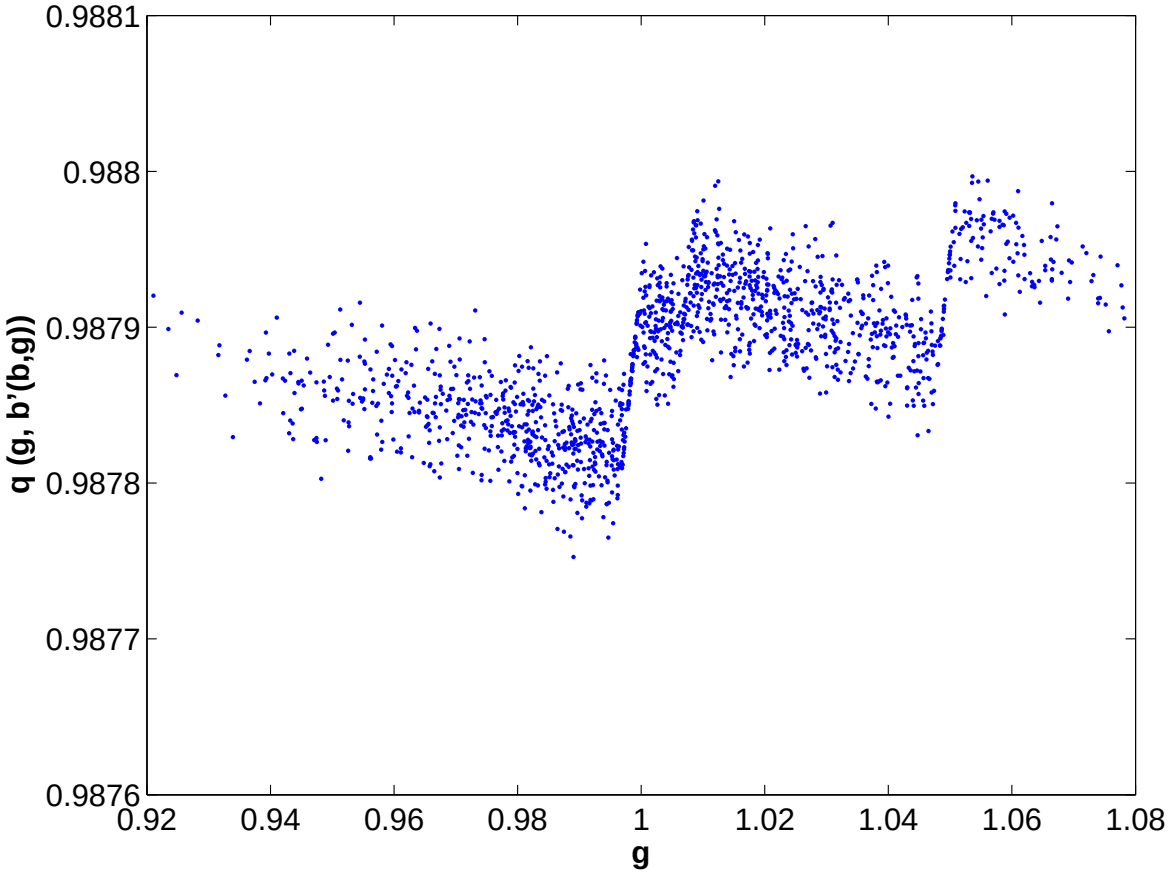
	Model I		Model II	
	As in AG	Without outliers	As in AG	Without outliers
$\sigma(y)$	4.34	4.13	4.42	4.16
$\sigma(c)$	4.47	4.25	4.67	4.39
$\sigma\left(TB/Y\right)$	0.48	0.24	0.92	0.65
$\sigma\left(R_s\right)$	0.012	0.007	0.07	0.014
$\rho\left(c,y\right)$	0.99	0.99	0.98	0.99
$\rho\left(TB/Y,y\right)$	-0.30	-0.43	-0.19	-0.29
$\rho\left(R_s,y\right)$	-0.53	-0.74	0.08	-0.09
$\rho\left(R_s,TB/Y\right)$	0.73	0.89	0.01	0.91
Mean debt output ratio (%)	25	25	19	19
Maximum R_s (basis point)	27	27	97	97

- Without outliers: first 400 samples of 72 periods before a default episode. The last exclusion period was observed at least 2 periods before the beginning of each sample.

Bonus track: why Chebychev polynomials?

- Need a procedure to interpolate and extrapolate.
  - Allow the agent to choose from saving levels that do not belong to the grid,
  - Can compute expectations evaluating functions at endowment shocks outside the grid.
  - More accurate approximation of price function $q(b', x)$
- Need the procedure to display "smooth" first order derivatives.
- Linear interpolation/extrapolation does not work.

Shocks to the trend and equilibrium bond price with a piece-wise linear interpolation scheme



- FOC if the country has not defaulted:

$$\underbrace{u'\left(y(x)+b-q\left(b',x\right)b'\right)\left(q\left(b',x\right)+\frac{\partial q\left(b',x\right)}{\partial b'}b'\right)}_{LHS}=\underbrace{\beta\int\frac{\partial V\left(b',x'\right)}{\partial b'}F_x\left(dx'\mid x\right)}_{RHS}$$

- We find that $q\left(b',x\right)$ can be written as:

$$q\left(b',x\right)=\frac{1-F\left(x^*\left(b'\right)\right)}{1+r},$$

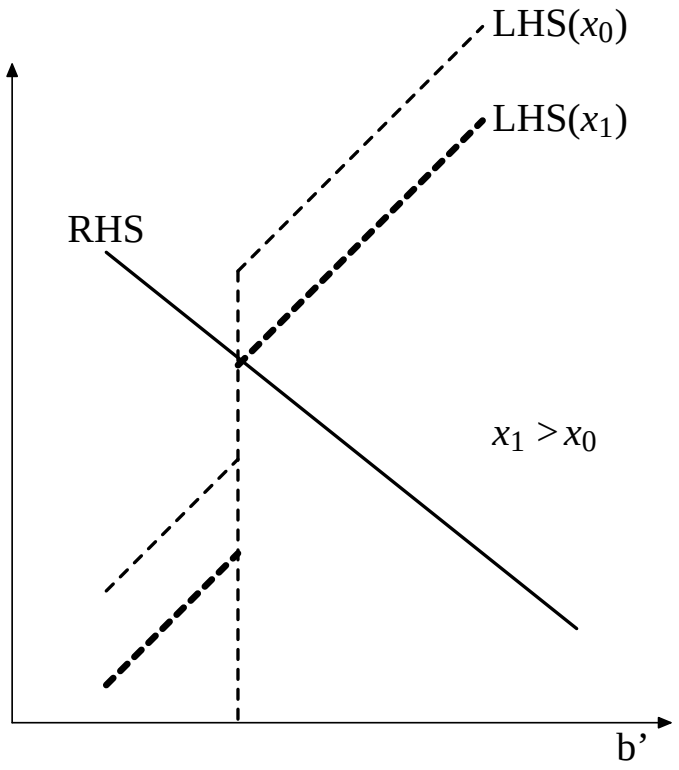
where $x^*\left(b'\right)$ is s.t. $V_0(b',x^*\left(b'\right))=V_1(x^*\left(b'\right))$

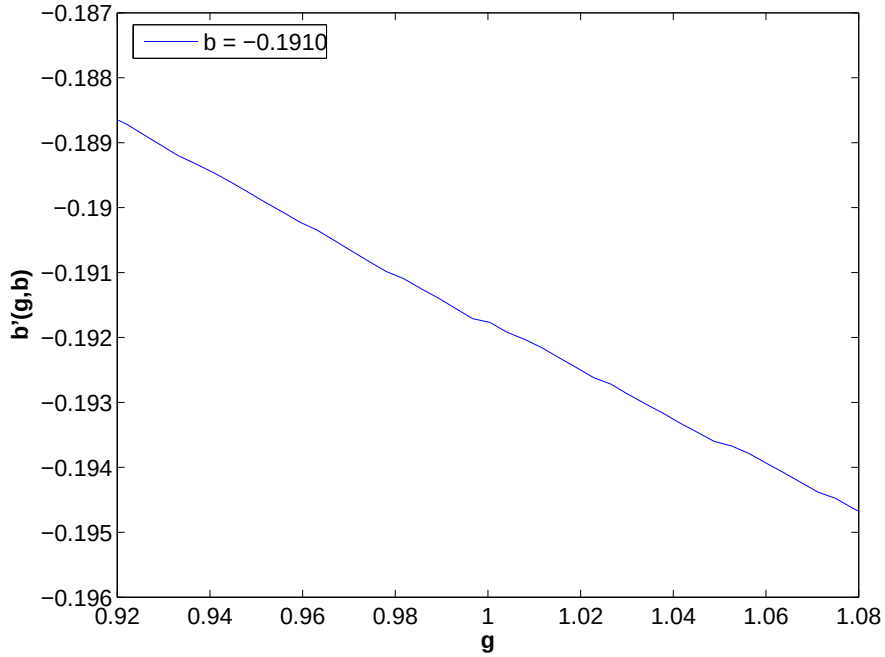
- Therefore:

$$\frac{\partial q\left(b',x^*\left(b'\right)\right)}{\partial b'}=\frac{-f\left(x^*\left(b'\right)\right)}{1+r}\times\underbrace{\frac{\partial x^*\left(b'\right)}{\partial b'}}_{\frac{\frac{\partial V_0(b',x^*(b'))}{\partial b}}{\frac{\partial V_1(x^*(b'))}{\partial x}-\frac{\partial V_0(b',x^*(b'))}{\partial x}}}$$

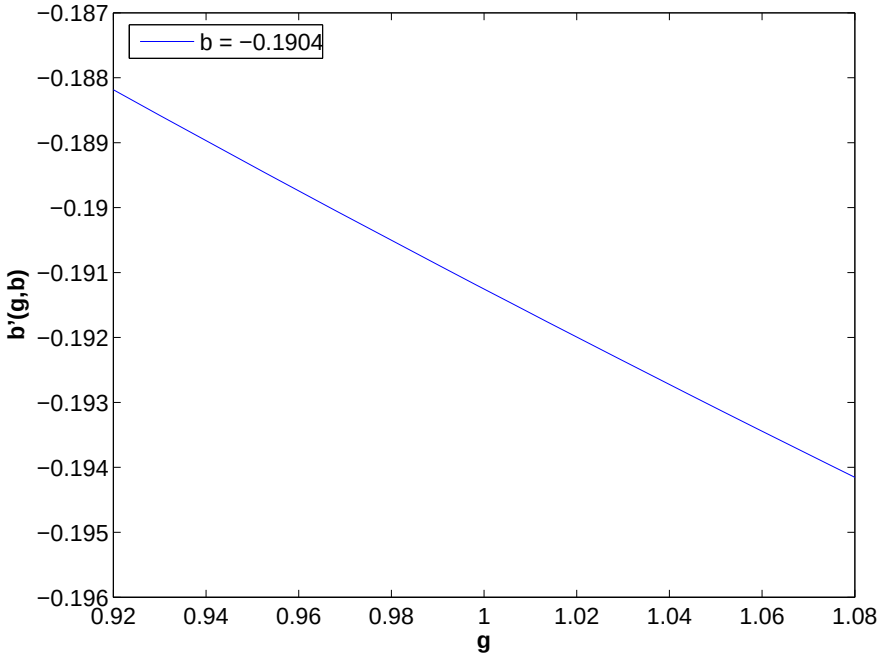
Example

- Nature of the problem with linear interpolation.

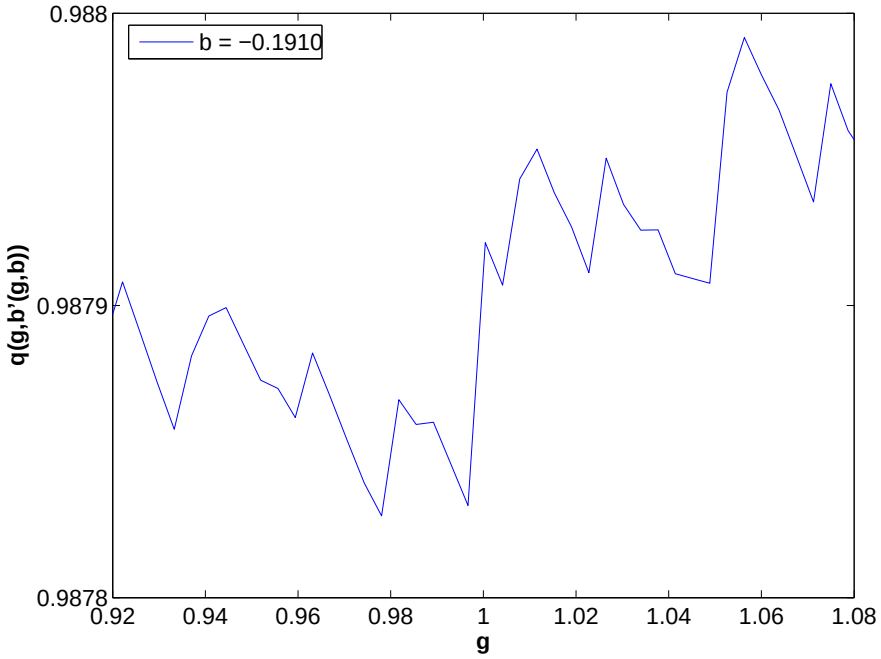




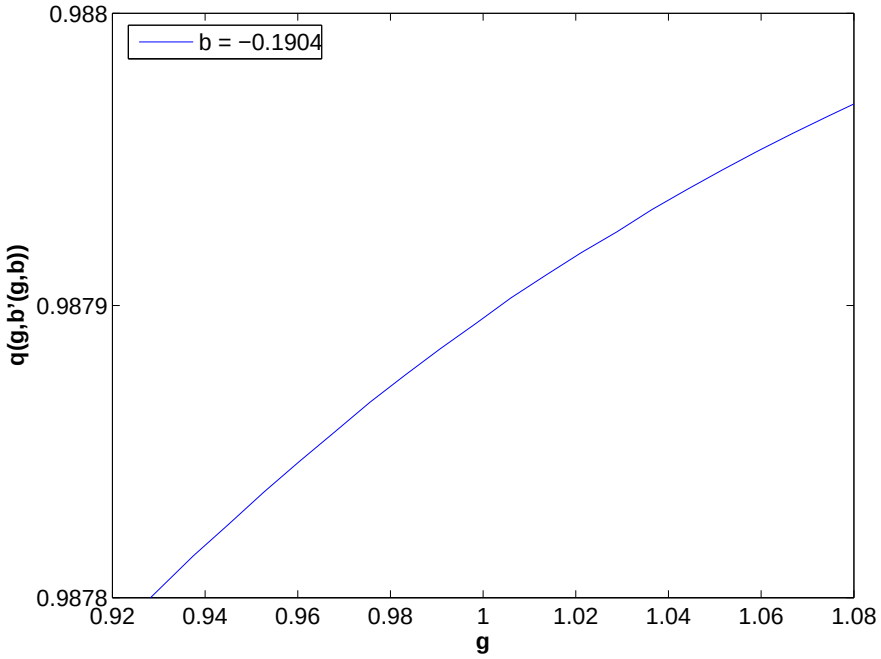
(a) Linear interpolation



(b) Chebychev

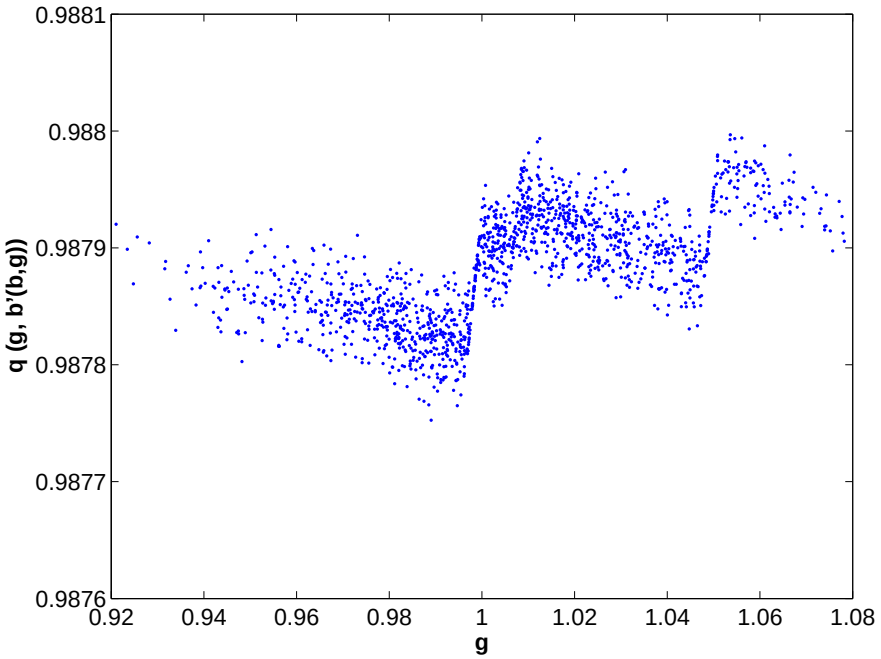


(c) Linear interpolation

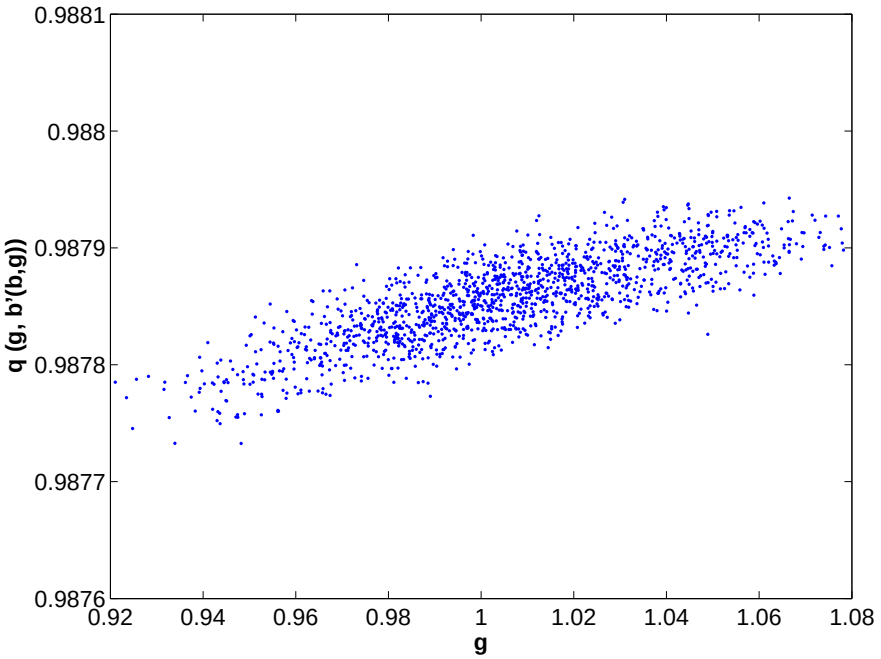


(d) Chebychev

Shocks to the trend and equilibrium bond price



(e) Linear interpolation



(f) Chebychev

CONCLUSIONS

- Discrete state space method may distort the behavior of the spread over the business cycle.
  - The discrete space technique overstates the spread volatility.
  - The correlations of the spread with output and trade balance are also distorted. In contrast with Aguiar and Gopinath (2006), the spread is countercyclical in Model I.
- Other business cycle statistics that do not involve the spread levels seem to be robust.

MODEL II SIMULATION RESULTS WITH DIFFERENT GRIDS

	AG	(400,75,8.3)	(600,25,8.3)	(400,25,12)	(400,75,12)	(600,25,12)
$\sigma(y)$	4.45	4.36	4.39	4.42	4.38	4.42
$\sigma(c)$	4.71	4.61	4.64	4.67	4.62	4.67
$\sigma\left(TB/Y\right)$	0.95	0.8	0.94	0.91	0.89	0.91
$\sigma\left(R_s\right)$	0.32	0.18	0.32	0.38	0.27	0.38
$\rho\left(c,y\right)$	0.98	0.98	0.98	0.98	0.98	0.98
$\rho\left(TB/Y,y\right)$	-0.19	-0.21	-0.19	-0.21	-0.20	-0.20
$\rho\left(R_s,y\right)$	-0.03	-0.12	0.07	-0.13	-0.04	-0.13
$\rho\left(R_s,TB/Y\right)$	0.11	0.46	0.04	0.34	0.14	0.34
Mean debt to Y	19	19	19	19	19	19
Rate of default	23	21	23	21	21	21

Sovereign Default and the Choice of Maturity

- Joint with EMIR Yurdagul and Horacio Sapriza

- Motivation:

  1. Sovereigns borrow from international markets at a duration that typically exceeds one year.

	Argentina	Brazil	Chile	Colombia	Mexico
Overall	6.11	6.47	7.11	3.96	6.69

Sovereign Default and the Choice of Maturity

- Joint with EMIR Yurdagul and Horacio Sapriza

- Motivation:

1. Sovereigns borrow from international markets at a duration that typically exceeds one year.

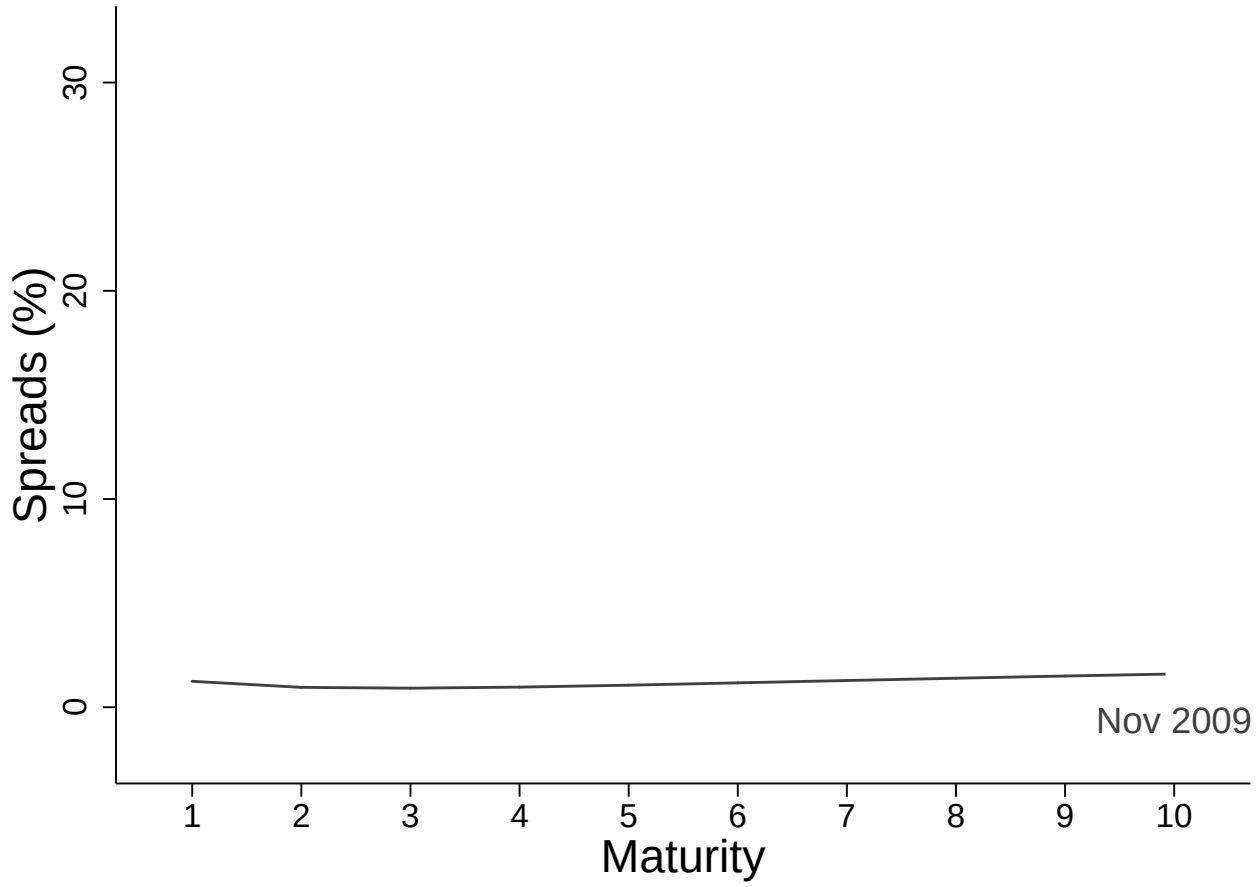
	Argentina	Brazil	Chile	Colombia	Mexico
Overall	6.11	6.47	7.11	3.96	6.69

2. The duration of debt varies over the business cycle.

	Argentina	Brazil	Chile	Colombia	Mexico
Good times	6.65	8.23	9.12	4.08	7.45
Bad times	4.86	5.82	6.87	3.85	6.79

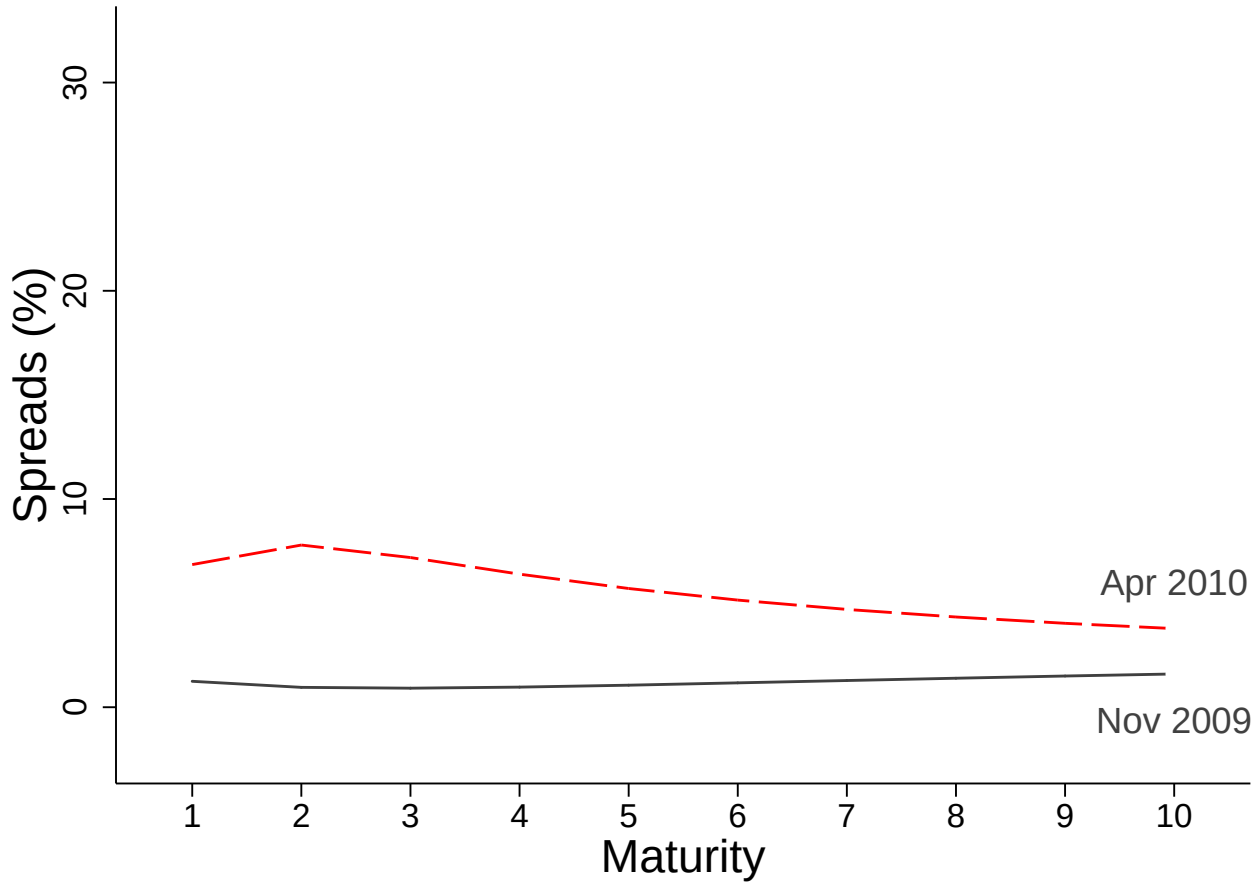
Motivation: Greek Spread curve

3. The level and shape of the yield curve changes near default.



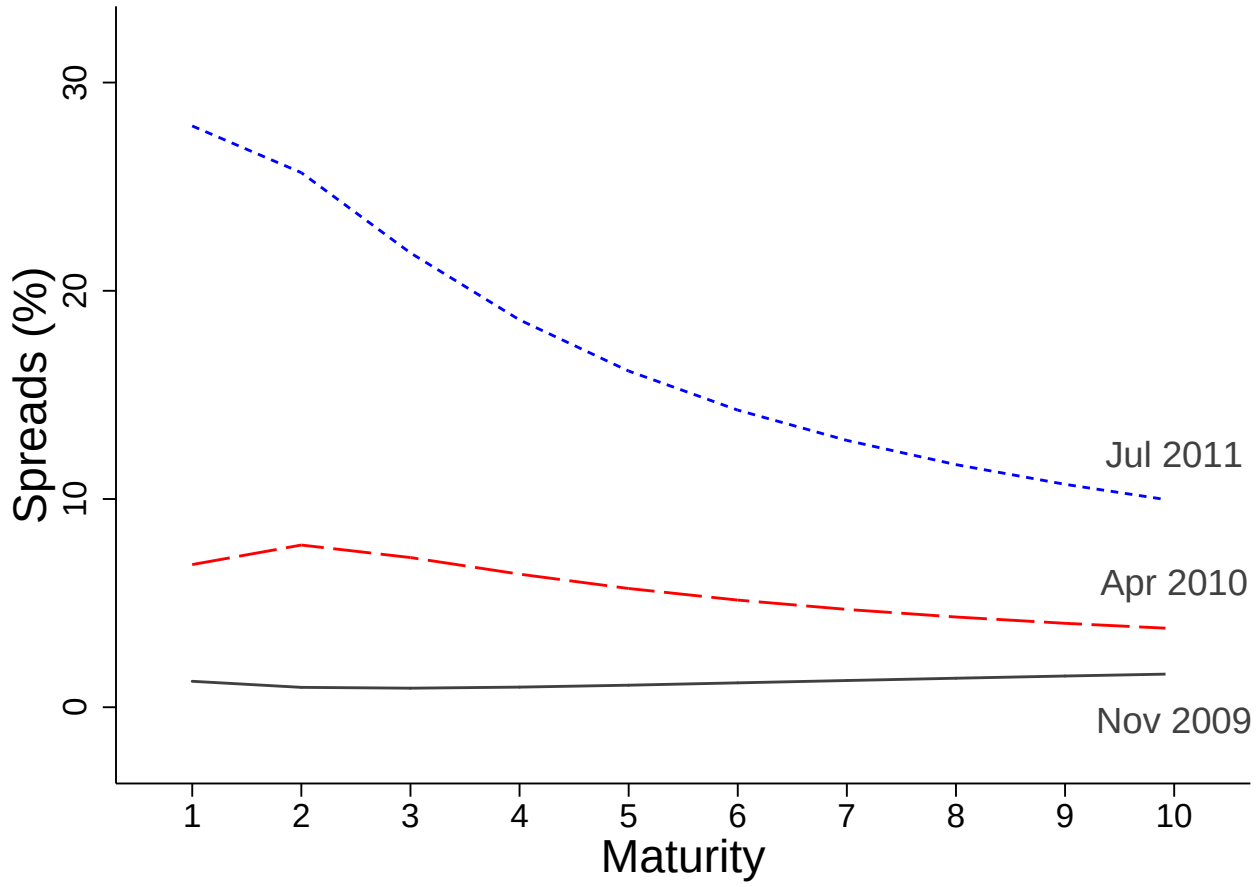
Motivation: Greek Spread curve

3. The level and shape of the yield curve changes near default.

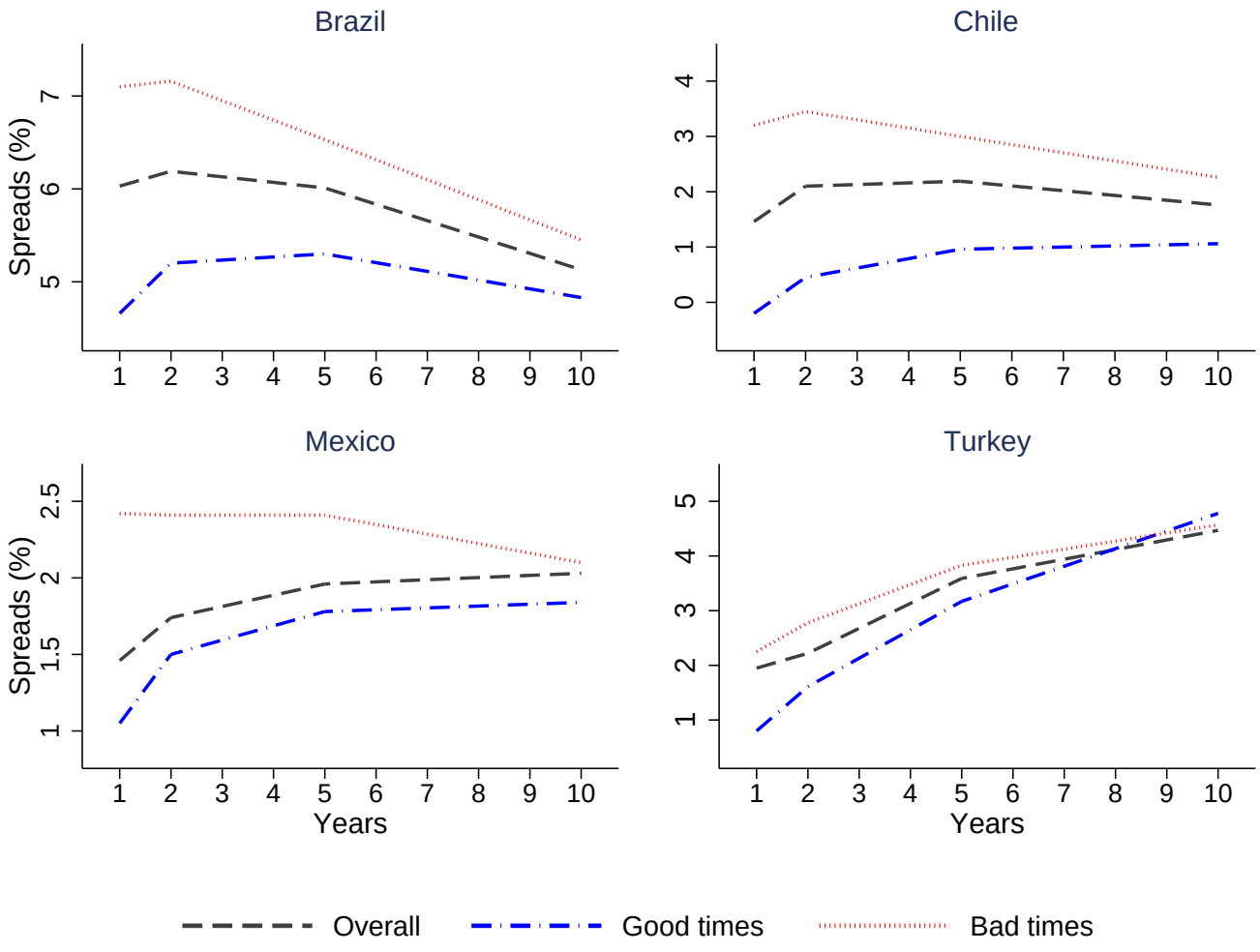


Motivation: Greek Spread curve

3. The level and shape of the yield curve changes near default.



Spread curves, several countries



Note: Good (bad) times are observations with 1-year spread lower (higher) than the median.

What we do

- Present a new model to study:

What we do

- Present a new model to study:
  1. Term structure of interest rate spreads.

What we do

- Present a new model to study:
  1. Term structure of interest rate spreads.
  2. Countries' choice of sovereign bonds maturity.

What we do

- Present a new model to study:
  1. Term structure of interest rate spreads.
  2. Countries' choice of sovereign bonds maturity.
- Find that there are three key features affecting maturity choice:

What we do

- Present a new model to study:
  1. Term structure of interest rate spreads.
  2. Countries' choice of sovereign bonds maturity.

- Find that there are three key features affecting maturity choice:
  1. Debt dilution. ⇓

What we do

- Present a new model to study:
  1. Term structure of interest rate spreads.
  2. Countries' choice of sovereign bonds maturity.

- Find that there are three key features affecting maturity choice:
  1. Debt dilution. $\Downarrow$
  2. Debt restructuring episodes. $\Uparrow$

What we do

- Present a new model to study:
  1. Term structure of interest rate spreads.
  2. Countries' choice of sovereign bonds maturity.

- Find that there are three key features affecting maturity choice:
  1. Debt dilution. $\Downarrow$
  2. Debt restructuring episodes. $\Uparrow$
  3. Sudden stops. $\Uparrow$

Related literature

- Arellano [2008] and Aguiar and Gopinath [2006]: Most quantitative models of sovereign default has used one-period debt.

Related literature

- Arellano [2008] and Aguiar and Gopinath [2006]: Most quantitative models of sovereign default has used one-period debt.
- Chatterjee and Eyigungor [2012] and Hatchondo and Martinez [2009]: long-term debt with exogenous maturity and front-loaded payments.

Related literature

- Arellano [2008] and Aguiar and Gopinath [2006]: Most quantitative models of sovereign default has used one-period debt.
- Chatterjee and Eyigungor [2012] and Hatchondo and Martinez [2009]: long-term debt with exogenous maturity and front-loaded payments.
- Arellano and Ramanarayanan [2012]: one-period bond and a perpetuity bond with decaying coupons.

Related literature

- Arellano [2008] and Aguiar and Gopinath [2006]: Most quantitative models of sovereign default has used one-period debt.
- Chatterjee and Eyigungor [2012] and Hatchondo and Martinez [2009]: long-term debt with exogenous maturity and front-loaded payments.
- Arellano and Ramanarayanan [2012]: one-period bond and a perpetuity bond with decaying coupons.
- More recently:

Related literature

- Arellano [2008] and Aguiar and Gopinath [2006]: Most quantitative models of sovereign default has used one-period debt.
- Chatterjee and Eyigungor [2012] and Hatchondo and Martinez [2009]: long-term debt with exogenous maturity and front-loaded payments.
- Arellano and Ramanarayanan [2012]: one-period bond and a perpetuity bond with decaying coupons.
- More recently:
  - Bai et al. [2014] study the profile of payments in a model with endogenous profile and maturity, but without debt dilution.

Related literature

- Arellano [2008] and Aguiar and Gopinath [2006]: Most quantitative models of sovereign default has used one-period debt.
- Chatterjee and Eyigungor [2012] and Hatchondo and Martinez [2009]: long-term debt with exogenous maturity and front-loaded payments.
- Arellano and Ramanarayanan [2012]: one-period bond and a perpetuity bond with decaying coupons.
- More recently:
  - Bai et al. [2014] study the profile of payments in a model with endogenous profile and maturity, but without debt dilution.
  - Under some specific assumptions, Aguiar and Amador [2014] prove that the country wants to eliminate holdings of legacy long term debt.

Benchmark Model, Main Features

- Time is discrete.
- Small open economy.
- Incomplete markets.
- Exogenous income given by AR(1) process.
- Country’s utility function is CRRA.
- Lenders are risk neutral.

Benchmark Model, Bonds

- In any period, a country portfolio is represented by a fixed coupon b and its maturity m (the number of periods the coupon will be paid for).
- Countries cannot commit to repay their debts—sovereign default is allowed.
- The portfolio's characteristics (b, m) and current income y determine the default probability.
- The unit price of a bond of maturity n for a country with portfolio (b, m) and income y is represented by $q(y, b, m; n)$.

Notation, Illustrative Example

- Suppose that the country has the portfolio $= \{b, b, b, b\}$.

$$\text{portfolio value} = b \times q(y, b, 4; 4).$$

Notation, Illustrative Example

- Suppose that the country has the portfolio $= \{b, b, b, b\}$.

$$\text{portfolio value} = b \times q(y, b, 4; 4).$$

- Imagine a promise to pay part of the coupon for the following 4 periods is holding part of the coupon $= \{\underline{b}, \underline{b}, \underline{b}, \underline{b}\}$.

$$\text{value} = \underline{b} \times q(y, b, 4; 4).$$

Notation, Illustrative Example

- Suppose that the country has the portfolio $= \{b, b, b, b\}$.

$$\text{portfolio value} = b \times q(y, b, 4; 4).$$

- Imagine a promise to pay part of the coupon for the following 4 periods is holding part of the coupon $= \{\underline{b}, \underline{b}, \underline{b}, \underline{b}\}$.

$$\text{value} = \underline{b} \times q(y, b, 4; 4).$$

- Also, the price of a subset of portfolio $= \{\underline{b}, \underline{b}, \underline{b}, 0\}$.

$$\text{value} = \underline{b} \times q(y, b, 4; 3).$$

Notation, Illustrative Example

- Suppose that the country has the portfolio $= \{b, b, b, b\}$.

$$\text{portfolio value} = b \times q(y, b, 4; 4).$$

- Imagine a promise to pay part of the coupon for the following 4 periods is holding part of the coupon $= \{\underline{b}, \underline{b}, \underline{b}, \underline{b}\}$.

$$\text{value} = \underline{b} \times q(y, b, 4; 4).$$

- Also, the price of a subset of portfolio $= \{\underline{b}, \underline{b}, \underline{b}, 0\}$.

$$\text{value} = \underline{b} \times q(y, b, 4; 3).$$

- Similarly, to obtain the price of a 4-period zero coupon bond $= \{0, 0, 0, \underline{b}\}$.

$$\text{value} = \underline{b} \times [q(y, b, 4; 4) - q(y, b, 4; 3)] .$$

Bellman Equations

- The country's choice to either pay or default is represented by

$$V(y, b, m) = \max\left[V^G(y, b, m), V^D(y)\right].$$

Policy function $D(y, b, m)$ is 1 if default and 0 otherwise.

Bellman Equations

- The country's choice to either pay or default is represented by

$$V(y, b, m) = \max\left[V^G(y, b, m), V^D(y)\right].$$

Policy function $D(y, b, m)$ is 1 if default and 0 otherwise.

- If the country chooses to default, lifetime utility can be represented by

$$V^D(y) = \frac{(y - \Phi(y))^{1-\gamma}}{1-\gamma} + \beta E_{y'|y}\left[(1-\lambda)V^D(y') + \lambda V(y', 0, 0)\right],$$

Bellman Equations

- Not defaulting involves paying and potentially changing both the portfolio coupon b' and maturity m':

$$V^G(y, b, m) = \max_{b', m'} \frac{c^{1-\gamma}}{1-\gamma} + \beta E_{y'|y} V(y', b', m')$$

subject to

$$c = y + b - \underbrace{q(y, b', m'; m')b'}_{\text{issue new debt}} + \underbrace{q(y, b', m'; m-1)b}_{\text{retire old debt}}$$

Bellman Equations

- Not defaulting involves paying and potentially changing both the portfolio coupon b' and maturity m':

$$V^G(y, b, m) = \max_{b', m'} \frac{c^{1-\gamma}}{1-\gamma} + \beta E_{y'|y} V(y', b', m')$$

subject to

$$c = y + b - \underbrace{q(y, b', m'; m')b'}_{\text{issue new debt}} + \underbrace{q(y, b', m'; m-1)b}_{\text{retire old debt}}$$

Bellman Equations

- Not defaulting involves paying and potentially changing both the portfolio coupon b' and maturity m':

$$V^G(y,b,m) = \max_{b',m'} \frac{c^{1-\gamma}}{1-\gamma} + \beta E_{y'|y} V(y',b',m')$$

subject to

$$c = y + b - \underbrace{q(y,b',m';m')b'}_{\text{issue new debt}} + \underbrace{q(y,b',m';m-1)b}_{\text{retire old debt}}$$

or

$$c = y + b - \underbrace{q(y,b',m';m')(b'-b)}_{\text{change coupon, given } m'} - \underbrace{[q(y,b',m';m') - q(y,b',m';m-1)]\, b}_{\text{change maturity, given } b \ (\textit{NEW TERM})}.$$

Bellman Equations

- Not defaulting involves paying and potentially changing both the portfolio coupon b' and maturity m':

$$V^G(y,b,m) = \max_{b',m'} \frac{c^{1-\gamma}}{1-\gamma} + \beta E_{y'|y} V(y',b',m')$$

subject to

$$c = y + b - \underbrace{q(y,b',m';m')b'}_{\text{issue new debt}} + \underbrace{q(y,b',m';m-1)b}_{\text{retire old debt}}$$

or

$$c = y + b - \underbrace{q(y,b',m';m')(b'-b)}_{\text{change coupon, given } m'} - \underbrace{[q(y,b',m';m') - q(y,b',m';m-1)]\, b}_{\text{change maturity, given } b \ (\textit{NEW TERM})}.$$

- Policy functions are $\underbrace{B(y,b,m)}_{\textit{coupon}}$ and $\underbrace{M(y,b,m)}_{\textit{maturity}}$.

Equilibrium prices

- Price of a bond of maturity $n > 0$ in a country with income y, debt $-b'$, and maturity $m' > 0$ is

$$q(y, b', m'; n) =$$

$$\frac{E_{y'|y}}{1+r}\{(1 - D(y', b', m'))[1 + q(y', B(y', b', m'), M(y', b', m'); n - 1)]\}.$$

Equilibrium prices

- Price of a bond of maturity $n > 0$ in a country with income y, debt $-b'$, and maturity $m' > 0$ is

$$q(y, b', m'; n) =$$

$$\frac{E_{y'|y}}{1+r}\{(1 - D(y', b', m'))[1 + q(y', \underbrace{B(y', b', m')}_{\text{new coupon}}, \underbrace{M(y', b', m')}_{\text{new maturity}}; n - 1)]\}.$$

Calibration

- Each period is one year.
- Most parameters taken from Arellano and Ramanarayanan [2012].
- Function controlling the income loss in default as in Arellano [2008].
  - Keep only income below threshold $\bar{\phi}$.
- Discount factor, β, and threshold in the income loss function, $\bar{\phi}$, calibrated to match debt-to-output ratio (25%) and default rate (2%).

Benchmark calibration results: optimal maturity is one

	Data	Benchmark
Duration	5.52	1.00
Maturity	8.78	1.00
$\sigma(log(c))/\sigma(log(y))$	1.32	1.39
$\rho(log(c), log(y))$	0.80	0.73
Default (%)	2.00	2.29
Value of debt / Income	0.25	0.24

Why? Debt dilution

- Why is optimal maturity so low?

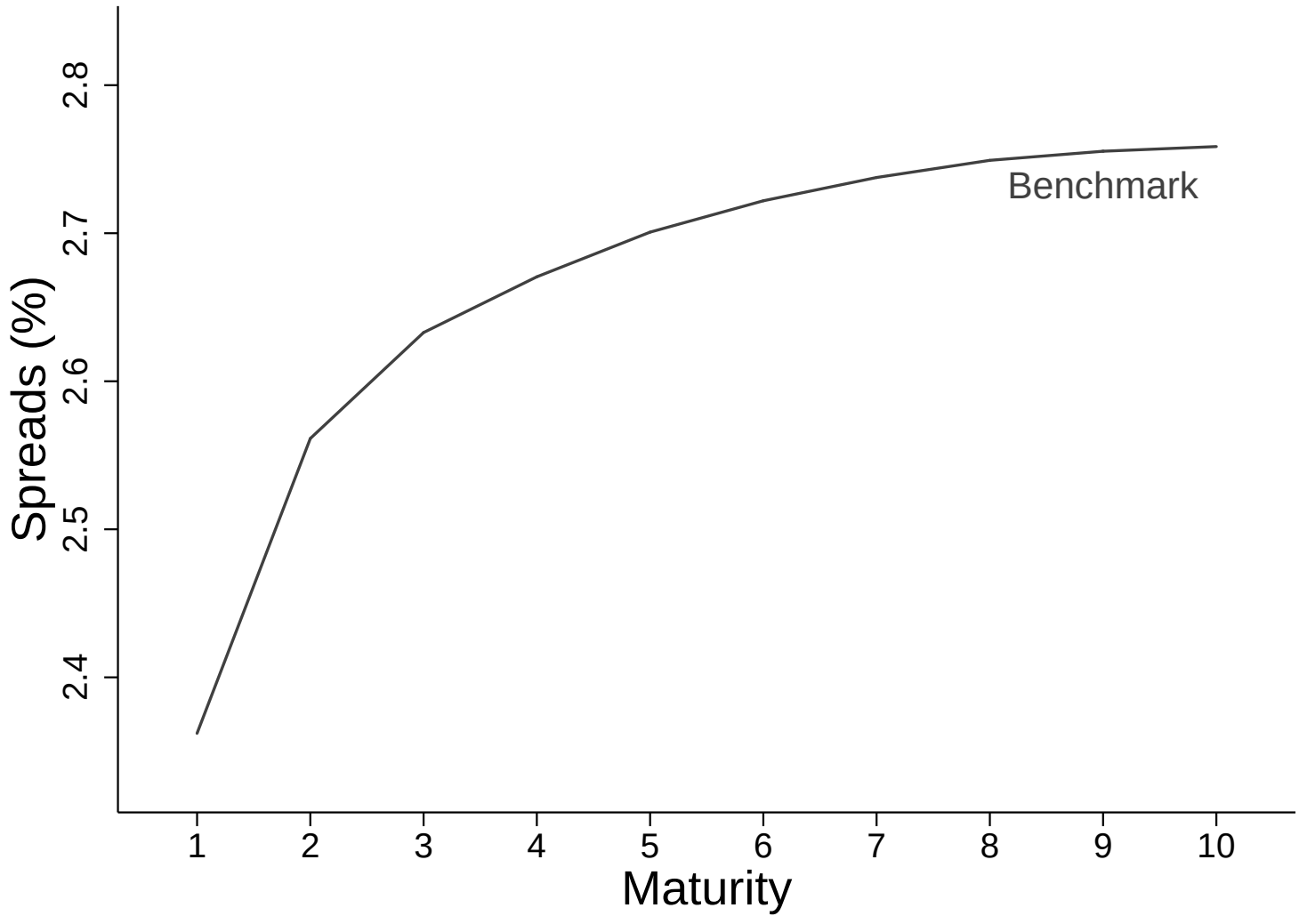
Why? Debt dilution

- Why is optimal maturity so low?
- Our model features debt dilution: the price of outstanding bonds decreases when new bonds are issued.

Why? Debt dilution

- Why is optimal maturity so low?
- Our model features debt dilution: the price of outstanding bonds decreases when new bonds are issued.
- This increases the yields of long-maturity bonds and reduces the optimal maturity.

Model with debt dilution: spread curve



Model without debt dilution

- Assume that to issue new debt, the country must retire previous issuances at the present value discounted at the risk free rate.

Model without debt dilution

- Assume that to issue new debt, the country must retire previous issuances at the present value discounted at the risk free rate.
- This discourages frequent issuances and increases optimal maturity.

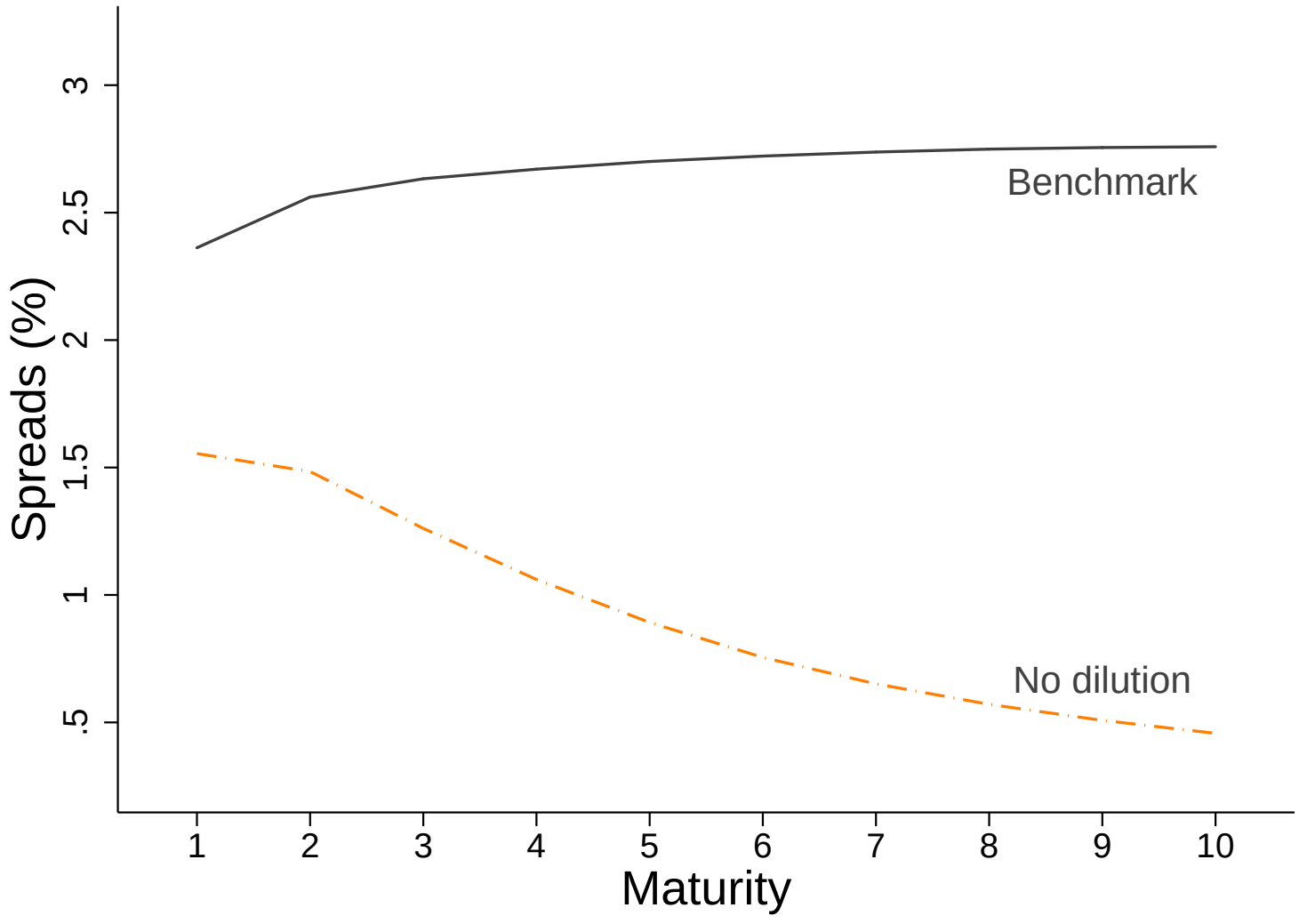
Model without debt dilution

- Assume that to issue new debt, the country must retire previous issuances at the present value discounted at the risk free rate.
- This discourages frequent issuances and increases optimal maturity.
- However, it increases maturity at the cost of generating a decreasing spread curve.

Model with no debt dilution: maturity is much longer

	Data	Benchmark	No dilution
Duration	5.52	1.00	3.50
Maturity	8.78	1.00	6.34
$\sigma(log(c))/\sigma(log(y))$	1.32	1.39	1.32
$\rho(log(c), log(y))$	0.80	0.73	0.76
Default (%)	2.00	2.29	2.27
Value of debt / Income	0.25	0.24	0.24

Model with no debt dilution: spread curve



What may account for long duration?

If the model must have debt dilution, what else can induce countries to choose long maturity bonds?

(1) Sudden stops

(2) High risk aversion

(3) Debt restructuring

What may account for long duration?

If the model must have debt dilution, what else can induce countries to choose long maturity bonds?

(1) Sudden stops

(2) High risk aversion

(3) Debt restructuring

Sudden stops

- Sudden stops: abrupt reduction in net capital inflows.

Sudden stops

- Sudden stops: abrupt reduction in net capital inflows.
- With long maturity debt, the country does not need to roll over all its obligations every period.

Sudden stops, Decision problem

- For a country that has not received a sudden stop shock ($a = 1$):

$$V^G(y, 1, b, m) = \max_{b', m'} \frac{c^{1-\gamma}}{1-\gamma} + \beta E_{y', a'|y} V(y', a', b', m')$$

subject to

$$c = y + b - q(y, b', m'; m')b' + q(y, b', m'; m-1)b$$

Sudden stops, Decision problem

- For a country that has not received a sudden stop shock ($a = 1$):

$$V^G(y, 1, b, m) = \max_{b', m'} \frac{c^{1-\gamma}}{1-\gamma} + \beta E_{y', a'|y} V(y', a', b', m')$$

subject to

$$c = y + b - q(y, b', m'; m')b' + q(y, b', m'; m-1)b$$

- For a country that has received a sudden stop shock ($a = 0$):

$$V^G(y, 0, b, m) = \frac{(y+b)^{1-\gamma}}{1-\gamma} + \beta E_{y', a'|y} V(y', a', b, m-1)$$

Sudden stops, Equilibrium prices

- Prices take into account the probability of sudden stops in the future:

$$q(y, b', m'; n) =$$

$$\frac{E_{y',a'|y}}{1+r}\{(1 - D(y', a', b', m'))[1 + q(y', B(y', a', b', m'), M(y', a', b', m'); n-1)]\}.$$

Sudden stops

- Probability of sudden stops set at 10%, close to the average estimates in Bianchi et al. [2013].

	Data	Benchmark	Sudden stops
Duration	5.52	1.00	2.48
Maturity	8.78	1.00	4.22
$\sigma(log(c))/\sigma(log(y))$	1.32	1.39	1.32
$\rho(log(c), log(y))$	0.80	0.73	0.74
Default (%)	2.00	2.29	1.94
Value of debt / Income	0.25	0.24	0.29

What may account for long duration?

If the model must have debt dilution, what else can induce countries to choose long maturity bonds?

(1) Sudden stops

(2) High risk aversion

(3) Debt restructuring

High risk aversion

- Assume higher risk aversion.

High risk aversion

- Assume higher risk aversion.
- Longer duration provides insurance.

High risk aversion

- Change risk aversion from $\gamma = 2$ to $\gamma = 5$.

	Data	Benchmark	High risk aversion
Duration	5.52	1.00	2.53
Maturity	8.78	1.00	4.21
$\sigma(log(c))/\sigma(log(y))$	1.32	1.39	1.11
$\rho(log(c), log(y))$	0.80	0.73	0.91
Default (%)	2.00	2.29	1.64
Value of debt / Income	0.25	0.24	0.25

What may account for long duration?

If the model must have debt dilution, what else can induce countries to choose long maturity bonds?

(1) Sudden stops

(2) High risk aversion

(3) Debt restructuring

Debt restructuring

- Assume that in cases of default, the sovereign faces two possible outcomes: disorderly default ($1 - \epsilon$) or debt rescheduling (ϵ).

Debt restructuring

- Assume that in cases of default, the sovereign faces two possible outcomes: disorderly default (1 – ϵ) or debt rescheduling (ϵ).

- If debt with level b and maturity m is rescheduled, the new maturity is given by:

$$n^R = \min\{m + \eta, N\}$$

where N is the maximum allowed maturity and $\eta > 0$.

Debt restructuring

- Assume that in cases of default, the sovereign faces two possible outcomes: disorderly default (1 – ϵ) or debt rescheduling (ϵ).

- If debt with level b and maturity m is rescheduled, the new maturity is given by:

$$n^R = \min\{m + \eta, N\}$$

  where N is the maximum allowed maturity and $\eta > 0$.

- The rescheduled debt level satisfies:

$$\underbrace{\psi}_{\text{haircut}} b^R n^R = bm$$

Debt restructuring

- Assume that in cases of default, the sovereign faces two possible outcomes: disorderly default (1 – ϵ) or debt rescheduling (ϵ).

- If debt with level b and maturity m is rescheduled, the new maturity is given by:

$$n^R = \min\{m + \eta, N\}$$

  where N is the maximum allowed maturity and $\eta > 0$.

- The rescheduled debt level satisfies:

$$\underbrace{\psi}_{\text{haircut}} b^R n^R = bm$$

- Hereafter we set $\psi = 1$.

Debt restructuring, Decision problem

- The country's choice to either pay or default is represented by

$$V(y, b, m) = \max\Big[V^G(y, b, m), V^D(y, b, m).\Big].$$

$$V^D(y, b, m) = \epsilon V^{D,o}(y, b, m) + (1 - \epsilon)V^{D,u}(y).$$

Debt restructuring, Decision problem

- The country's choice to either pay or default is represented by

$$V(y,b,m) = \max\Big[V^G(y,b,m), V^D(y,b,m).\Big].$$

$$V^D(y,b,m) = \epsilon V^{D,o}(y,b,m) + (1-\epsilon)V^{D,u}(y).$$

- Value of orderly default:

$$V^{D,o}(y,b,m) = \frac{(y-\phi(y))^{1-\gamma}}{1-\gamma} + \beta E_{y'|y} V(y', b^R, n^R)$$

where

$$n^R = min(n+\eta, N), \quad b^R = b\frac{m}{n^R}$$

Debt restructuring, Decision problem

- The country's choice to either pay or default is represented by

$$V(y,b,m) = \max\Big[V^G(y,b,m), V^D(y,b,m).\Big].$$

$$V^D(y,b,m) = \epsilon V^{D,o}(y,b,m) + (1-\epsilon)V^{D,u}(y).$$

- Value of orderly default:

$$V^{D,o}(y,b,m) = \frac{(y-\phi(y))^{1-\gamma}}{1-\gamma} + \beta E_{y'|y} V(y', b^R, n^R)$$

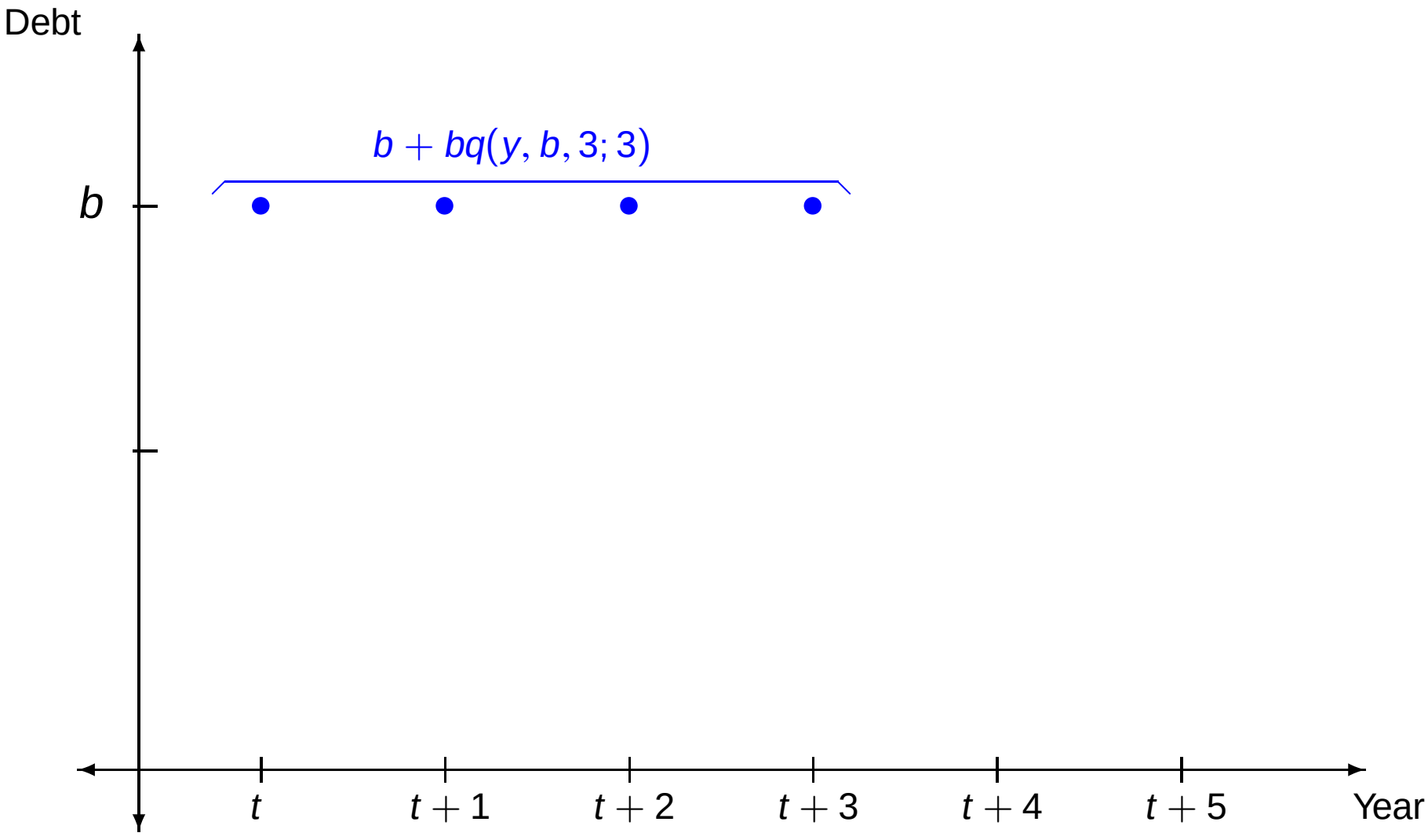
where

$$n^R = min(n+\eta, N), \quad b^R = b\frac{m}{n^R}$$

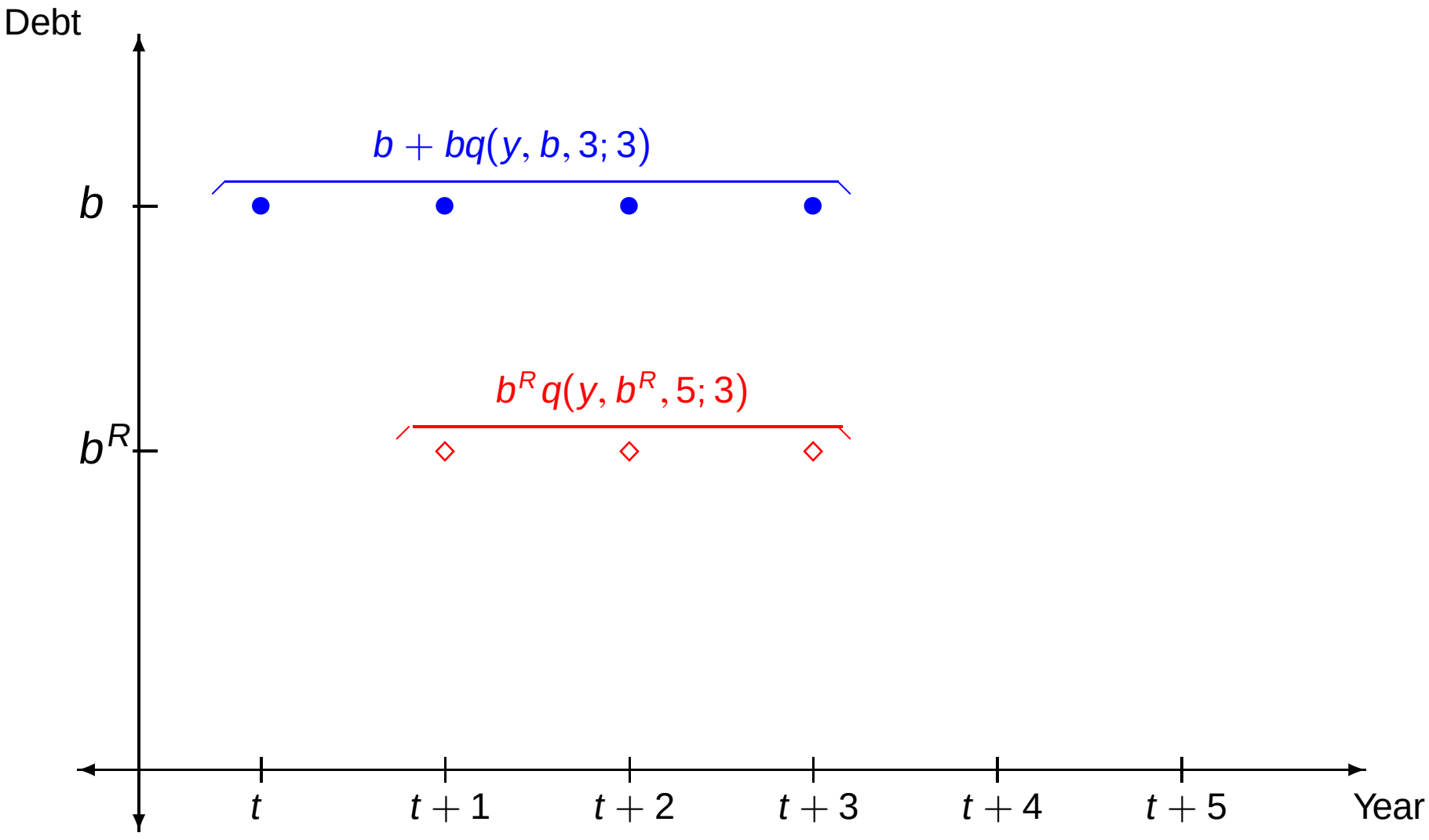
- Value of disorderly default as before.

$$V^{D,u}(y) = \frac{(y-\widetilde{\phi}(y))^{1-\gamma}}{1-\gamma} + \beta E_{y'|y}\Big[(1-\lambda)V^D(y',0,0) + \lambda V(y',0,0)\Big]$$

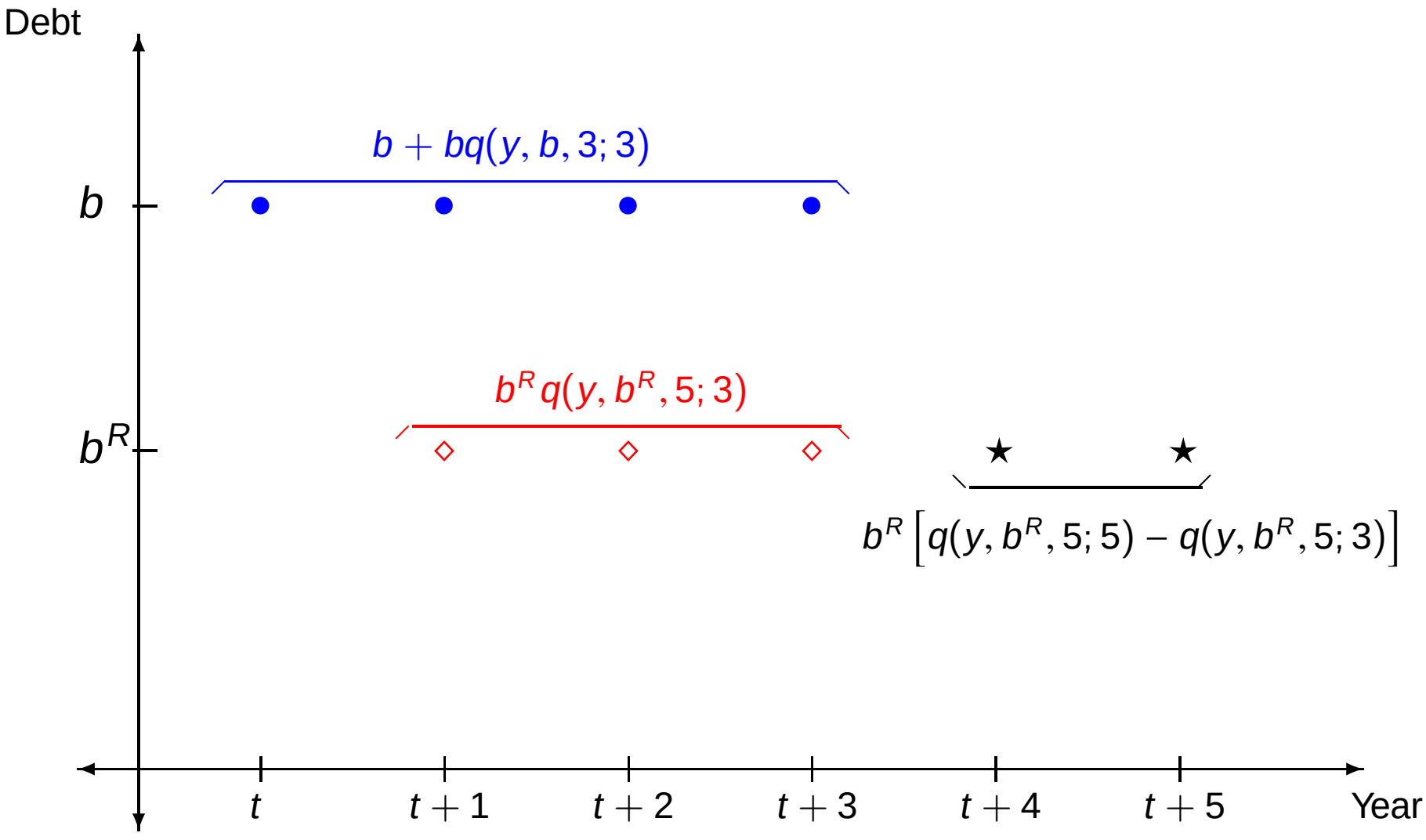
Debt restructuring, Simple example



Debt restructuring, Simple example



Debt restructuring, Simple example



Debt restructuring, Equilibrium prices

- Prices take into account the probability of restructuring in the future:

$$q(y, b', m'; n) =$$

$$\frac{E_{y'|y}}{1+r}\{(1 - D(y', b', m'))[1 + q(B(y', b', m'), y', M(y', b', m'); n-1)]$$

$$+ \underbrace{\epsilon D(y', b', m')}_{\text{debt rescheduling}} \underbrace{\times \frac{b^R}{b'} q(y', b^R, n^R; n-1)}_{\text{recovery on remaining coupons}}$$

$$+ \underbrace{\epsilon D(y', b', m')}_{\text{debt rescheduling}} \underbrace{\times \frac{b^R \times (n-1)}{b' \times (m'-1)}}_{\text{share of face value of debt}} \times \underbrace{[q(y', b^R, n^R; n^R) - q(y', b^R, n^R; m'-1)]}_{\text{unit value of extended maturity}}\}.$$

Debt restructuring, Equilibrium prices

- Prices take into account the probability of restructuring in the future:

$$q(y, b', m'; n) =$$

$$\frac{E_{y'|y}}{1+r}\{(1 - D(y', b', m'))[1 + q(B(y', b', m'), y', M(y', b', m'); n-1)]$$

$$+ \underbrace{\epsilon D(y', b', m')}_{\text{debt rescheduling}} \times \underbrace{\frac{b^R}{b'} q(y', b^R, n^R; n-1)}_{\text{recovery on remaining coupons}}$$

$$+ \underbrace{\epsilon D(y', b', m')}_{\text{debt rescheduling}} \underbrace{\times \frac{b^R \times (n-1)}{b' \times (m'-1)}}_{\text{share of face value of debt}} \times \underbrace{[q(y', b^R, n^R; n^R) - q(y', b^R, n^R; m'-1)]}_{\text{unit value of extended maturity}}\}.$$

Debt restructuring, Equilibrium prices

- Prices take into account the probability of restructuring in the future:

$$q(y, b', m'; n) =$$

$$\frac{E_{y'|y}}{1+r}\{(1 - D(y', b', m'))[1 + q(B(y', b', m'), y', M(y', b', m'); n-1)]$$

$$+ \underbrace{\epsilon D(y', b', m')}_{\text{debt rescheduling}} \times \underbrace{\frac{b^R}{b'} q(y', b^R, n^R; n-1)}_{\text{recovery on remaining coupons}}$$

$$+ \underbrace{\epsilon D(y', b', m')}_{\text{debt rescheduling}} \underbrace{\times \frac{b^R \times (n-1)}{b' \times (m'-1)}}_{\text{share of face value of debt}} \times \underbrace{[q(y', b^R, n^R; n^R) - q(y', b^R, n^R; m'-1)]}_{\text{unit value of extended maturity}}\}.$$

Debt restructuring

- Calibrate $\epsilon = 0.5$ and the number of years that the debt maturity is extended $\eta = 2$ (see Cruces and Trebesch [2011]).

	Data	Benchmark	Restructuring
Duration	5.52	1.00	1.67
Maturity	8.78	1.00	2.35
$\sigma(log(c))/\sigma(log(y))$	1.32	1.39	1.25
$\rho(log(c), log(y))$	0.80	0.73	0.79
Reprofiling (%)		0.00	1.97
Default (%)	2.00	2.29	2.01
Value of debt / Income	0.25	0.24	0.34

Comprehensive specification

Combine the three features:

(1) Probability of sudden stops set at 10%

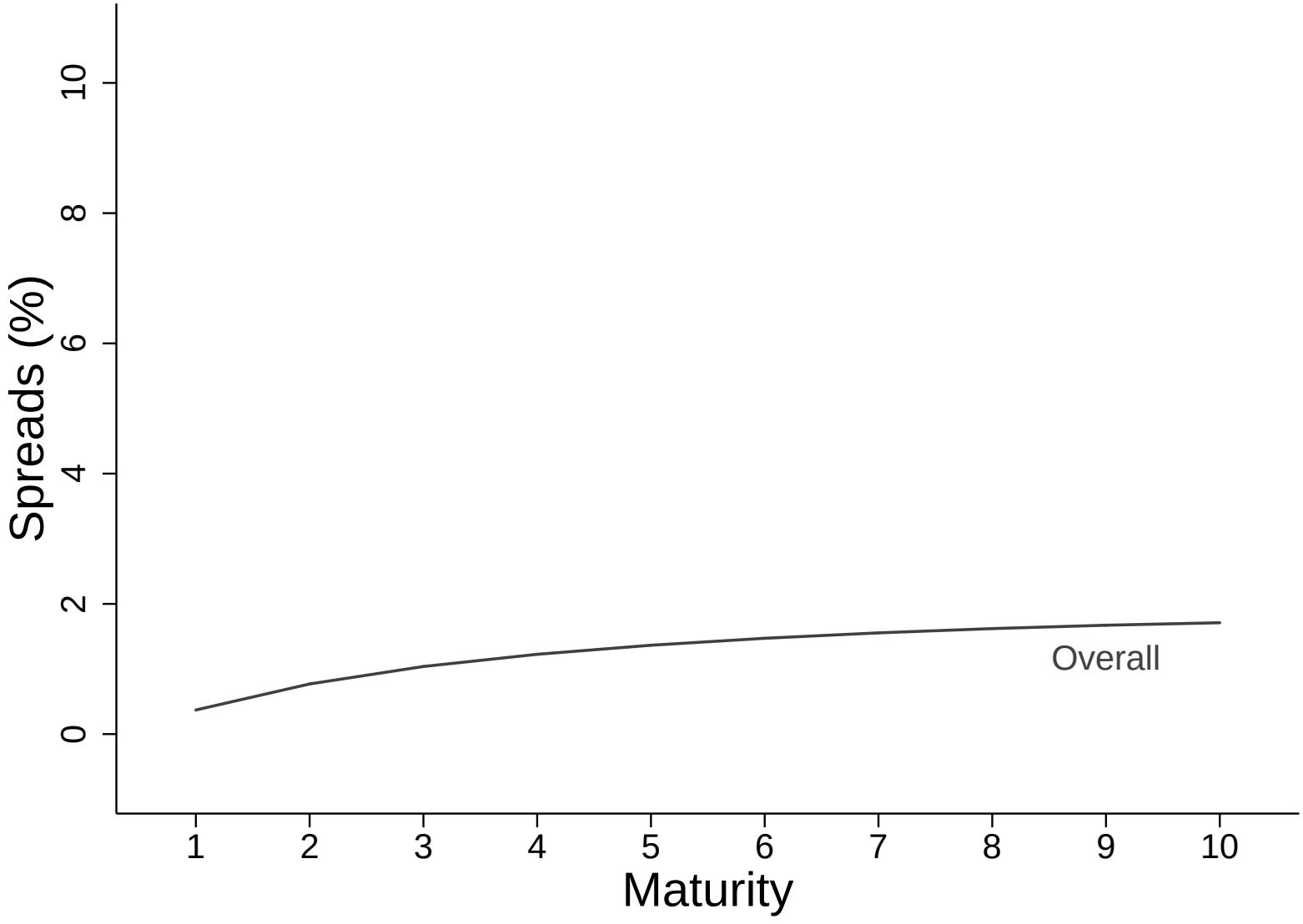
(2) Debt restructuring with probability $\epsilon = 0.5$ extending the maturity by $\eta = 2$.

(3) Risk aversion at $\gamma = 5$.

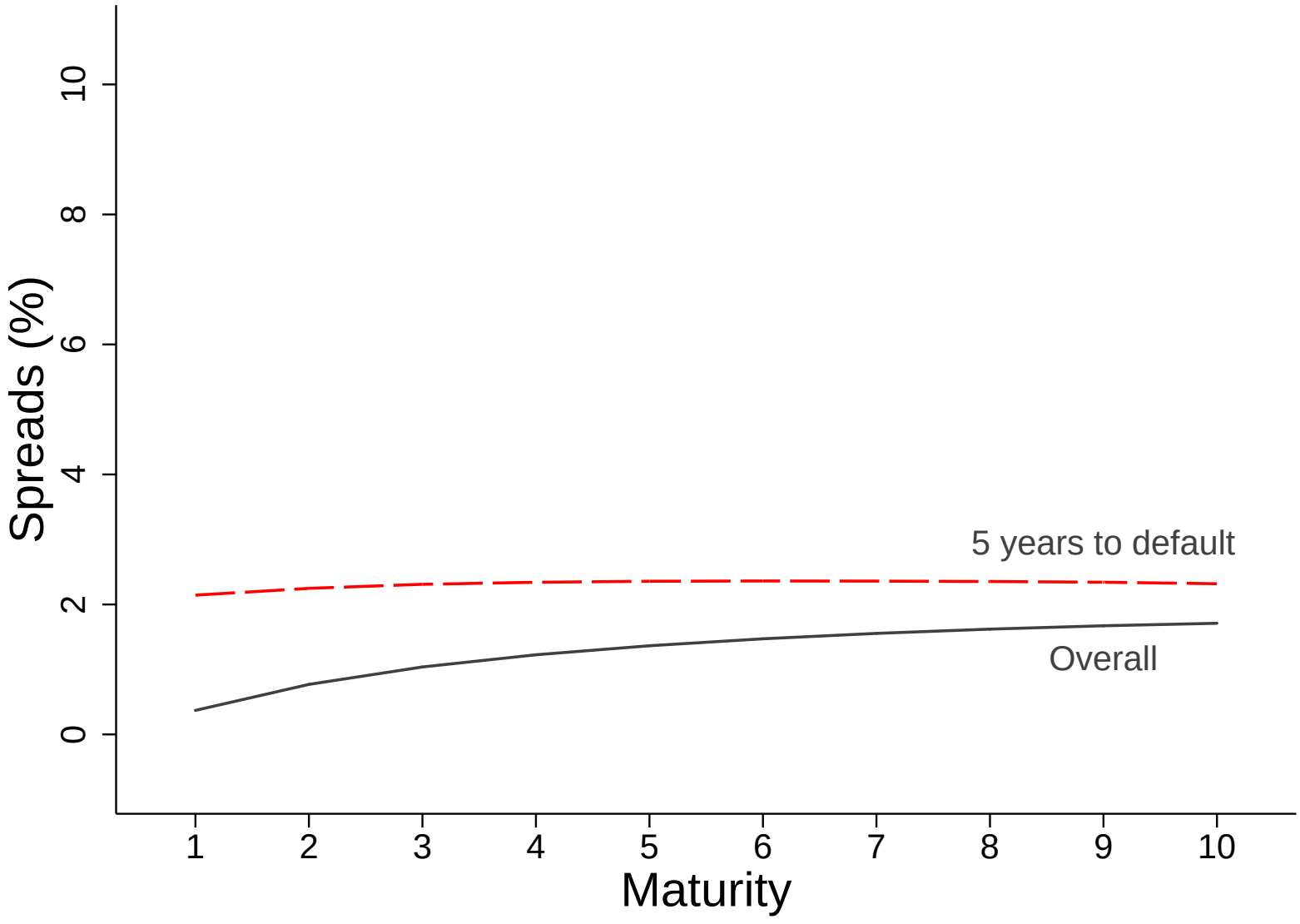
Comprehensive specification

	Data	Benchmark	Comprehensive specification
Duration	5.52	1.00	4.82
Maturity	8.78	1.00	9.72
Duration (good times)	6.99	1.00	4.98
Duration (bad times)	5.97	1.00	4.73
Maturity (good times)	10.00	1.00	9.72
Maturity (bad times)	8.71	1.00	9.67
$\sigma(log(c))/\sigma(log(y))$	1.32	1.39	1.11
$\rho(log(c), log(y))$	0.80	0.73	0.84
Reprofiling (%)		0.00	1.79
Default (%)	2.00	2.29	1.76
Value of debt / Income	0.25	0.24	0.36

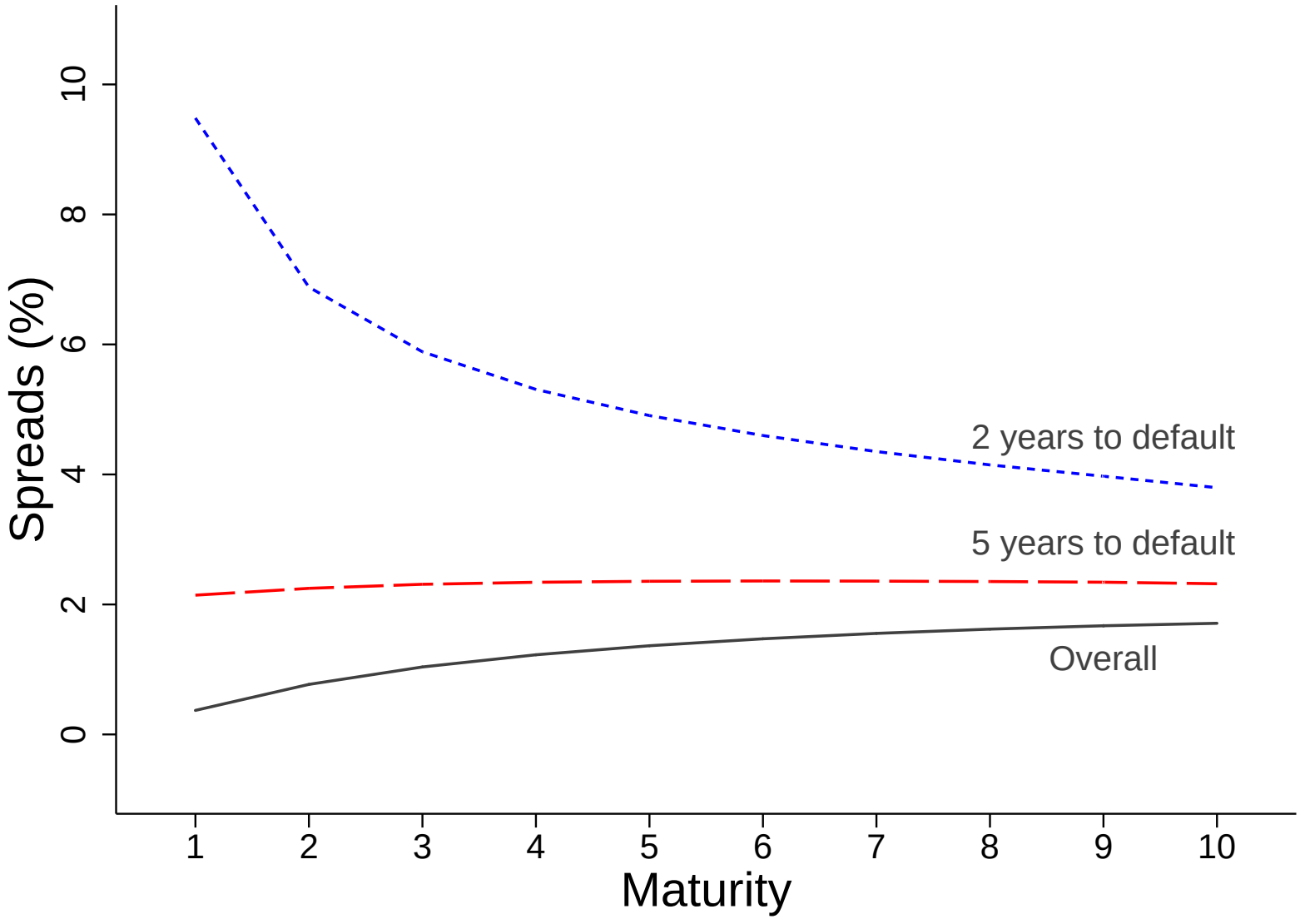
Spread curves, Comprehensive specification



Spread curves, Comprehensive specification



Spread curves, Comprehensive specification



Conclusion

- Presented a novel quantitative model of endogenous sovereign default and maturity choice.

Conclusion

- Presented a novel quantitative model of endogenous sovereign default and maturity choice.
- Identify debt dilution as a key factor reducing debt duration.

Conclusion

- Presented a novel quantitative model of endogenous sovereign default and maturity choice.
- Identify debt dilution as a key factor reducing debt duration.
- The presence of sudden stops and debt restructurings can go a long way in reconciling the model with the data.

Conclusion

- Presented a novel quantitative model of endogenous sovereign default and maturity choice.
- Identify debt dilution as a key factor reducing debt duration.
- The presence of sudden stops and debt restructurings can go a long way in reconciling the model with the data.

→ In progress: using our framework to analyze debt crisis resolution mechanisms as those proposed recently by the IMF.

Sovereign Debt Restructuring

A Dynamic Discrete Choice Approach

Maximiliano Dvorkin
FRB of St. Louis

Juan M. Sánchez
FRB of St. Louis

Horacio Sapriza
FRB of Governors

Emircan Yurdagul
Universidad Carlos III

Types of Sovereign Debt Restructurings

Type	Typical Features	Recent Episodes	NPV Haircut
Reprofiling	3-5-year maturity extension. No payments reduction.	Uruguay, 2003, Dominican Republic, 2005, Pakistan, 1999.	5 – 15% Allows full recovery of par/market value if no subsequent restructuring.
Soft Restructuring	5-10-year maturity extension. Coupons reduced 30%. No principal reduction.	Ukraine, 2000.	15 – 30% May allow full recovery of par/market value after interim loss of income.
Medium Restructuring	10-20-year maturity extension. Large reduction in coupons and moderate principal reduction.	Brady restructurings for Mexico, Brazil, Philippines and Uruguay. Ecuador, 2000.	30 – 50% Moderate permanent impairment of capital.
Hard Restructuring	20-30-year maturity extension. Large reduction in coupons and principal reduction.	Argentina, 2005 and 2010. Greece, 2012.	50 – 75% Deep permanent impairment of capital.

Source: G. Makoff (CIGI Papers No.63, April 2015). Notes

What we do (I): Endogenous Restructurings

- Model maturity choice and debt restructuring
  - Maturity is endogenous (as in Sánchez et. al, 2017)
- Lender optimally chooses the restructuring conditions
  - But the country chooses when to accept the deal
  - and which new bonds to issue
- Intervention may alleviate market failure
  - Lack of commitment (borrowers and lenders)
  - Incomplete markets / lack of insurance

Questions

1. How do market restructurings look like?
2. Is there a role for policy intervention?

What we do (II): Methodological Contribution

- Hard to solve model. Convergence is hard to obtain
- Why? Many discrete choices
  - Default or Repay
  - Maturity (e.g. years from 1 to 30)
  - Accept or reject restructuring deal
  - Debt, which is chosen from a grid
- We add shocks to choices as in the dynamic discrete choice literature and take advantage of extreme value distributions.
  - McFadden (1978), Rust (1987), among others.
- Why it works?
  It assigns similar probability to actions that give similar payoff.

Model

- Small open endowment economy à la Eaton and Gersovitz (1981)
- Lenders are risk neutral (to be changed for robustness analysis)
- The government is benevolent and households have CRRA utility
- The country cannot commit to repay its debts
- Debt is non-state-contingent with
  - A flat profile of payments b (coupon)
  - To be paid for m periods (maturity)

Decision problem

- The sovereign's choice to either pay or default is represented by

$$V(y,b,m,\epsilon) = \max \left\{ \begin{array}{l} \underbrace{\max_{b',m'} \left[\frac{c^{1-\gamma}}{1-\gamma} + \epsilon_{b',m'} \right] + \beta E_{y',\epsilon'|y} \left[V(y',b',m',\epsilon') \right]}_{\textit{repay}}; \\ \\ \underbrace{\left[\frac{(y-\Phi(y))^{1-\gamma}}{1-\gamma} + \epsilon_D \right] + \beta E_{y',\epsilon'|y} \left[V^D(y',b,m,\epsilon') \right]}_{\textit{default}} \end{array} \right\}$$

- States: income level (y), portfolio yearly payment (b), maturity of the portfolio (m), and vector of shocks to default and maturity-debt decisions (ϵ).

- If default is preferred $D(y,b,m,\epsilon) = 1$ (and otherwise is 0). Default cost is $\Phi(y)$

Decision problem: Repayment

- If the country does not default, it chooses b' and m'

$$\max_{b',m'} \left[\frac{c^{1-\gamma}}{1-\gamma} + \epsilon_{b',m'} \right] + \beta E_{\epsilon',y'|y} V(y', b', m', \epsilon'),$$

subject to

$$c = y + b - q(y, b', m'; m')b' + q(y, b', m'; m-1)b$$
$$b' \in \mathbb{R}, m' \in \mathbf{M}.$$

Debt Restructuring

- The lender makes a take-it-or-leave-it offer to the borrower.
  - It specifies a value W for the restructured portfolio.
- The borrower decides whether to accept the offer and
  - the new debt level, b^R
  - the new maturity, m^R
  - a transfer of "fresh money" from the lender, τ^R
  - such that $W = b^R \tau^R - b^R\, q(y, b^R, m^R, m^R)$

$$V^D(y,b,m,\epsilon_D) = (1-\lambda)\left[\frac{(y-\Pi(y))^{1-\gamma}}{1-\gamma}+\epsilon_D\right]+\beta E_{y',\epsilon'|y}\left[V^D(y',b,m,\epsilon'_D)\right]+$$

$$+\lambda \max\left\{\begin{array}{l}\underbrace{\left[\frac{(y-\Pi(y))^{1-\gamma}}{1-\gamma}+\epsilon_D\right]+\beta E_{y',\epsilon'|y}\left[V^D(y',b,m,\epsilon'_D)\right]}_{\text{stay in default}};\\ \underbrace{\max_{b^R,m^R,\tau^R}\left[\frac{(y-\Pi(y)-\tau^R b^R)^{1-\gamma}}{1-\gamma}+\epsilon_R\right]+\beta\, E_{y',\epsilon'|y}\left[V(y',b^R,m^R,\epsilon')\right]}_{\text{restructure the debt}}\end{array}\right\}$$

Value and Policy functions

- Simplify value functions by taking expectations over ϵ

$$V(y, b, m) = E_\epsilon[V(y, b, m, \epsilon)]$$

- This is the *ex-ante* value function (before observing ϵ)
- This problem is smooth and has fewer state variables
- The policy functions corresponding to $V(y, b, m)$ are
  - $D(y, b, m) = E_\epsilon[D(y, b, m, \epsilon)]$ is the default probability
  - $G_{y,b,m}(b', m')$ is the probability of choosing b' and m'.
  - $H(y, b, m)$ is the probability of restructuring (if possible, λ)

Prices & Lender's problem

The price of debt with maturity n if the country is not in default is

$$q(y, b', m'; n) = \frac{E_{y',|y}}{1+r} \times$$

$$\left\{ (1 - D(y', b', m')) \left[1 + \int \int q(y', b'', m''; n-1) \, \mathrm{d}G_{y',b',m'}(b'', m'') \right] + \right.$$

$$\left. D(y', b', m') \, q^D(y', b', m'; n) \right\},$$

The price of a bond with maturity n that is in default is

$$q^D(y, b, m; n) = \frac{E_{y'|y}}{1+r} \left\{ (1 - \lambda) q^D(y', b, m; n) + \right.$$

$$\left. \lambda \left[\max_W \; H(y', b, m, W) \frac{1}{-b} \frac{q^*(n)}{q^*(m)} W + \left(1 - H(y', b, m, W)\right) q^D(y', b, m; n) \right] \right\}$$

where $W = b^R \, \tau^R - b^R \, q(y, b^R, m^R; m^R)$.

Solution method

To make the problem tractable we assume

1. Portfolio maturity is a natural number: $m' \in \{1, 2, \ldots, M\}$
2. Debt can only take a value in a discrete grid of N points.
3. ϵ is a vector of size $J = M \times N$ (all possible debt and maturity combinations (b, m)), plus one more element for default and another for restructuring.
4. ϵ is i.i.d over time with a Generalized Extreme Value distribution (McFadden, 1978.)

Solution: Value functions

By properties of the distribution, the *expected* value can be computed as

$$
\begin{aligned}
V(y, b_j, m_j) &= E_\epsilon[V(y, b_j, m_j, \epsilon)] \\
&= \sigma \log \left[\begin{array}{c} \left(\sum_{\ell=1}^{J} \exp\left(\frac{c_{j\ell}^{\mathbf{1}-\gamma}}{1-\gamma} + \beta E_{y'|y}\left[V(y', b_\ell, m_\ell)\right] \right)^{1/\rho\sigma} \right)^{\rho} \\ + \exp\left(\left[\frac{(y-\Phi(y))^{\mathbf{1}-\gamma}}{1-\gamma} \right] + \beta E_{y'|y}\left[V^D(y', b_j, m_j)\right] \right)^{1/\sigma} \end{array} \right]
\end{aligned}
$$

where

$$
c_{j\ell} = y + b_j - q(y, b_\ell, m_\ell; m_j) b_\ell + q(y, b_\ell, m_\ell; m_j - 1) b_j
$$

Solution: Value functions

Similarly, for the *expected* value of being in default, the expression is

$$V^D(y, b_j, m_j) = (1-\lambda)\left(\frac{(y-\Pi(y))^{1-\gamma}}{1-\gamma} + \beta E_{y'|y}\left[V^D(y', b_j, m_j)\right]\right) +$$

$$\lambda\sigma \log\left[\begin{array}{l} \exp\left(\frac{(y-\Pi(y))^{1-\gamma}}{1-\gamma} + \beta E_{y'|y}\left[V^D(y', b_j, m_j)\right]\right)^{1/\sigma} + \\ +\left(\sum_{\ell\in\Omega} \exp\left(\frac{(y-\Pi(y)-\tau_\ell^R b_\ell^R)^{1-\gamma}}{1-\gamma} + \beta\, E_{y'|y}\left[V(y', b_\ell^R, n_\ell^R)\right]\right)^{1/\rho\sigma}\right)^{\rho} \end{array}\right]$$

where set Ω contains elements ℓ such that

$$W = b_\ell^R\, \tau_\ell^R - b_\ell^R\, q(y, b_\ell^R, m_\ell^R; m_\ell^R).$$

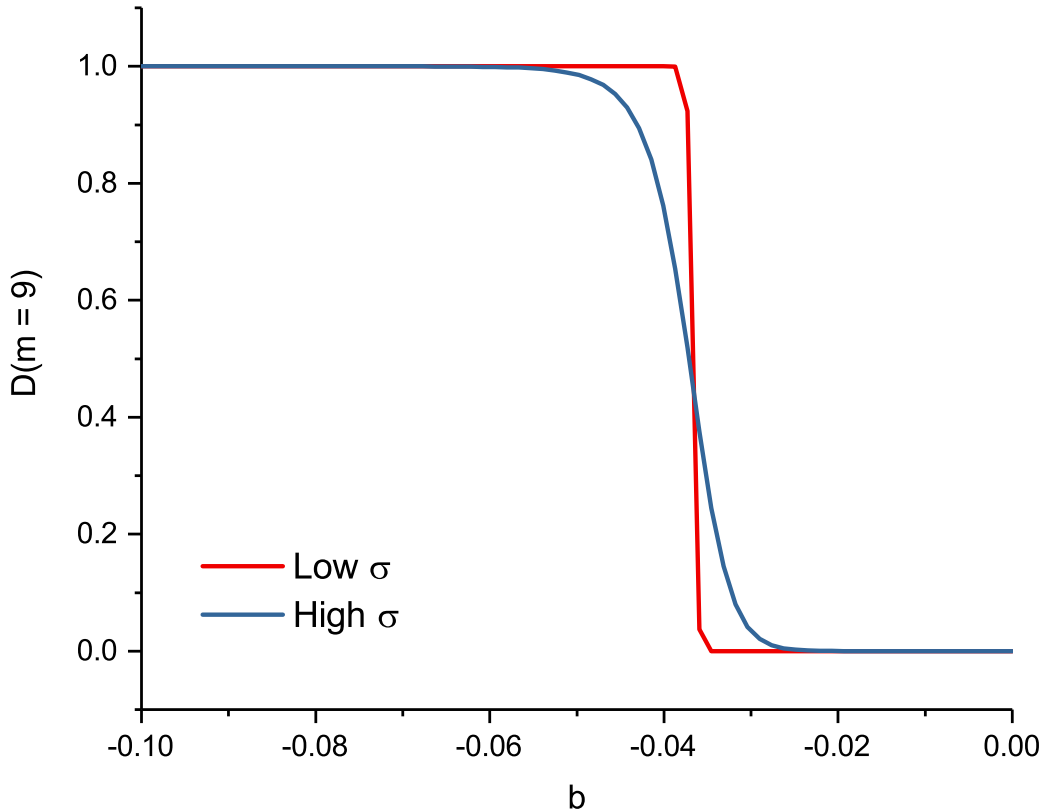
Solution: Default policy

The default probability is $D(y, b_j, m_j) =$

$$\frac{\exp\left(\left[\frac{(y-\Phi(y))^{1-\gamma}}{1-\gamma}\right]+\beta E_{y'|y}\left[V^D(y', b_j, m_j)\right]\right)^{1/\sigma}}{\left(\sum_{\ell=1}^{J}\exp\left(\frac{c_{j\ell}^{1-\gamma}}{1-\gamma}+\beta E_{y'|y}[V(y', b_\ell, m_\ell)]\right)^{\frac{1}{\rho\sigma}}\right)^{\rho}+\exp\left(\left[\frac{(y-\Phi(y))^{1-\gamma}}{1-\gamma}\right]+\beta E_{y'|y}\left[V^D(y', b_j, m_j)\right]\right)^{1/\sigma}}$$

- Default is more likely the larger is the value of default
- Probability takes a logistic structure (usual in dynamic discrete choice literature)

Policy function D: Default probability



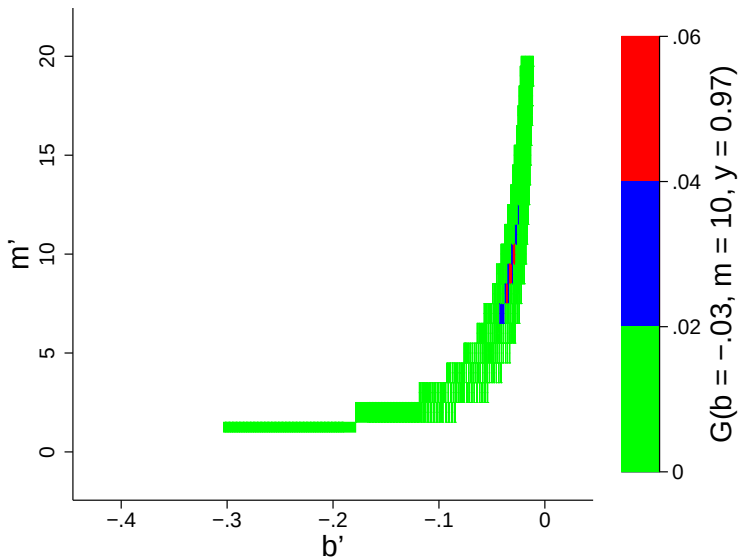
Solution: Policy on maturity and borrowing

Probability of choosing b_ℓ and m_ℓ conditional on not defaulting,

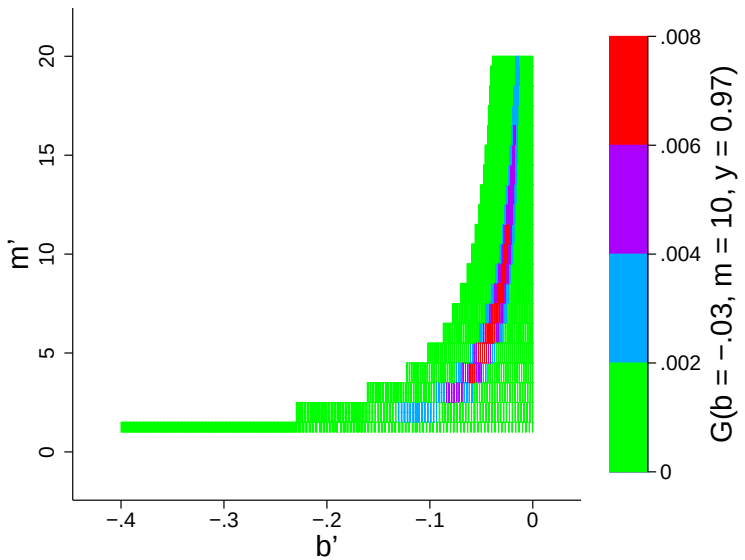
$$G(b_\ell, m_\ell | y, b_j, m_j) = \frac{\exp\left(\frac{c_{j\ell}^{1-\gamma}}{1-\gamma} + \beta E_{y'|y}\left[V(y', b_\ell, m_\ell)\right]\right)^{\frac{1}{\rho\sigma}}}{\sum_{k=1}^{J} \exp\left(\frac{c_{jk}^{1-\gamma}}{1-\gamma} + \beta E_{y'|y}\left[V(y', b_k, m_k)\right]\right)^{\frac{1}{\rho\sigma}}}$$

- The larger the value of portfolio ℓ, more likely to select it

Policy function G: Debt-maturity portfolio probability



Lower effective variance of ϵ



Higher effective variance of ϵ

Solution: Restructuring policy

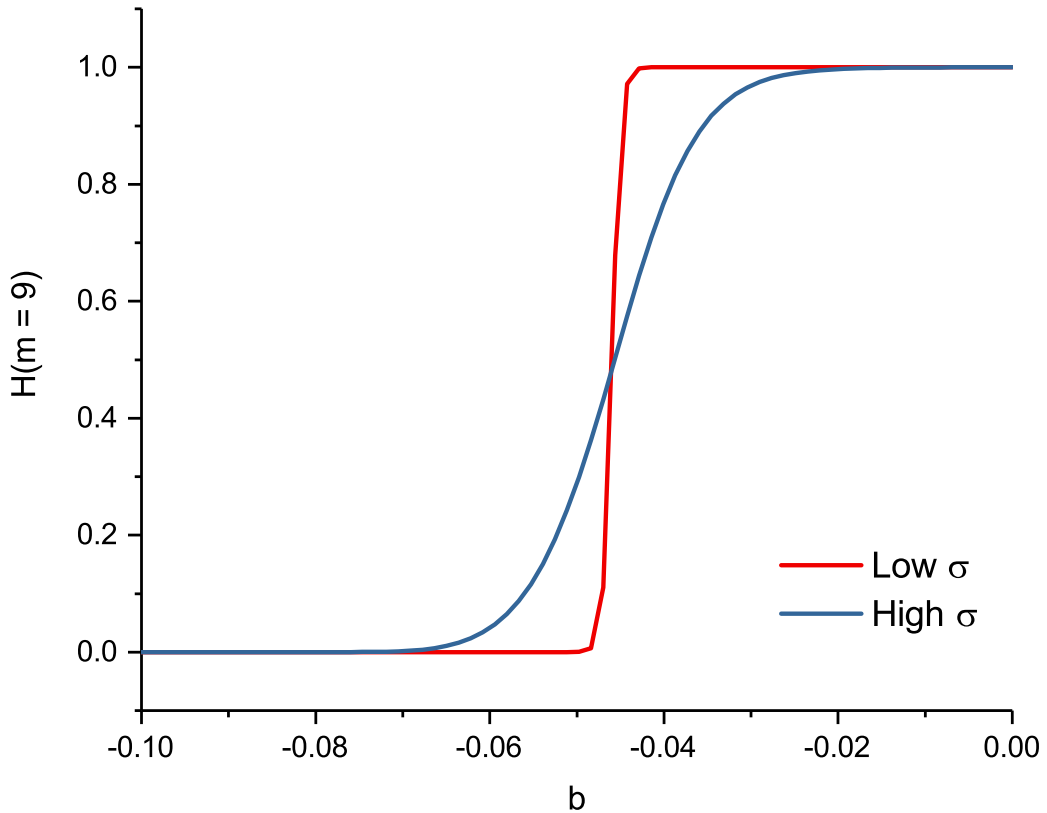
The probability that the sovereign decides to restructure if given the choice is

$$H(y, b_j, m_j, W) =$$

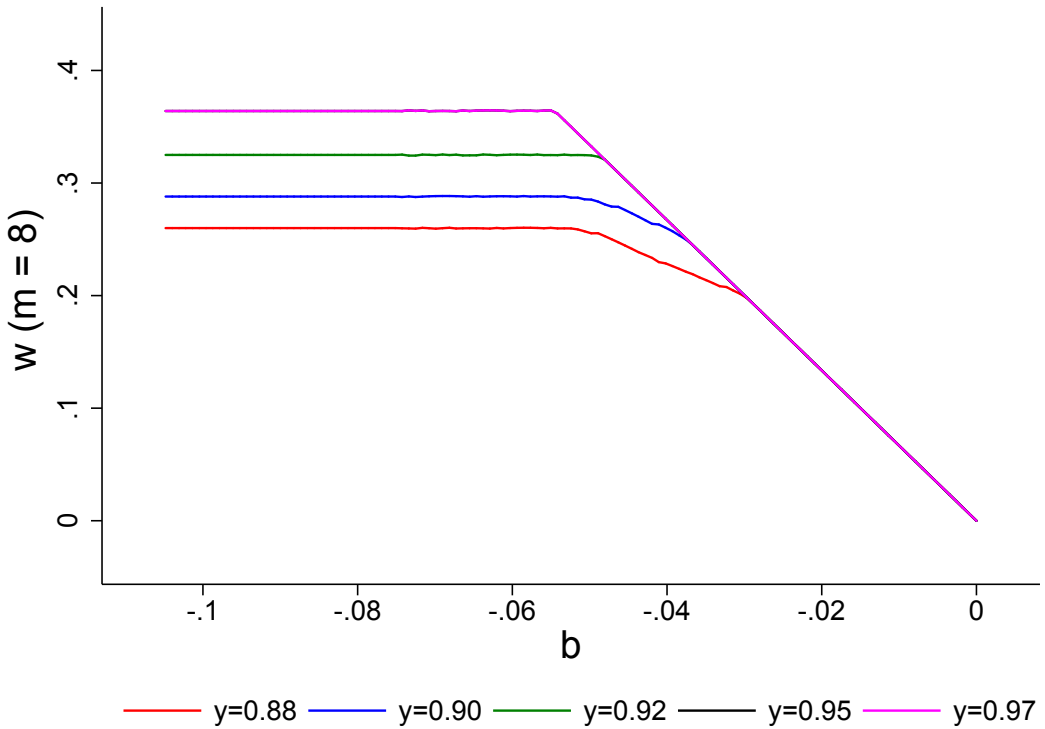
$$\frac{\left[\sum_{\ell \in \Omega} \exp\left(\frac{(y-\Pi(y)-\tau_\ell^R\, b_\ell^R)^{\mathbf{1}-\gamma}}{\mathbf{1}-\gamma} + \beta E_{y'|y}\left[V(y', b_\ell^R, m_\ell^R)\right]\right)^{\mathbf{1}/\rho\sigma}\right]^{\rho}}{\exp\left(\frac{(y-\Pi(y))^{\mathbf{1}-\gamma}}{\mathbf{1}-\gamma} + \beta E_{y'|y}\left[V^D(y', b_j, m_j)\right]\right)^{\frac{\mathbf{1}}{\sigma}} + \left[\sum_{\ell \in \Omega} \exp\left(\frac{(y-\Pi(y)-\tau_\ell^R\, b_\ell^R)^{\mathbf{1}-\gamma}}{\mathbf{1}-\gamma} + \beta E_{y'|y}\left[V(y', b_\ell^R, m_\ell^R)\right]\right)^{\mathbf{1}/\rho\sigma}\right]^{\rho}}$$

- Restructuring is more likely the larger is the value of returning to financial markets

Policy function H: Restructuring probability



Lender's decision rule: Recovery increases with income



- We impose a constraint on W: no negative haircuts

Calibration: parameters calibrated outside the model

- Introduce sudden stop shocks
  - Prevent the sovereign from issuing new debt
  - 2x2 transition matrix estimated from Comelli (2015)
  - Probability of transition from non-SS $\rightarrow$ SS is 11.8%
  - Probability of transition from SS $\rightarrow$ SS is 46.8%

- Parameters on ϵ shocks
  - Variance parameter, $\sigma = 0.003$
  - Correlation parameter, $\rho = 1$ ▸ preliminary

- Interest rate, $r = 4\%$, and risk aversion, $\gamma = 2$, standard in the literature

- Income shock std dev, $\sigma_y = 0.019$, and autocorrelation, $\rho_y = 0.86$, estimated for Colombia.

Calibration: parameters jointly calibrated

- Output losses in default and exclusion
  - In default: $\Phi(y) = \max\{0, y - \phi\}$
  - In exclusion: $\Pi(y) = \max\{0, y - \pi\}$
  - We make $\phi = \pi$

Parameter	Value	Basis	Data	Model
Default cost, ϕ	0.945	Default rate	3%	2.9%
Restruc. offer prob, λ	0.20	Exclusion duration	4 yrs	3.9 yrs
Discount factor, β	0.92	Debt/output	35%	38%

Model Evaluation: Fit of non-targeted moments

- The model fits well various types of non-targeted statistics
  - business cycles
  - maturity and duration
  - spreads and yield curve

Table: Business Cycles

Statistics	Brazil	Chile	Colombia	Mexico	Model
$\sigma(log(c))/\sigma(log(y))$	1.15	1.74	1.15	1.65	1.42
$\sigma(log(TB/y))/\sigma(log(y))$	0.57	0.92	1.36	1.35	1.05
$\rho(log(c), log(y))$	0.75	0.85	0.90	0.69	0.66
$\rho(TB/y, log(y))$	0.16	-0.29	-0.08	-0.09	0.06

Model Evaluation: Fit of non-targeted moments

Table: Maturity and Duration

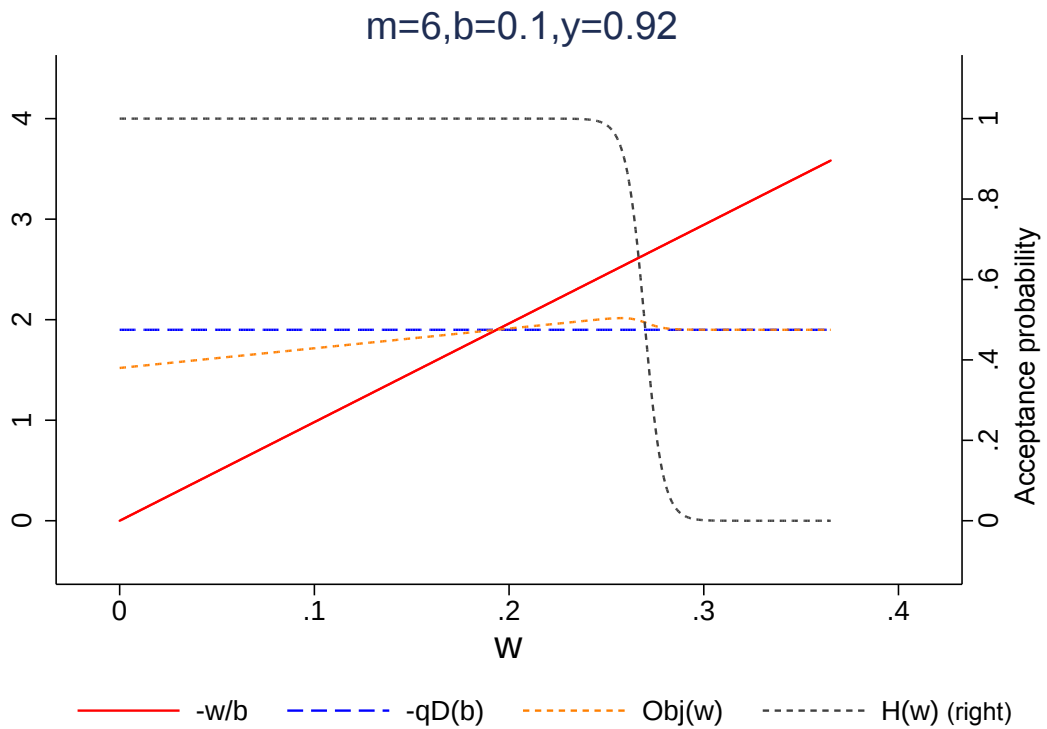
Statistics	Brazil	Chile	Colombia	Mexico	Model
Duration	3.41	4.31	5.42	5.61	3.71
Duration (bad times)	3.18	4.31	4.26	5.57	3.30
Maturity	5.98	5.22	9.69	10.90	7.13
Maturity (bad times)	5.64	5.22	7.34	10.68	6.41
$\rho(mat, log(y))$	0.65	0.58	0.93	0.03	0.24
$\rho(dur, log(y))$	0.69	0.53	0.93	-0.20	0.33
St. dev Duration	0.27	1.66	0.89	0.66	1.14
St. dev Maturity	0.46	2.34	1.80	1.67	2.62
Autocorr duration	0.96	0.83	0.92	0.38	0.84
Autocorr maturity	0.93	0.84	0.96	0.56	0.82

Model Evaluation: Fit of non-targeted moments

Table: Spreads and the Yield Curve

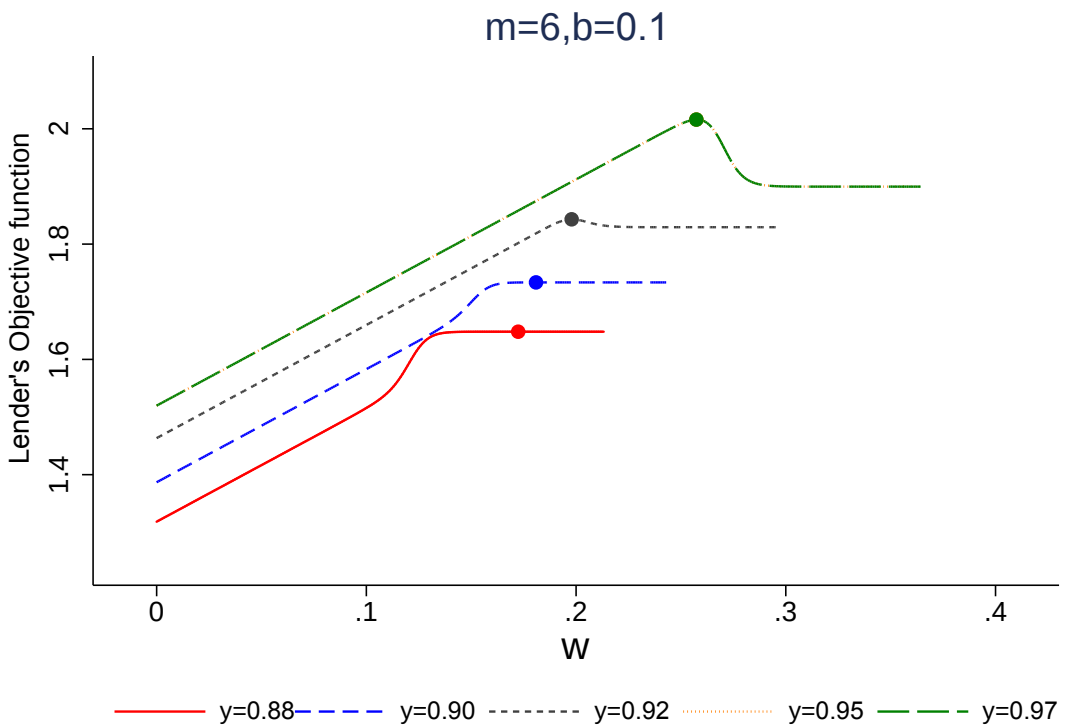
Statistics	Brazil	Chile	Colombia	Mexico	Model
1-year spread	1.01	2.31	0.96	0.64	0.00
1-year spread (bad times)	1.78	3.31	1.42	0.80	0.04
10-year spread	2.19	0.72	2.33	1.38	1.15
10-year spread (bad times)	7.08	0.84	4.32	1.78	1.43
$\rho(1YS, log(y))$	-0.43	0.07	-0.61	-0.14	-0.22
$\rho(10YS, log(y))$	-0.74	0.13	-0.89	-0.17	-0.33
$10YS - 1YS$	1.07	-1.52	1.59	0.79	0.96
$\sigma(10YS - 1YS)$	1.84	1.30	1.37	0.71	1.13

Endogenous restructurings: lender's objective

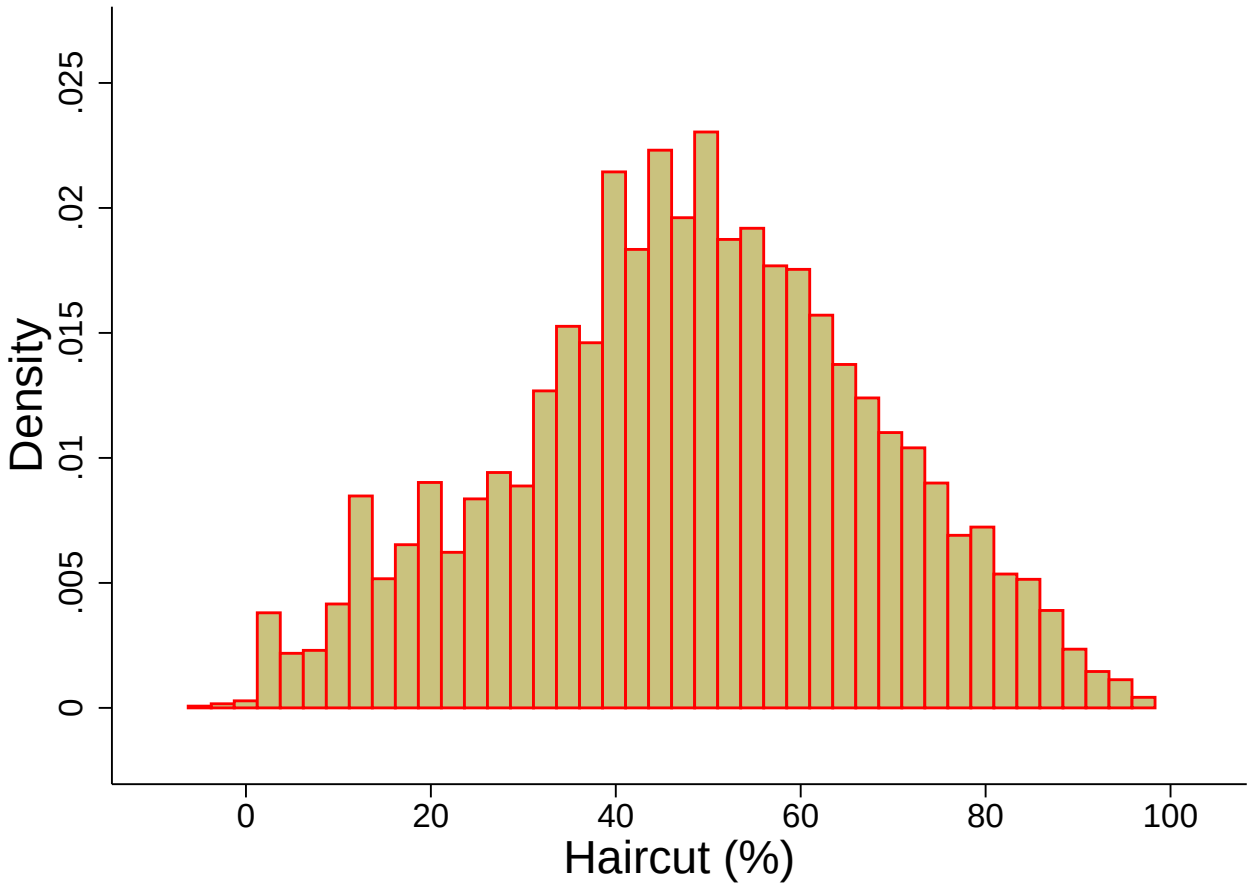


$$Obj(W) = -b\, q^D(y', b, m; n) + \lambda \left[\max_W \; H(y', b, m, W) \left(\frac{q^*(n)}{q^*(m)} \frac{W}{-b} - q^D(y', b, m; n) \right) \right]$$

Endogenous restructurings: lender's objective & GDP

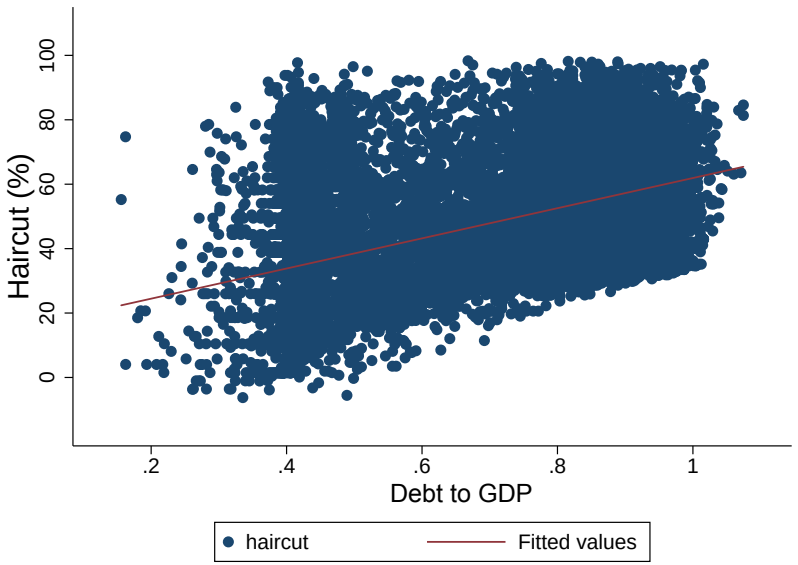


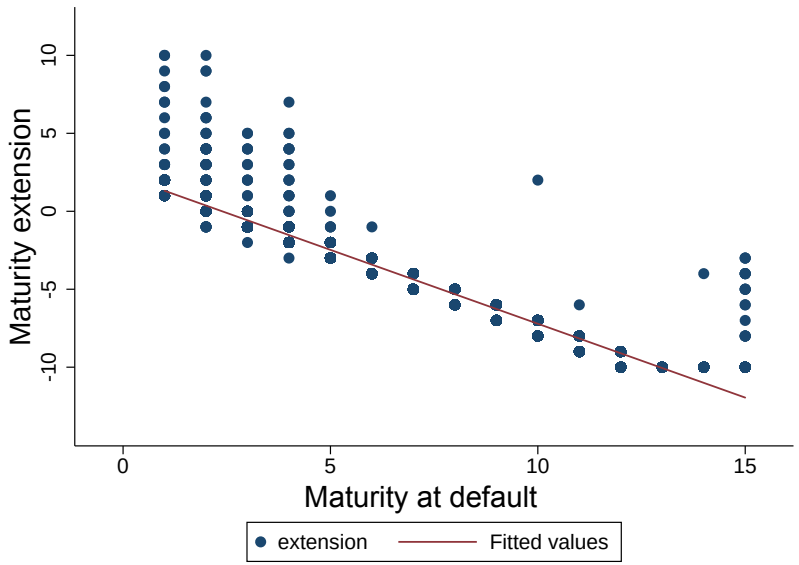
Endogenous restructurings: haircut heterogeneity



- Haircut varies with:
  - current income, debt and maturity at default, duration of default

Characteristics of Restructurings

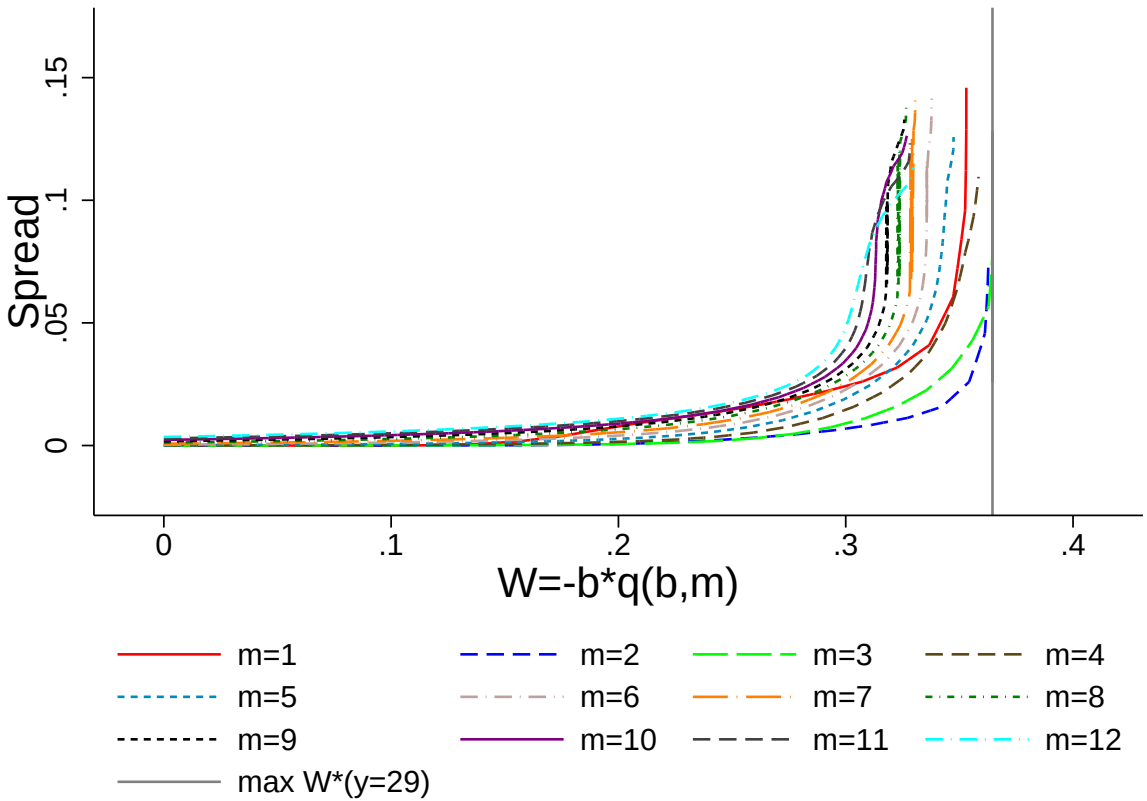




- Haircuts are larger for high debt to income at default
- New maturity is much shorter after restructuring

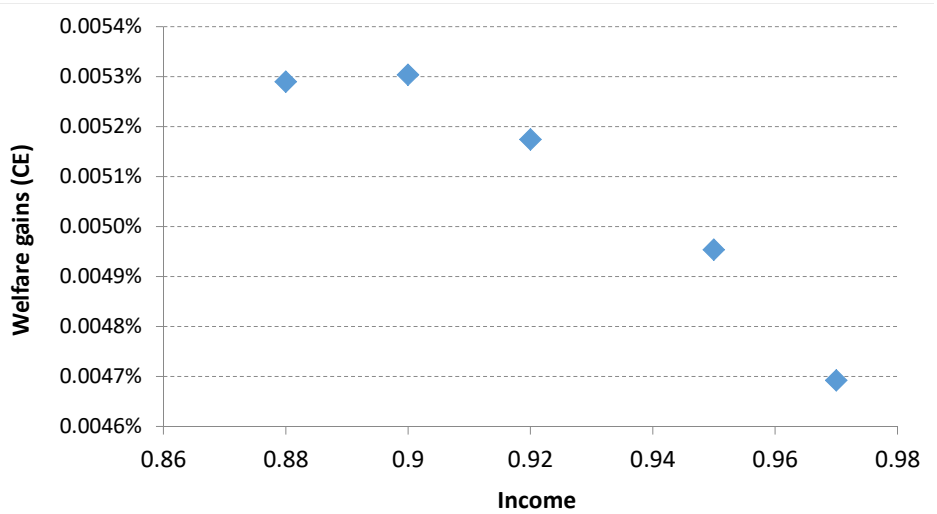
Policy Intervention

- The market alone wants high recovery, which can only be achieved with low maturity



Policy Intervention

- We study the effects of a policy that imposes a minimum haircut
  - Haircut must be 10% or more
  - Higher ex-ante welfare (debt=0) and increases maturity extensions



Model	Debt / GDP	Default rate	Median Haircut	Median maturity after restructuring
Free market	38%	2.9%		2.5 yrs
Small intervention	38%	3.0%		2.9 yrs

Conclusions

- Present a novel model of sovereign debt restructuring
  - Maturity is endogenously chosen
  - Restructuring conditions are endogenously chosen
- Incorporate features of dynamic discrete choice literature
  - random component adds smoothness
  - intuitive economic interpretation of these shocks
  - in the ex-ante problem they are not state variables (tractable)
- Model replicates facts about maturity, spreads, and business cycles
- Model captures the heterogeneity in haircuts
- Market restructuring pushes for high recovery and low extensions
- Policies that impose some minimum haircuts may be desirable and can account for longer maturity extensions

In progress

- ϵ shocks have an economic interpretation
- Their size (variance) affect decisions
  - The larger the shocks, less important market forces for the default and maturity/debt decisions
- Calibration strategy
  - in a panel of similar countries, σ and ρ are related to the dispersion in the default and maturity decisions
  - i.e. how likely is that 2 identical countries make different decisions
  - For default: run a Logit regression on income, debt and maturity. Use the result to plot the policy function
  - For debt/maturity: how often do countries change their portfolio?

back

Notes

1. The Appendix presents a range of stylized debt restructurings to illustrate the options available and the NPV effect.
2. Data sources: Sturzenegger and Zettlemeyer (2006,194, 323, 213, 236, 285, 333); Zettlemeyer, Trebesch and Gulati (2013,19); and Cline (1995, 234-35).
3. Most market analysts use exit yields in the 15–20% range to calculate NPV effect of changes in cash flows in a distressed exchange offer. NPV effects ranges are calculated here based on a common 10% discount rate to allow inter-transaction comparability as suggested by Das, Papaioannou and Trebesch (2012a, 10; 2012b).
4. Debt relief enjoyed by an issuer is typically much less than the market NPV loss, as pointed out by Sturzenegger and Zettlemyer (2007). A 6% discount rate is used here to reasonably reflect the cost of servicing debt in normal times. The expected inflation rate or the cost of any available concessional financing might be alternatives.

Back